Three-Month Student Project

AI/ML-Augmented Black–Scholes Framework

for Option Pricing and Market-Risk Analysis

Mathematical Modelling

$\longrightarrow$

Black–Scholes Pricing

$\longrightarrow$

Numerical Validation

$\longrightarrow$

Machine Learning

$\longrightarrow$

Hybrid Pricing

$\longrightarrow$

Market-Risk Analysis

Project Duration: 3 Months

Primary Reference

Carol Alexander
Market Risk Analysis, Volume I: Quantitative Methods in Finance
Wiley Finance Series

Department of Mathematics

BITS Pilani, Hyderabad Campus

Contents

1 Project Motivation **12**

1.1 Classical Mathematical Modelling of Financial Markets . . . 12
1.2 Black–Scholes Option Pricing . . . 12
1.3 Risk-Neutral Valuation and Numerical Verification . . . 13
1.4 The Role of Volatility . . . 14
1.5 From Black–Scholes to Model Risk . . . 15
1.6 Why Machine Learning? . . . 16
1.7 Residual-Learning Black–Scholes Model . . . 16
1.8 Machine Learning for Implied Volatility . . . 17
1.9 From Pricing to Market-Risk Analysis . . . 17
1.10 Overall Philosophy of the Project . . . 18
1.11 Central Research Question . . . 19

2 Project Objectives **19**

2.1 Overall Objective . . . 19
2.2 Specific Objective 1: Develop the Mathematical Black–Scholes Framework . . . 20
2.3 Specific Objective 2: Validate Black–Scholes Using Independent Numerical Methods 21
2.4 Specific Objective 3: Compute Option Sensitivities and Implied Volatility . . . 22
2.5 Specific Objective 4: Construct a Machine-Learning Dataset . . . 23
2.6 Specific Objective 5: Develop and Compare Selected Machine-Learning Models . 24
2.7 Specific Objective 6: Develop a Hybrid Black–Scholes–ML Model . . . 24
2.8 Specific Objective 7: Evaluate Pricing Accuracy and Generalization . . . 25
2.9 Specific Objective 8: Extend the Framework to Market-Risk Analysis . . . 26
2.10 Final Project Success Criteria . . . 27

3 Mathematical Foundation **28**

3.1 Simple Returns and Log Returns . . . 28
3.2 Brownian Motion . . . 29
3.3 Geometric Brownian Motion . . . 30
3.4 From Geometric Brownian Motion to Log Prices . . . 31
3.5 Itô's Lemma . . . 32
3.6 Construction of a Delta-Hedged Portfolio . . . 33
3.7 Derivation of the Black–Scholes PDE . . . 34
3.8 An Important Consequence of the PDE . . . 34
3.9 Risk-Neutral Dynamics . . . 35

3.10 European Call and Put Payoffs . . . 35
3.11 Black–Scholes Analytical Solution . . . 36
3.12 Put–Call Parity . . . 36
3.13 Moneyness . . . 37
3.14 Option Greeks . . . 37
3.14.1 Delta . . . 38
3.14.2 Gamma . . . 38
3.14.3 Vega . . . 38
3.15 Taylor Approximation and Local Market Risk . . . 38
3.16 Implied Volatility as an Inverse Problem . . . 39
3.17 Newton–Raphson Method for Implied Volatility . . . 40
3.18 From Constant Volatility to a Volatility Surface . . . 40
3.19 Mathematical Inputs to the Machine-Learning Model . . . 41
3.20 Required Mathematical Competency Before ML . . . 42

4 Computational Black–Scholes Framework 44
4.1 Computational Environment . . . 44
4.2 Algorithm 1: Black–Scholes Analytical Pricer . . . 45
4.2.1 Basic Computational Checks . . . 45
4.3 Algorithm 2: Analytical Greeks . . . 46
4.4 Algorithm 3: Numerical Greeks . . . 47
4.4.1 Step-Size Study . . . 47
4.5 Algorithm 4: Implied-Volatility Solver . . . 48
4.5.1 Newton–Raphson Iteration . . . 48
4.5.2 Robustness Check . . . 49
4.6 Algorithm 5: Binomial Option-Pricing Model . . . 49
4.6.1 Asset-Price Lattice . . . 50
4.6.2 Terminal Payoff . . . 50
4.6.3 Backward Induction . . . 50
4.7 Binomial Convergence Study . . . 51
4.8 Algorithm 6: Monte Carlo Pricing . . . 51
4.9 Monte Carlo Statistical Error . . . 52
4.10 Monte Carlo Convergence Experiment . . . 52
4.10.1 Repeated Monte Carlo Runs . . . 52
4.11 Algorithm 7: Simulation of Complete GBM Paths . . . 53
4.12 Unified Cross-Validation of the Three Pricing Methods . . . 54

4.13 Parameter-Sensitivity Experiments . . . 54
4.13.1 Underlying Asset Price . . . 54
4.13.2 Strike Price . . . 55
4.13.3 Volatility . . . 55
4.13.4 Time to Maturity . . . 55
4.13.5 Interest Rate . . . 55
4.14 Moneyness-Based Computational Study . . . 56
4.15 Required Computational Plots . . . 56
4.16 Computational Validation Gate . . . 57
4.17 Output of the Computational Stage . . . 58

5 AI/ML Formulation 58
5.1 Supervised-Learning Framework . . . 58
5.2 Choice of Financial Features . . . 59
5.3 Learning Problem 1: Direct Option-Price Prediction . . . 60
5.4 Learning Problem 2: Black–Scholes Residual Learning . . . 60
5.5 Interpretation of Residual Learning . . . 61
5.6 Residual-Learning Architecture . . . 62
5.7 Learning Problem 3: Implied-Volatility Prediction . . . 62
5.8 Pricing Through Predicted Implied Volatility . . . 62
5.9 Comparison of the Three Learning Formulations . . . 63
5.10 Training, Validation and Test Sets . . . 63
5.10.1 Training Set . . . 64
5.10.2 Validation Set . . . 64
5.10.3 Test Set . . . 64
5.11 Time-Aware Data Splitting for Market Data . . . 64
5.12 Feature Scaling . . . 65
5.13 Avoiding Data Leakage . . . 65
5.14 Loss Function . . . 66
5.15 Linear Regression as the Baseline Model . . . 67
5.16 Ridge Regression . . . 67
5.17 Nonlinear Machine-Learning Models . . . 68
5.18 Tree-Based Regression . . . 68
5.19 Small Feed-Forward Neural Network . . . 68
5.20 Restriction on Neural-Network Complexity . . . 69
5.21 Bias–Variance Trade-Off . . . 69

5.22 Training Error and Validation Error . . . 70
5.23 Hyperparameter Selection . . . 70
5.24 Model Comparison . . . 71
5.25 Primary Research Comparison . . . 71
5.26 Generalization Beyond the Training Region . . . 72
5.27 Controlled Extrapolation Experiment . . . 72
5.28 Financial Consistency Checks for ML Predictions . . . 73
5.29 Model Error as a Function of Market State . . . 73
5.30 Research Hypotheses . . . 74
5.30.1 Hypothesis 1 . . . 74
5.30.2 Hypothesis 2 . . . 74
5.30.3 Hypothesis 3 . . . 74
5.30.4 Hypothesis 4 . . . 74
5.30.5 Hypothesis 5 . . . 74
5.31 AI/ML Development Gate . . . 75
5.32 Output of the AI/ML Formulation . . . 76

6 Dataset Generation and Feature Engineering 76
6.1 Stage I: Synthetic Black–Scholes Dataset . . . 77
6.2 Selection of Parameter Ranges . . . 77
6.3 Sampling Strategy . . . 78
6.4 Importance of Dimensionless Variables . . . 78
6.5 Basic Synthetic Feature Set . . . 79
6.6 Extended Synthetic Feature Set . . . 79
6.7 Normalization by Strike . . . 80
6.8 Synthetic Dataset Validation . . . 80
6.9 Distributional Inspection of Synthetic Features . . . 81
6.10 Coverage of the Option Space . . . 81
6.11 Maturity Buckets . . . 82
6.12 Stage II: Real Market Option Dataset . . . 82
6.13 Choice of Market Price . . . 83
6.14 Time to Maturity . . . 83
6.15 Risk-Free Interest Rate . . . 84
6.16 Dividend Treatment . . . 84
6.17 Filtering Invalid or Unusable Market Observations . . . 84
6.18 No-Arbitrage Screening . . . 85

6.19 Computing Market Implied Volatility . . . 85
6.20 Computing the Black–Scholes Benchmark Price . . . 86
6.21 Historical Volatility Feature . . . 86
6.22 Black–Scholes Residual Dataset . . . 87
6.23 Implied-Volatility Dataset . . . 87
6.24 Volatility Surface Representation . . . 88
6.25 Feature Matrix . . . 88
6.25.1 Direct Price Learning . . . 89
6.25.2 Residual Learning . . . 89
6.25.3 Implied-Volatility Learning . . . 89
6.26 Feature Correlation Analysis . . . 89
6.27 Optional Principal Component Analysis . . . 89
6.28 Training, Validation and Test Split . . . 90
6.29 Standardization of Features . . . 90
6.30 Missing Data . . . 91
6.31 Outlier Inspection . . . 91
6.32 Final Dataset Structure . . . 92
6.33 Three Final ML Datasets . . . 92
6.33.1 Dataset A: Direct Pricing . . . 92
6.33.2 Dataset B: Residual Learning . . . 92
6.33.3 Dataset C: Implied Volatility . . . 92
6.34 Required Exploratory Plots . . . 92
6.35 Dataset Quality Gate . . . 93
6.36 Output of the Data Preparation Stage . . . 94

7 Machine-Learning Algorithms 94
7.1 Common Supervised-Learning Formulation . . . 94
7.2 Algorithm 1: Ordinary Least-Squares Regression . . . 95
7.3 Role of Linear Regression . . . 96
7.4 Algorithm 2: Ridge Regression . . . 96
7.5 Interpretation of the Regularization Parameter . . . 97
7.6 Algorithm 3: Tree-Based Ensemble Model . . . 98
7.7 Random Forest Regression . . . 98
7.8 Why Random Forest is Useful Here . . . 98
7.9 Principal Random-Forest Hyperparameters . . . 99
7.10 Alternative: Gradient-Boosted Trees . . . 99

7.11 Algorithm 4: Feed-Forward Neural Network . . . 100
7.12 Activation Function . . . 100
7.13 Recommended Neural-Network Architecture . . . 101
7.14 Neural-Network Loss Function . . . 101
7.15 Gradient-Based Training . . . 102
7.16 Epochs and Mini-Batches . . . 102
7.17 Early Stopping . . . 102
7.18 Algorithm Training Sequence . . . 103
7.19 Common Evaluation Metrics . . . 103
7.20 Model-Complexity Comparison . . . 104
7.21 Feature Importance . . . 104
7.22 Residual Analysis . . . 104
7.23 Model Performance by Moneyness . . . 105
7.24 Model Performance by Maturity . . . 106
7.25 Training Stability . . . 106
7.26 Model Selection Principle . . . 106
7.27 Required Plots for Each Model . . . 107
7.28 Algorithmic Comparison Table . . . 107
7.29 Machine-Learning Algorithm Gate . . . 108

8 Hybrid Black–Scholes–ML Framework 109
8.1 Model 1: Classical Black–Scholes Benchmark . . . 109
8.2 Model 2: Pure Machine-Learning Pricing . . . 109
8.3 Model 3: Black–Scholes Residual Learning . . . 110
8.4 Interpretation of the Residual Model . . . 110
8.5 Residual-Learning Architecture . . . 111
8.6 Choice of Residual-Learning Features . . . 111
8.7 Model 4: ML Implied Volatility + Black–Scholes . . . 112
8.8 Implied-Volatility Hybrid Architecture . . . 112
8.9 Why the Two Hybrid Models are Different . . . 113
8.10 Critical Separation of the Two Experiments . . . 113
8.11 Training Architecture for the Four Models . . . 114
8.11.1 Black–Scholes . . . 114
8.11.2 Pure ML . . . 115
8.11.3 Residual ML . . . 115
8.11.4 Implied-Volatility ML . . . 115

8.12 Unified Pricing Comparison . . . 115
8.13 Pricing Errors . . . 116
8.14 Improvement Relative to Black–Scholes . . . 116
8.15 Error Reduction by Market Region . . . 117
8.16 Residual Structure Before and After ML Correction . . . 117
8.17 Financial Consistency of the Hybrid Price . . . 118
8.18 Monotonicity Check . . . 118
8.19 Volatility Monotonicity . . . 119
8.20 Hybrid Greeks . . . 119
8.21 Numerical Hybrid Delta . . . 119
8.22 Risk Interpretation of the IV-ML Model . . . 120
8.23 Out-of-Sample Hybrid Evaluation . . . 120
8.24 Stress Testing the Four Pricing Models . . . 120
8.25 Hybrid Model Selection Criteria . . . 121
8.26 Required Hybrid-Model Plots . . . 122
8.27 Hybrid Framework Gate . . . 122

9 Model Validation and Performance Metrics 123
9.1 Models to be Evaluated . . . 123
9.2 Prediction Error . . . 124
9.3 Mean Error . . . 124
9.4 Mean Absolute Error . . . 125
9.5 Mean Squared Error . . . 125
9.6 Root Mean Square Error . . . 125
9.7 Coefficient of Determination . . . 125
9.8 Normalized Pricing Error . . . 126
9.9 Error Relative to Bid–Ask Spread . . . 126
9.10 Improvement Relative to Black–Scholes . . . 127
9.11 Percentage Error Reduction . . . 127
9.12 Predicted Versus Observed Price Plot . . . 127
9.13 Residual Diagnostics . . . 128
9.14 Desired Residual Behaviour . . . 129
9.15 Comparison of Black–Scholes and Hybrid Residuals . . . 129
9.16 Performance by Moneyness . . . 129
9.17 Performance by Maturity . . . 130
9.18 Two-Dimensional Error Map . . . 131

9.19 Implied-Volatility Model Validation . . . 131
9.20 Volatility-Surface Error . . . 132
9.21 Training Versus Validation Performance . . . 132
9.22 Final Test Evaluation . . . 133
9.23 Chronological Out-of-Sample Validation . . . 133
9.24 Rolling Validation as an Optional Extension . . . 134
9.25 Financial No-Arbitrage Checks . . . 134
9.26 Monotonicity with Respect to the Underlying . . . 134
9.27 Convexity with Respect to the Underlying . . . 135
9.28 Volatility Sensitivity Check . . . 135
9.29 Stress-Test Validation . . . 135
9.30 Model Response Under Stress . . . 136
9.31 Interpolation and Extrapolation Tests . . . 136
9.32 Stability Across Random Seeds . . . 137
9.33 Model Ranking . . . 137
9.34 Acceptance Criterion for ML Improvement . . . 138
9.35 Required Validation Figures . . . 138
9.36 Validation Gate . . . 139

10 Market-Risk Analysis 140

10.1 Construction of a Simple Option Portfolio . . . 140
10.2 Portfolio Delta . . . 141
10.3 Interpretation of Portfolio Delta . . . 142
10.4 Portfolio Gamma . . . 142
10.5 Portfolio Vega . . . 142
10.6 Delta–Gamma–Vega Approximation . . . 143
10.7 Full Repricing . . . 143
10.8 Taylor Approximation Error . . . 144
10.9 Local Versus Nonlocal Risk . . . 144
10.10 Underlying-Price Stress Scenarios . . . 145
10.11 Volatility Stress Scenarios . . . 145
10.12 Combined Price–Volatility Stress . . . 146
10.13 Stress Matrix . . . 146
10.14 Black–Scholes Portfolio Risk . . . 146
10.15 Hybrid Portfolio Risk . . . 147
10.16 Comparison of Classical and Hybrid Sensitivities . . . 147

10.17Risk Difference Between Models . . . 148
10.18Risk Contribution of Individual Positions . . . 148
10.19Risk Decomposition . . . 149
10.20Delta-Hedged Portfolio Experiment . . . 149
10.21Comparison Before and After Delta Hedging . . . 150
10.22Optional Monte Carlo Portfolio-Risk Experiment . . . 150
10.23Empirical P&L Distribution . . . 151
10.24Why Full VaR is Not the Core Objective . . . 151
10.25Stress Behaviour of the ML Model . . . 151
10.26Stress Distance from the Training Domain . . . 152
10.27Model-Risk Map . . . 153
10.28Required Market-Risk Figures . . . 153
10.29Market-Risk Comparison Table . . . 154
10.30Central Risk Question . . . 154
10.31Market-Risk Analysis Gate . . . 154

11 Three-Month Work Plan **155**

11.1 Week 1: Financial Returns, Probability and Data Handling . . . 156
11.2 Week 2: Brownian Motion and Geometric Brownian Motion . . . 157
11.3 Week 3: Black–Scholes Mathematics and Analytical Pricing . . . 158
11.4 Week 4: Greeks and Implied Volatility . . . 159
11.5 Week 5: Binomial and Monte Carlo Validation . . . 160
11.6 Week 6: Synthetic Dataset and Feature Engineering . . . 161
11.7 Week 7: Regression Baselines . . . 162
11.8 Week 8: Tree-Based Model and Neural Network . . . 164
11.9 Week 9: Real Market Data and Hybrid Models . . . 165
11.10Week 10: Out-of-Sample Model Validation . . . 166
11.11Week 11: Portfolio Risk and Stress Testing . . . 167
11.12Week 12: Final Integration and Project Completion . . . 168
11.13Monthly Stage Gates . . . 169
11.13.1 Stage Gate 1: End of Week 4 . . . 169
11.13.2 Stage Gate 2: End of Week 8 . . . 169
11.13.3 Stage Gate 3: End of Week 12 . . . 170
11.14Weekly Supervisor Review . . . 170
11.15Work-Plan Discipline . . . 170

12 Expected Deliverables **171**

12.1 Deliverable 1: Mathematical Foundation Note . . . 172
12.2 Deliverable 2: Core Python Pricing Library . . . 172
12.3 Deliverable 3: Black–Scholes Pricing Engine . . . 172
12.4 Deliverable 4: Implied-Volatility Solver . . . 173
12.5 Deliverable 5: Numerical Pricing Validation . . . 173
12.6 Deliverable 6: Synthetic Option Dataset . . . 173
12.7 Deliverable 7: Cleaned Real-Market Dataset . . . 174
12.8 Deliverable 8: Machine-Learning Models . . . 174
12.9 Deliverable 9: Pure ML Pricing Model . . . 175
12.10 Deliverable 10: Residual Hybrid Model . . . 175
12.11 Deliverable 11: Implied-Volatility Hybrid Model . . . 175
12.12 Deliverable 12: Unified Model Comparison . . . 176
12.13 Deliverable 13: Mandatory Validation Figures . . . 176
12.14 Deliverable 14: Market-Risk Analysis Package . . . 177
12.15 Deliverable 15: Stress-Test Results . . . 177
12.16 Deliverable 16: Mandatory Risk Figures . . . 178
12.17 Deliverable 17: Jupyter Notebooks . . . 178
12.18 Deliverable 18: Reproducibility Package . . . 179
12.19 Deliverable 19: Final Technical Report . . . 179
12.20 Deliverable 20: Final Presentation . . . 179
12.21 Minimum Acceptable Final Package . . . 180
12.22 Deliverable Acceptance Principle . . . 180

13 Assessment and Project Milestones 181

13.1 Assessment Components . . . 181
13.2 Component 1: Mathematical Foundation . . . 182
13.3 Component 2: Black–Scholes and Numerical Implementation . . . 182
13.4 Component 3: Data Preparation and Feature Engineering . . . 183
13.5 Component 4: Machine-Learning Models . . . 183
13.6 Component 5: Hybrid Black–Scholes–ML Framework . . . 184
13.7 Component 6: Validation and Market-Risk Analysis . . . 184
13.8 Formal Review Gate 1: End of Week 4 . . . 185
13.9 Formal Review Gate 2: End of Week 8 . . . 185
13.10 Formal Review Gate 3: End of Week 12 . . . 185
13.11 Final Viva Questions . . . 186
13.12 Criteria for Successful Completion . . . 186

13.13 Final Project Success Criterion . . . 187

14 Expected Outcomes, Limitations and Future Extensions 187

14.1 Expected Outcome 1: Validated Black–Scholes Framework . . . 188
14.2 Expected Outcome 2: Understanding the Black–Scholes Error . . . 188
14.3 Expected Outcome 3: Pure Machine-Learning Benchmark . . . 189
14.4 Expected Outcome 4: Residual Hybrid Model . . . 189
14.5 Expected Outcome 5: ML Implied-Volatility Model . . . 189
14.6 Expected Outcome 6: Identification of Where ML Helps . . . 190
14.7 Expected Outcome 7: Financial Consistency Assessment . . . 190
14.8 Expected Outcome 8: Portfolio-Risk Comparison . . . 191
14.9 Expected Outcome 9: Quantification of Model Risk . . . 191
14.10 Expected Outcome 10: Understanding ML Extrapolation Risk . . . 192
14.11 Limitations of the Study . . . 192
14.11.1 Black–Scholes Assumptions . . . 192
14.11.2 Constant-Volatility Benchmark . . . 192
14.11.3 Limited Market Dataset . . . 193
14.11.4 Transaction Costs and Liquidity . . . 193
14.11.5 Restricted Machine-Learning Complexity . . . 193
14.11.6 Limited Hyperparameter Search . . . 194
14.11.7 Stress Scenarios are Not Forecasts . . . 194
14.11.8 No Complete Regulatory Risk System . . . 194
14.12 Future Extension 1: Volatility-Surface Modelling . . . 194
14.13 Future Extension 2: Stochastic Volatility . . . 195
14.14 Future Extension 3: Jump Models . . . 195
14.15 Future Extension 4: Physics-Informed or Finance-Informed Neural Networks . . . 195
14.16 Future Extension 5: No-Arbitrage-Constrained ML . . . 196
14.17 Future Extension 6: Dynamic Hedging . . . 196
14.18 Future Extension 7: Value-at-Risk and Expected Shortfall . . . 197
14.19 Future Extension 8: Regime-Aware Machine Learning . . . 197
14.20 Future Extension 9: Model Uncertainty . . . 198
14.21 Future Extension 10: Explainable AI . . . 198
14.22 Research Path Beyond the Three-Month Project . . . 198
14.23 Final Expected Research Outcome . . . 199
14.24 Closing Perspective . . . 199

1. Project Motivation

1.1 Classical Mathematical Modelling of Financial Markets

Mathematical models provide a systematic framework for describing the evolution of financial assets, valuing derivative securities and quantifying market risk. Among these models, the Black–Scholes–Merton framework occupies a central position in quantitative finance.

The starting point is the stochastic evolution of an underlying asset. Let S_t denote the price of an asset at time t. A standard model assumes that S_t follows a geometric Brownian motion,

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t,$$

where μ denotes the drift, σ denotes volatility and W_t is a standard Brownian motion.

Equivalently,

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

The solution of this stochastic differential equation is

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right].$$

Consequently,

$$\ln\left(\frac{S_t}{S_0}\right)$$

is normally distributed, while S_t itself follows a lognormal distribution.

This lognormal asset-price assumption forms an important mathematical foundation of the Black–Scholes–Merton option-pricing framework.

Key Idea

The Black–Scholes model is not simply an option-pricing formula. It is the consequence of a mathematical model for the stochastic evolution of the underlying asset together with no-arbitrage and risk-neutral valuation principles.

1.2 Black–Scholes Option Pricing

Consider a European call option written on an underlying asset with

$$S_0 = \text{current asset price},$$

$$K = \text{strike price},$$

$$T = \text{time to maturity},$$

$$r = \text{risk-free interest rate},$$

and

$$\sigma = \text{volatility of the underlying asset.}$$

Under the classical Black–Scholes assumptions, the price of a European call option is

$$C_{\mathrm{BS}} = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

Here $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

Similarly, the corresponding European put price is

$$P_{\mathrm{BS}} = K e^{-rT} N(-d_2) - S_0 N(-d_1).$$

Thus, the Black–Scholes model provides an explicit mapping

$$(S_0, K, T, r, \sigma) \longrightarrow C_{\mathrm{BS}}.$$

This analytical structure makes Black–Scholes an excellent benchmark against which numerical and data-driven pricing methods can be compared.

1.3 Risk-Neutral Valuation and Numerical Verification

Under risk-neutral valuation, the expected return of the underlying asset is replaced by the risk-free rate. Hence, under the risk-neutral measure,

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t.$$

The corresponding log return over a time interval Δt is normally distributed with mean

$$\left(r - \frac{1}{2}\sigma^2\right)\Delta t$$

and variance

$$\sigma^2 \Delta t.$$

This provides an important connection between the continuous Black–Scholes model and discrete numerical models such as the binomial lattice.

For a sufficiently fine discretization, a properly constructed binomial model should approach the Black–Scholes price,

$$C_{\text{Binomial}} \longrightarrow C_{\text{BS}}$$

as the number of time steps increases.

Similarly, under risk-neutral Monte Carlo simulation,

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\right],$$

where

$$Z \sim N(0,1).$$

For a European call option, the Monte Carlo estimator is

$$C_{\text{MC}} = e^{-rT}\frac{1}{M}\sum_{j=1}^{M}\max\left(S_T^{(j)} - K, 0\right).$$

Therefore, before introducing machine learning, the student will establish the numerical consistency

$$\boxed{C_{\text{BS}} \approx C_{\text{Binomial}} \approx C_{\text{MC}}.}$$

Project Milestone

The first computational milestone of the project is a validated Black–Scholes implementation. Machine learning will not be introduced until the analytical, binomial and Monte Carlo implementations have been cross-verified.

1.4 The Role of Volatility

Volatility is one of the most important inputs to the Black–Scholes model. The classical formulation assumes that volatility is constant over the life of the option.

In practice, however, volatility is not directly observable in the same manner as the asset price. An important quantity obtained from market option prices is therefore the *implied volatility*.

Suppose that the observed market price of a European call option is

$$C_{\text{market}}.$$

The implied volatility σ_{IV} is defined implicitly through

$$C_{\text{market}} = C_{\text{BS}}\left(S_0, K, T, r, \sigma_{\text{IV}}\right).$$

Thus,

$$\sigma_{\mathrm{IV}}$$

is the volatility that forces the Black–Scholes model to reproduce the observed market price.

In general, the Black–Scholes formula cannot simply be rearranged to give an explicit analytical expression for implied volatility. Consequently, σ_{IV} must be obtained through a numerical root-finding procedure.

For example, define

$$F(\sigma) = C_{\mathrm{BS}}(\sigma) - C_{\mathrm{market}}.$$

The implied-volatility problem becomes

$$F(\sigma_{\mathrm{IV}}) = 0.$$

This apparently simple inverse problem provides one of the natural entry points for numerical methods and subsequently for machine learning in the present project.

1.5 From Black–Scholes to Model Risk

The analytical tractability of Black–Scholes is obtained through a set of simplifying assumptions. In particular, the standard framework assumes a geometric Brownian motion with constant volatility.

Observed financial markets need not behave exactly according to these assumptions.

Consequently,

$$C_{\mathrm{market}} \neq C_{\mathrm{BS}}$$

in general.

Define the Black–Scholes pricing residual by

$$\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}.$$

If the Black–Scholes assumptions described market behaviour perfectly, these residuals would contain no systematic structure.

However, if

$$\varepsilon_{\mathrm{BS}}$$

depends systematically on variables such as moneyness, maturity, volatility or other observable market conditions, the residual itself contains useful information about the limitations of the classical model.

This observation motivates the AI/ML component of the project.

Model-Risk Perspective

The purpose of machine learning in this project is not to demonstrate that a sufficiently flexible neural network can reproduce a known Black–Scholes formula. Instead, the objective is to investigate whether systematic structure remains in the pricing error after the classical mathematical model has already been applied.

1.6 Why Machine Learning?

A supervised machine-learning model attempts to learn a relationship

$$\widehat{y} = f_\theta(\mathbf{x}),$$

where $\mathbf{x}$ is a vector of explanatory variables and θ represents model parameters learned from data.

A direct, purely data-driven approach could attempt to construct

$$\widehat{C}_{\mathrm{ML}} = f_\theta\left(S_0, K, T, r, \sigma\right).$$

Although this model will be studied as a benchmark, it ignores much of the mathematical information already contained in Black–Scholes.

A more interesting approach is therefore to combine the mathematical and machine-learning models.

1.7 Residual-Learning Black–Scholes Model

Let

$$\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}.$$

Instead of asking the machine-learning algorithm to learn the complete option-pricing function, it is trained to approximate the residual,

$$\widehat{\varepsilon}_{\mathrm{ML}} = f_\theta(\mathbf{x}).$$

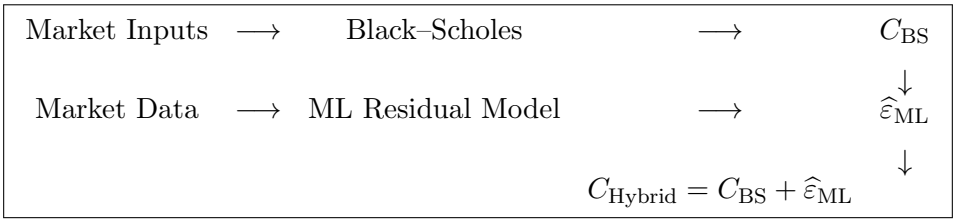
The feature vector may contain quantities such as

$$\mathbf{x} = \left[\ln\left(\frac{S_0}{K}\right), T, r, \sigma, \Delta, \Gamma, \nu\right].$$

The corrected price is then defined by

$$\boxed{C_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}.}$$

The resulting architecture is



This approach retains the financial structure of Black–Scholes while allowing the data-driven model to learn systematic discrepancies.

1.8 Machine Learning for Implied Volatility

A second hybrid approach will treat volatility rather than option price as the quantity to be learned.

The machine-learning model will estimate

$$\widehat{\sigma}_{\mathrm{IV}} = g_\theta(\mathbf{x}),$$

where the input variables may include moneyness,

$$m = \ln\left(\frac{S_0}{K}\right),$$

time to maturity,

$$T,$$

and appropriate market variables.

The estimated implied volatility is subsequently inserted into the Black–Scholes model,

$$\widehat{C} = C_{\mathrm{BS}}\left(S_0, K, T, r, \widehat{\sigma}_{\mathrm{IV}}\right).$$

Hence,

$$\boxed{\text{Market Features} \longrightarrow \text{ML} \longrightarrow \widehat{\sigma}_{\mathrm{IV}} \longrightarrow \text{Black–Scholes} \longrightarrow \widehat{C}.}$$

This provides a second form of hybrid modelling in which machine learning estimates a financially meaningful latent quantity rather than directly replacing the option-pricing equation.

1.9 From Pricing to Market-Risk Analysis

Option pricing alone is not sufficient for understanding the risk of a derivative position. The sensitivity of the option price to changes in the underlying risk factors is equally important.

For a call option,

$$\Delta = \frac{\partial C}{\partial S},$$

measures first-order sensitivity to the underlying asset price,

$$\Gamma = \frac{\partial^2 C}{\partial S^2},$$

measures the curvature of the option value with respect to the underlying price, and

$$\nu = \frac{\partial C}{\partial \sigma},$$

measures sensitivity to volatility.

For sufficiently small changes in the market variables, the change in option value may be approximated using a Taylor expansion,

$$\Delta C \approx \Delta\, \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \nu\, \Delta\sigma.$$

Thus the project will not terminate with the question

"Which model gives the smallest pricing error?"

Instead, it will proceed to the more important question

How does the model behave when the market is stressed?

This connects option pricing, machine learning and market-risk analysis within a single computational framework.

1.10 Overall Philosophy of the Project

The project is deliberately structured so that machine learning is introduced only after the mathematical and numerical foundations have been established.

The learning sequence is

Probability and Stochastic Modelling
↓
Geometric Brownian Motion
↓
Black–Scholes Model
↓
Binomial and Monte Carlo Validation
↓
Machine Learning
↓
Hybrid Black–Scholes–ML Model
↓
Market-Risk and Stress Analysis.

Project Idea

The central philosophy of the project is

Do not replace the mathematical model with AI; use AI to identify and learn its residual limitations.

This distinction converts the project from a simple neural-network curve-fitting exercise into a study of hybrid mathematical and data-driven modelling.

1.11 Central Research Question

The complete three-month project will therefore address the following research question.

Central Research Question

Can machine-learning methods augment the classical Black–Scholes framework to improve option-pricing accuracy while preserving the mathematical interpretability and market-risk sensitivities of the traditional model?

To answer this question, the student will construct and compare

Classical Black–Scholes Model
Pure Machine-Learning Model
Black–Scholes + ML Residual Model
ML Implied Volatility + Black–Scholes Model

and subsequently evaluate their behaviour under changes in

$$S, \qquad \sigma, \qquad T, \qquad \text{and moneyness.}$$

The final objective is therefore not merely improved prediction, but a computational framework in which classical quantitative finance and modern machine learning can be studied together in a transparent and systematic manner.

2. Project Objectives

2.1 Overall Objective

The overall objective of this three-month project is to develop, implement and evaluate a computational framework that combines the classical Black–Scholes option-pricing model with selected machine-learning algorithms for option pricing, implied-volatility estimation and market-risk analysis.

The project will investigate whether data-driven models can improve upon the classical Black–Scholes benchmark while retaining the financial interpretability provided by the underlying mathematical model.

The overall computational philosophy is

Mathematical Model ⟶ Numerical Validation ⟶ Machine Learning ⟶ Hybrid Model ⟶ Risk Analysis.

The project is therefore not intended to construct an unrestricted black-box option-pricing model. Instead, Black–Scholes will provide the mathematical backbone against which all machine-learning models will be developed and evaluated.

2.2 Specific Objective 1: Develop the Mathematical Black–Scholes Framework

The first objective is to establish the mathematical foundation required for the project.

The student shall study the geometric Brownian motion model

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t,$$

and its solution

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right].$$

Under risk-neutral valuation, the model becomes

$$\frac{dS_t}{S_t} = r\, dt + \sigma\, dW_t.$$

The student shall understand the relationship between this stochastic asset-price model and the Black–Scholes pricing framework.

The Black–Scholes partial differential equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

shall be introduced and related to the analytical European-option pricing formula.

For a European call option,

$$C_{\mathrm{BS}} = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The corresponding put price shall also be implemented.

Project Milestone

At the completion of this objective, the student must have an independently implemented Black–Scholes call and put pricing code that accepts

$$S_0, \qquad K, \qquad T, \qquad r, \qquad \sigma$$

as inputs and returns the corresponding option prices.

2.3 Specific Objective 2: Validate Black–Scholes Using Independent Numerical Methods

The second objective is to ensure that the analytical implementation is computationally correct before introducing machine learning.

Two independent numerical approaches shall be implemented:

Binomial Lattice

and

Monte Carlo Simulation.

For the binomial method, the student shall investigate convergence of

$$C_{\text{Binomial}}^{(N)}$$

toward the analytical Black–Scholes solution as the number of time steps N increases.

The expected convergence behaviour is

$$C_{\text{Binomial}}^{(N)} \longrightarrow C_{\text{BS}} \qquad \text{as} \qquad N \longrightarrow \infty.$$

For the Monte Carlo method, risk-neutral terminal asset prices shall be generated using

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z\right],$$

where

$$Z \sim N(0,1).$$

The European call price shall then be estimated from

$$C_{\text{MC}} = e^{-rT} \frac{1}{M} \sum_{j=1}^{M} \max\left(S_T^{(j)} - K, 0\right).$$

The student shall investigate the effect of increasing M on the Monte Carlo pricing error.

The numerical validation criterion is

$$\boxed{C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}.}$$

Project Milestone

Machine-learning development shall begin only after the student has demonstrated numerical agreement between the analytical Black–Scholes, binomial and Monte Carlo implementations for a representative set of European options.

2.4 Specific Objective 3: Compute Option Sensitivities and Implied Volatility

The third objective is to extend the pricing framework to quantities required for market-risk analysis.

The student shall calculate the principal Black–Scholes Greeks, including

$$\Delta = \frac{\partial C}{\partial S},$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2},$$

and

$$\nu = \frac{\partial C}{\partial \sigma}.$$

Where appropriate, analytical Greeks shall be compared with numerical finite-difference approximations.

For example,

$$\Delta \approx \frac{C(S+h) - C(S-h)}{2h},$$

and

$$\Gamma \approx \frac{C(S+h) - 2C(S) + C(S-h)}{h^2}.$$

The student shall also implement an implied-volatility solver.

Given an observed option price

$$C_{\mathrm{market}},$$

the implied volatility shall satisfy

$$C_{\mathrm{BS}}\left(S_0, K, T, r, \sigma_{\mathrm{IV}}\right) = C_{\mathrm{market}}.$$

Thus, defining

$$F(\sigma) = C_{\mathrm{BS}}(\sigma) - C_{\mathrm{market}},$$

the computational problem becomes

$$F(\sigma_{\mathrm{IV}}) = 0.$$

A suitable numerical root-finding algorithm shall be implemented to obtain

$$\sigma_{\mathrm{IV}}.$$

Project Milestone

At the completion of this objective, the computational code must return not only an option price but also

$$\Delta, \qquad \Gamma, \qquad \nu, \qquad \sigma_{\mathrm{IV}}.$$

2.5 Specific Objective 4: Construct a Machine-Learning Dataset

The fourth objective is to transform the option-pricing problem into a well-defined supervised machine-learning problem.

The initial dataset shall be generated by sampling combinations of

$$S_0, \qquad K, \qquad T, \qquad r, \qquad \sigma.$$

For every combination, the validated Black–Scholes program shall calculate

$$C_{\mathrm{BS}}, \qquad \Delta, \qquad \Gamma, \qquad \nu.$$

Rather than relying only on the dimensional quantities S_0 and K, the student shall construct financially meaningful features.

An important feature is log-moneyness,

$$m = \ln\left(\frac{S_0}{K}\right).$$

Thus, a basic machine-learning feature vector may be written as

$$\mathbf{x} = [m, T, r, \sigma, \Delta, \Gamma, \nu].$$

The dataset shall subsequently be divided into

$$\text{Training Data}, \qquad \text{Validation Data}, \qquad \text{Test Data}.$$

Care shall be taken to ensure that the test dataset is not used during model training or hyperparameter selection.

Key Idea

The student must understand the financial meaning of every feature before using it in a machine-learning algorithm. Feature engineering shall therefore be based on the structure of the option-pricing problem rather than on arbitrary data transformations.

2.6 Specific Objective 5: Develop and Compare Selected Machine-Learning Models

The fifth objective is to construct a controlled hierarchy of machine-learning models.

The project shall begin with a simple regression benchmark,

$$\widehat{C} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j.$$

Regularized regression may subsequently be introduced in the form

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^T \boldsymbol{\beta} \right)^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right].$$

A nonlinear tree-based model shall then be implemented, followed by a small feed-forward neural network.

The model hierarchy shall therefore be

$$\boxed{\text{Linear/Ridge Regression} \longrightarrow \text{Tree-Based Model} \longrightarrow \text{Neural Network.}}$$

The purpose is not simply to identify the model with the smallest error. The student shall investigate whether the additional complexity of each model produces a meaningful improvement in predictive performance.

Model-Risk Perspective

Deep architectures such as LSTMs, Transformers, reinforcement-learning models and large neural networks are deliberately excluded from the three-month project.
The objective is controlled comparison and understanding rather than unnecessary algorithmic complexity.

2.7 Specific Objective 6: Develop a Hybrid Black–Scholes–ML Model

The sixth objective constitutes the central AI/ML contribution of the project.

For observed market option prices, define

$$\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}.$$

A machine-learning algorithm shall be trained to estimate

$$\widehat{\varepsilon}_{\mathrm{ML}} = f_\theta(\mathbf{x}).$$

The resulting hybrid price is

$$\boxed{C_{\text{Hybrid}} = C_{\text{BS}} + \widehat{\varepsilon}_{\text{ML}}.}$$

This model will be compared with a purely data-driven model

$$\widehat{C}_{\text{ML}} = f_\theta(\mathbf{x}).$$

A second hybrid strategy shall estimate implied volatility,

$$\widehat{\sigma}_{\text{IV}} = g_\theta(\mathbf{x}),$$

followed by Black–Scholes pricing,

$$\widehat{C}_{\text{IV}-\text{ML}} = C_{\text{BS}}\left(S_0, K, T, r, \widehat{\sigma}_{\text{IV}}\right).$$

The final comparison will therefore involve

Black–Scholes
Pure Machine Learning
Black–Scholes + ML Residual
ML Implied Volatility + Black–Scholes.

2.8 Specific Objective 7: Evaluate Pricing Accuracy and Generalization

The seventh objective is to perform a quantitative comparison of all pricing models.

The principal error measures shall include the mean absolute error,

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|C_i - \widehat{C}_i\right|,$$

and the root mean square error,

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(C_i - \widehat{C}_i\right)^2}.$$

The coefficient of determination may also be reported as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(C_i - \widehat{C}_i\right)^2}{\sum_{i=1}^{n}\left(C_i - \overline{C}\right)^2}.$$

However, aggregate accuracy alone will not be considered sufficient.

Errors shall also be analysed according to moneyness,

$$\text{ITM}, \qquad \text{ATM}, \qquad \text{OTM},$$

and according to maturity,

Short Maturity, Medium Maturity, Long Maturity.

The objective is to determine not only

which model performs better,

but also

where and under what market conditions each model fails.

2.9 Specific Objective 8: Extend the Framework to Market-Risk Analysis

The final objective is to demonstrate that the project extends beyond option-price prediction. Consider an option portfolio

$$V = \sum_{i=1}^{N} n_i C_i,$$

where n_i denotes the position in option i.
The portfolio delta is

$$\Delta_P = \sum_{i=1}^{N} n_i \Delta_i,$$

the portfolio gamma is

$$\Gamma_P = \sum_{i=1}^{N} n_i \Gamma_i,$$

and the portfolio vega is

$$\nu_P = \sum_{i=1}^{N} n_i \nu_i.$$

For changes in the underlying asset price and volatility, the delta–gamma–vega approximation is

$$\Delta V \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.$$

The portfolio shall subsequently be subjected to controlled stress scenarios such as

$$S \longrightarrow S(1 + \delta_S)$$

and

$$\sigma \longrightarrow \sigma + \delta_\sigma.$$

The student shall compare

Taylor Risk Approximation

with

Full Option Repricing.

The analysis shall investigate how the classical and hybrid models behave when the underlying market conditions move away from the region represented strongly in the training dataset.

Model-Risk Perspective

A machine-learning model with a low test RMSE is not automatically a reliable financial model.
The final assessment must therefore include sensitivity, stress and out-of-sample behaviour in addition to conventional machine-learning accuracy measures.

2.10 Final Project Success Criteria

The project shall be considered successfully completed only when the student can demonstrate the complete computational chain

GBM and Black–Scholes Mathematics
↓
Analytical Black–Scholes Implementation
↓
Binomial and Monte Carlo Validation
↓
Greeks and Implied Volatility
↓
Dataset and Feature Engineering
↓
Regression / Tree Model / Neural Network
↓
Hybrid Black–Scholes–ML Model
↓
Model Comparison
↓
Market-Risk and Stress Analysis.

At the end of the three-month period, the student must therefore be able to answer three fundamental questions:

1. How accurately does the classical Black–Scholes model price the options considered?

2. Can machine learning identify systematic structure that is not captured by the classical model?

3. Does the resulting hybrid model remain reliable when used for risk and stress analysis?

These three questions define the principal criteria against which the final project will be evaluated.

3. Mathematical Foundation

The purpose of this section is to establish the minimum mathematical framework required before the student proceeds to numerical implementation and machine learning.

The mathematical progression of the project is

Returns $\longrightarrow$ Brownian Motion $\longrightarrow$ Geometric Brownian Motion $\longrightarrow$ Itô's Lemma $\longrightarrow$ Black–Scholes PDE $\longrightarrow$

The student is expected to understand the meaning of each stage and its role in the final computational framework.

3.1 Simple Returns and Log Returns

Let S_t denote the price of a financial asset at time t. If the asset price changes from S_t to $S_{t+\Delta t}$, the simple return is

$$R_t = \frac{S_{t+\Delta t} - S_t}{S_t}.$$

Equivalently,

$$R_t = \frac{S_{t+\Delta t}}{S_t} - 1.$$

The continuously compounded or logarithmic return is defined as

$$r_t = \ln\left(\frac{S_{t+\Delta t}}{S_t}\right).$$

For relatively small price changes,

$$\ln(1 + R_t) \approx R_t,$$

and therefore

$$r_t \approx R_t.$$

Log returns are particularly useful because returns over successive time intervals are additive.

For example,

$$\ln\left(\frac{S_2}{S_0}\right) = \ln\left(\frac{S_1}{S_0}\right) + \ln\left(\frac{S_2}{S_1}\right).$$

More generally,

$$\ln\left(\frac{S_T}{S_0}\right) = \sum_{i=1}^{N} \ln\left(\frac{S_{t_i}}{S_{t_{i-1}}}\right).$$

This additive property is important when constructing stochastic models of continuously evolving asset prices.

3.2 Brownian Motion

A standard Brownian motion W_t is a continuous-time stochastic process satisfying

$$W_0 = 0,$$

with independent increments and normally distributed changes.

For

$$0 \leq s < t,$$

the increment

$$W_t - W_s$$

satisfies

$$W_t - W_s \sim N(0, t-s).$$

Consequently,

$$\mathbb{E}[W_t - W_s] = 0$$

and

$$\mathrm{Var}(W_t - W_s) = t - s.$$

Over a small time interval Δt,

$$\Delta W \sim N(0, \Delta t).$$

Hence it may be represented computationally as

$$\Delta W = \sqrt{\Delta t}\, Z,$$

where

$$Z \sim N(0,1).$$

This representation will later be used directly in Monte Carlo simulation.

Key Idea

Brownian motion provides the random component of the asset-price model. Over a short interval Δt, the stochastic increment has magnitude proportional to

$$\sqrt{\Delta t},$$

rather than Δt.

3.3 Geometric Brownian Motion

The Black–Scholes framework assumes that the underlying asset follows a geometric Brownian motion.

The stochastic differential equation is

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

Dividing by S_t,

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.$$

The first term,

$$\mu\, dt,$$

represents the deterministic drift component, while

$$\sigma\, dW_t$$

represents the stochastic component.

Here,

$$\mu = \text{expected continuously compounded rate of return},$$

and

$$\sigma = \text{volatility}.$$

A crucial feature of geometric Brownian motion is that the random component is proportional to the current asset price S_t.

3.4 From Geometric Brownian Motion to Log Prices

To determine the distribution of S_t, consider

$$f(S) = \ln S.$$

The derivatives are

$$\frac{\partial f}{\partial S} = \frac{1}{S},$$

and

$$\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S^2}.$$

Applying Itô's lemma gives

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2}\frac{1}{S_t^2}(dS_t)^2.$$

Since

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t,$$

and using the Itô multiplication rules

$$(dW_t)^2 = dt,$$

$$dt\, dW_t = 0,$$

and

$$(dt)^2 = 0,$$

we obtain

$$(dS_t)^2 = \sigma^2 S_t^2\, dt.$$

Therefore,

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dW_t.$$

Integrating from 0 to T,

$$\ln S_T - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T.$$

Thus,

$$\ln\left(\frac{S_T}{S_0}\right) = \left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T.$$

Since

$$W_T \sim N(0, T),$$

we obtain

$$\ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right).$$

Equivalently,

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\right],$$

where

$$Z \sim N(0, 1).$$

Therefore,

$$\boxed{S_T \text{ is lognormally distributed.}}$$

3.5 Itô's Lemma

Itô's lemma is the stochastic-calculus counterpart of the chain rule.

Suppose

$$dS = \mu S\,dt + \sigma S\,dW,$$

and let

$$V = V(S, t).$$

Then Itô's lemma gives

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2.$$

Since

$$(dS)^2 = \sigma^2 S^2 dt,$$

we obtain

$$dV = \left[\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right]dt + \sigma S\frac{\partial V}{\partial S}dW.$$

This equation provides the direct mathematical bridge between the stochastic dynamics of the underlying asset and the dynamics of a derivative written on that asset.

3.6 Construction of a Delta-Hedged Portfolio

Consider a derivative with value

$$V = V(S, t).$$

Construct a portfolio consisting of one derivative and a short position in Δ units of the underlying asset,

$$\Pi = V - \Delta S.$$

The change in the portfolio value is

$$d\Pi = dV - \Delta\, dS.$$

Choose

$$\Delta = \frac{\partial V}{\partial S}.$$

Substituting the Itô expansion for dV,

$$d\Pi = \left[\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right] dt$$

$$+\sigma S\frac{\partial V}{\partial S}dW - \frac{\partial V}{\partial S}\left(\mu S\, dt + \sigma S\, dW\right).$$

The stochastic terms cancel, giving

$$d\Pi = \left[\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right] dt.$$

Thus, over the infinitesimal time interval, the delta-hedged portfolio contains no Brownian-motion term.

Key Idea

The central Black–Scholes argument is not that risk disappears from the market. Rather, a particular combination of the derivative and the underlying asset eliminates the instantaneous stochastic term

$$dW.$$

The resulting locally riskless portfolio must earn the risk-free rate under the no-arbitrage argument.

3.7 Derivation of the Black–Scholes PDE

Since the delta-hedged portfolio is locally riskless, its return must equal the risk-free rate. Therefore,

$$d\Pi = r\Pi\, dt.$$

Since

$$\Pi = V - S\frac{\partial V}{\partial S},$$

we have

$$d\Pi = r\left(V - S\frac{\partial V}{\partial S}\right) dt.$$

Equating this expression with the previous expression for $d\Pi$,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = r\left(V - S\frac{\partial V}{\partial S}\right).$$

Rearranging gives

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.}$$

This is the Black–Scholes partial differential equation.

3.8 An Important Consequence of the PDE

The drift parameter

$$\mu$$

does not appear in the Black–Scholes PDE.

The final equation contains

$$S, \qquad t, \qquad r, \qquad \sigma,$$

but not

$$\mu.$$

This is an important result.

The expected return of the underlying asset is not required for Black–Scholes option pricing.

This observation leads naturally to the risk-neutral representation of the pricing problem.

3.9 Risk-Neutral Dynamics

Under the risk-neutral probability measure, the underlying asset is modelled as

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t.$$

Therefore,

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\right],$$

where

$$Z \sim N(0,1).$$

For a derivative with terminal payoff

$$\Phi(S_T),$$

its value may be represented as the discounted risk-neutral expectation

$$\boxed{V_0 = e^{-rT}\mathbb{E}^Q\left[\Phi(S_T)\right].}$$

For a European call,

$$\Phi(S_T) = \max(S_T - K, 0),$$

and hence

$$C_0 = e^{-rT}\mathbb{E}^Q\left[\max(S_T - K, 0)\right].$$

This representation will later provide the mathematical basis for the Monte Carlo pricing algorithm.

3.10 European Call and Put Payoffs

The terminal payoff of a European call option is

$$C_T = \max(S_T - K, 0).$$

Thus,

$$C_T = \begin{cases} 0, & S_T \le K, \\ S_T - K, & S_T > K. \end{cases}$$

The terminal payoff of a European put option is

$$P_T = \max(K - S_T, 0).$$

Thus,

$$P_T = \begin{cases} K - S_T, & S_T < K, \\ 0, & S_T \geq K. \end{cases}$$

These payoff functions provide the terminal conditions for the Black–Scholes PDE.

3.11 Black–Scholes Analytical Solution

For a European call option without dividends, the Black–Scholes solution is

$$\boxed{C = SN(d_1) - Ke^{-rT}N(d_2),}$$

where

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

For a European put,

$$\boxed{P = Ke^{-rT}N(-d_2) - SN(-d_1).}$$

The analytical solution therefore defines the nonlinear mapping

$$\boxed{(S, K, T, r, \sigma) \longrightarrow V.}$$

This mapping will later serve both as a pricing model and as a reference benchmark for the machine-learning algorithms.

3.12 Put–Call Parity

European call and put prices satisfy the no-arbitrage relationship

$$C - P = S - Ke^{-rT}.$$

Equivalently,

$$C + Ke^{-rT} = P + S.$$

This relationship provides a useful computational verification test.

After implementing the Black–Scholes call and put functions, the student shall verify numerically that

$$\left|C - P - S + Ke^{-rT}\right|$$

is close to zero within numerical precision.

Project Milestone

The Black–Scholes implementation will not be accepted solely because it produces plausible option prices.
The code must also satisfy internal financial consistency tests such as put–call parity.

3.13 Moneyness

The relationship between the current underlying price and the strike price is one of the principal determinants of option value.

A useful dimensionless quantity is

$$\frac{S}{K}.$$

For machine-learning purposes, the project will primarily use log-moneyness,

$$\boxed{m = \ln\left(\frac{S}{K}\right).}$$

For a call option,

$$m > 0$$

corresponds approximately to an in-the-money option,

$$m = 0$$

corresponds to an at-the-money option, and

$$m < 0$$

corresponds approximately to an out-of-the-money option.

Using moneyness rather than the raw values of S and K will later provide a more meaningful feature for machine-learning models.

3.14 Option Greeks

The Black–Scholes model provides not only an option price but also analytical sensitivities of that price to the underlying market variables.

These sensitivities are known as the Greeks.

3.14.1 Delta

For a European call,

$$\Delta_C = \frac{\partial C}{\partial S} = N(d_1).$$

For a European put,

$$\Delta_P = N(d_1) - 1.$$

Delta measures the first-order sensitivity of option value to the underlying asset price.

3.14.2 Gamma

Gamma measures the rate at which delta changes as the underlying asset price changes,

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

For European calls and puts,

$$\boxed{\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}},}$$

where $\phi(\cdot)$ is the standard normal probability density function,

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

3.14.3 Vega

Vega measures sensitivity to volatility,

$$\nu = \frac{\partial V}{\partial \sigma}.$$

For European calls and puts,

$$\boxed{\nu = S\phi(d_1)\sqrt{T}.}$$

Vega will be particularly important in the implied-volatility and machine-learning components of the project.

3.15 Taylor Approximation and Local Market Risk

For a derivative

$$V = V(S, \sigma, t),$$

a local Taylor expansion gives approximately

$$\Delta V \approx \frac{\partial V}{\partial S}\Delta S + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(\Delta S)^2 + \frac{\partial V}{\partial \sigma}\Delta\sigma.$$

Therefore,

$$\boxed{\Delta V \approx \Delta\,\Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \nu\,\Delta\sigma.}$$

The three terms have distinct interpretations:

$$\Delta\,\Delta S = \text{first-order price risk},$$

$$\frac{1}{2}\Gamma(\Delta S)^2 = \text{nonlinear price risk},$$

and

$$\nu\,\Delta\sigma = \text{volatility risk}.$$

This approximation will later be compared against complete option repricing during the market-risk component of the project.

3.16 Implied Volatility as an Inverse Problem

Suppose that

$$C_{\text{market}}$$

is an observed market price.

The Black–Scholes implied volatility is defined as the value of σ satisfying

$$C_{\text{BS}}\,(S, K, T, r, \sigma) = C_{\text{market}}.$$

Define

$$F(\sigma) = C_{\text{BS}}(\sigma) - C_{\text{market}}.$$

Then implied volatility is the root

$$\boxed{F(\sigma_{\text{IV}}) = 0.}$$

Unlike the Black–Scholes price, implied volatility does not generally have a simple closed-form solution.

It must therefore be obtained numerically.

3.17 Newton–Raphson Method for Implied Volatility

Starting with an initial estimate

$$\sigma^{(0)},$$

Newton–Raphson iteration gives

$$\sigma^{(k+1)} = \sigma^{(k)} - \frac{F(\sigma^{(k)})}{F'(\sigma^{(k)})}.$$

Since

$$F'(\sigma) = \frac{\partial C_{\mathrm{BS}}}{\partial \sigma} = \nu,$$

the iteration becomes

$$\boxed{\sigma^{(k+1)} = \sigma^{(k)} - \frac{C_{\mathrm{BS}}\left(\sigma^{(k)}\right) - C_{\mathrm{market}}}{\nu\left(\sigma^{(k)}\right)}.}$$

The iteration is continued until

$$\left|C_{\mathrm{BS}}\left(\sigma^{(k)}\right) - C_{\mathrm{market}}\right| < \varepsilon,$$

where ε is a prescribed numerical tolerance.

Key Idea

Implied volatility is especially useful for this project because it converts an observed option price into a quantity that can be compared across strikes and maturities.
It therefore provides a natural bridge between

Black–Scholes and Machine Learning.

3.18 From Constant Volatility to a Volatility Surface

Under the classical Black–Scholes model,

$$\sigma = \text{constant}.$$

If this assumption described market prices exactly, options written on the same underlying asset would yield approximately the same implied volatility after accounting for the relevant contract variables.

In actual option data, the implied volatility extracted from market prices may depend on strike and maturity.

Conceptually,

$$\sigma_{\mathrm{IV}} = \sigma_{\mathrm{IV}}(K, T).$$

Using log-moneyness,

$$m = \ln\left(\frac{S}{K}\right),$$

the same relationship may be represented as

$$\boxed{\sigma_{\mathrm{IV}} = f(m, T).}$$

This motivates one of the principal machine-learning problems in the project,

$$(m, T, \text{market features}) \longrightarrow \widehat{\sigma}_{\mathrm{IV}}.$$

The predicted volatility can then be returned to the mathematical pricing model,

$$\widehat{\sigma}_{\mathrm{IV}} \longrightarrow C_{\mathrm{BS}}\left(S, K, T, r, \widehat{\sigma}_{\mathrm{IV}}\right).$$

Thus, machine learning need not replace Black–Scholes. It can instead be used to estimate one of the quantities for which the classical constant-volatility assumption is restrictive.

3.19 Mathematical Inputs to the Machine-Learning Model

At the completion of the mathematical stage, the student will have constructed the quantities

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma,$$

together with derived financial variables

$$m = \ln\left(\frac{S}{K}\right),$$

$$C_{\mathrm{BS}},$$

$$\Delta, \qquad \Gamma, \qquad \nu,$$

and, where market prices are available,

$$\sigma_{\mathrm{IV}}.$$

The mathematical layer therefore produces the feature set

$$\boxed{\mathbf{x} = [m, T, r, \sigma, \Delta, \Gamma, \nu].}$$

Depending on the particular experiment, the machine-learning target may be

$$y = C_{\text{market}},$$

or

$$y = C_{\text{market}} - C_{\text{BS}},$$

or

$$y = \sigma_{\text{IV}}.$$

This gives three distinct supervised-learning problems:

$$\begin{aligned} \mathbf{x} &\longrightarrow C_{\text{market}}, \\ \mathbf{x} &\longrightarrow C_{\text{market}} - C_{\text{BS}}, \\ \mathbf{x} &\longrightarrow \sigma_{\text{IV}}. \end{aligned}$$

3.20 Required Mathematical Competency Before ML

Before beginning the machine-learning component, the student should be able to explain and implement the following chain without treating the equations as black-box formulas:

Geometric Brownian Motion

$$\frac{dS_t}{S_t} = \mu\,dt + \sigma\,dW_t$$

Derivative Value and Itô's Lemma

$$V = V(S,t)$$

$$dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S\,dW_t$$

Delta-Hedged Portfolio

$$\Pi = V - \Delta S, \qquad \Delta = V_S$$

The stochastic dW_t term is eliminated.

No-Arbitrage Condition

The locally riskless portfolio must earn the risk-free rate:

$$d\Pi = r\Pi\,dt$$

Black–Scholes Partial Differential Equation

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$$

Terminal Payoff Condition

$$V(S,T) = \max(S-K,0) \qquad \text{for a European call}$$

Black–Scholes Analytical Price

$$C_{\mathrm{BS}} = SN(d_1) - Ke^{-rT}N(d_2)$$

Numerical Validation, Risk Quantities and AI/ML

$$C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}$$

$$\Delta, \qquad \Gamma, \qquad \nu, \qquad \sigma_{\mathrm{IV}}$$

$$\mathbf{x} = [m, T, r, \sigma, \Delta, \Gamma, \nu]$$

$$y \in \{C,\ C_{\mathrm{market}} - C_{\mathrm{BS}},\ \sigma_{\mathrm{IV}}\}$$

Project Milestone

The student shall proceed to the AI/ML stage only after being able to

1. explain the GBM asset-price model;
2. generate GBM sample paths computationally;
3. explain the role of Itô's lemma;

4. explain the delta-hedging argument;
5. implement Black–Scholes call and put pricing;
6. verify put–call parity;
7. compute delta, gamma and vega; and
8. recover implied volatility from a given option price.

This mathematical checkpoint prevents the machine-learning stage from becoming a purely black-box exercise.

4. Computational Black–Scholes Framework

The computational stage translates the mathematical framework developed in the previous section into a sequence of independently testable algorithms.

The student shall not begin with a single large program.

Instead, the computational framework shall be developed in modules,

Black–Scholes Pricer $\longrightarrow$ Greeks $\longrightarrow$ Implied Volatility $\longrightarrow$ Binomial Model $\longrightarrow$ Monte Carlo Model $\longrightarrow$ Vali

Each module must be validated independently before it is integrated into the final AI/ML workflow.

4.1 Computational Environment

The project shall be implemented primarily in Python.

The recommended computational environment is

$$\text{Python} + \text{NumPy} + \text{SciPy} + \text{Pandas} + \text{Matplotlib} + \text{scikit-learn}.$$

A neural-network library may be introduced later if required.

The initial phase should not depend on complex software frameworks. The emphasis is on transparent implementation of the mathematical algorithms.

The student shall maintain a modular project structure such as

```
black_scholes.py
greeks.py
implied_volatility.py
binomial.py
monte_carlo.py
validation.py
data_generation.py
ml_models.py
```

or an equivalent notebook-based structure during the initial development stage.

4.2 Algorithm 1: Black–Scholes Analytical Pricer

The first computational task is the direct implementation of the Black–Scholes analytical formula. For a European call,

$$C = SN(d_1) - Ke^{-rT}N(d_2),$$

and for a European put,

$$P = Ke^{-rT}N(-d_2) - SN(-d_1),$$

where

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The corresponding computational procedure is

$$\boxed{(S, K, T, r, \sigma) \longrightarrow (d_1, d_2) \longrightarrow (C, P).}$$

The student shall implement separate functions for calls and puts.

A generic program structure may be represented as follows:

Input: S, K, T, r, σ
Compute: d_1, d_2
Evaluate: $N(d_1), N(d_2)$
Return: $C_{\mathrm{BS}}, P_{\mathrm{BS}}$.

4.2.1 Basic Computational Checks

The code shall be tested for financially sensible behaviour.

For example,

$$C_{\mathrm{BS}} \geq 0,$$

and

$$P_{\mathrm{BS}} \geq 0.$$

For a call option,

$$C_{\mathrm{BS}} \leq S.$$

The student shall also verify put–call parity,

$$C - P = S - Ke^{-rT}.$$

Define the numerical parity error

$$E_{\mathrm{PCP}} = \left| C - P - S + Ke^{-rT} \right|.$$

The code shall satisfy

$$E_{\mathrm{PCP}} < \varepsilon_{\mathrm{num}},$$

where $\varepsilon_{\mathrm{num}}$ is an appropriately small numerical tolerance.

Project Milestone

The first program is accepted only when the student can demonstrate correct prices and put–call parity over multiple combinations of

$$S, \quad K, \quad T, \quad r, \quad \sigma.$$

4.3 Algorithm 2: Analytical Greeks

The next module shall compute the principal Black–Scholes Greeks.

For a European call,

$$\Delta_C = N(d_1),$$

and for a European put,

$$\Delta_P = N(d_1) - 1.$$

Gamma is

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}},$$

and vega is

$$\nu = S\phi(d_1)\sqrt{T}.$$

The computational map is therefore

$$\boxed{(S, K, T, r, \sigma) \longrightarrow (\Delta, \Gamma, \nu).}$$

The student shall implement these quantities as separate reusable functions.

4.4 Algorithm 3: Numerical Greeks

Analytical Greeks shall be verified using finite differences.

For a small perturbation h,

$$\Delta_{\mathrm{FD}} = \frac{C(S+h) - C(S-h)}{2h}.$$

Gamma may be approximated by

$$\Gamma_{\mathrm{FD}} = \frac{C(S+h) - 2C(S) + C(S-h)}{h^2}.$$

Similarly, vega may be approximated by

$$\nu_{\mathrm{FD}} = \frac{C(\sigma+h_\sigma) - C(\sigma-h_\sigma)}{2h_\sigma}.$$

The corresponding relative errors may be defined as

$$E_\Delta = \frac{|\Delta_{\mathrm{FD}} - \Delta_{\mathrm{analytic}}|}{|\Delta_{\mathrm{analytic}}| + \epsilon},$$

with analogous expressions for

$$E_\Gamma$$

and

$$E_\nu.$$

Here ϵ is a small number introduced to avoid division by zero.

4.4.1 Step-Size Study

The student shall investigate the effect of the finite-difference step size.

The experiment shall use several values such as

$$h, \quad \frac{h}{2}, \quad \frac{h}{4}, \quad \frac{h}{8}.$$

The objective is to observe the competing effects of

truncation error

and

round-off error.

Key Idea

Finite-difference Greeks are not introduced because analytical Black–Scholes Greeks are unavailable.
They are introduced to teach the student how sensitivities can be computed when an analytical derivative is unavailable, which becomes important later for more general numerical and ML-based models.

4.5 Algorithm 4: Implied-Volatility Solver

Given a market option price

$$C_{\text{market}},$$

the implied volatility satisfies

$$C_{\text{BS}}\left(S, K, T, r, \sigma_{\text{IV}}\right) = C_{\text{market}}.$$

Define

$$F(\sigma) = C_{\text{BS}}(\sigma) - C_{\text{market}}.$$

The numerical objective is

$$F(\sigma_{\text{IV}}) = 0.$$

4.5.1 Newton–Raphson Iteration

Using

$$F'(\sigma) = \nu,$$

the Newton–Raphson iteration is

$$\boxed{\sigma^{(k+1)} = \sigma^{(k)} - \frac{C_{\text{BS}}\left(\sigma^{(k)}\right) - C_{\text{market}}}{\nu\left(\sigma^{(k)}\right)}.}$$

The iteration terminates when either

$$|C_{\text{BS}} - C_{\text{market}}| < \varepsilon$$

or

$$\left|\sigma^{(k+1)} - \sigma^{(k)}\right| < \varepsilon_{\sigma}.$$

4.5.2 Robustness Check

Newton–Raphson may become unstable when vega is very small.

Therefore, the student should also implement a bracketing method such as bisection.

The bisection algorithm starts from a volatility interval

$$[\sigma_L, \sigma_U]$$

such that

$$F(\sigma_L)F(\sigma_U) < 0.$$

At every iteration,

$$\sigma_M = \frac{\sigma_L + \sigma_U}{2}.$$

The interval containing the root is retained and the procedure is repeated.

Thus, the implied-volatility module shall contain

$$\boxed{\text{Newton–Raphson} + \text{Bisection Fallback.}}$$

Project Milestone

For a selected value of

$$\sigma_{\text{true}},$$

the student shall first compute

$$C_{\text{market}} = C_{\text{BS}}(\sigma_{\text{true}})$$

and then verify that the implied-volatility solver recovers

$$\sigma_{\text{IV}} \approx \sigma_{\text{true}}.$$

4.6 Algorithm 5: Binomial Option-Pricing Model

The next independent pricing method is the binomial lattice.

Divide the maturity interval T into N equal steps,

$$\Delta t = \frac{T}{N}.$$

At each time step, the asset price either moves upward or downward.

Define

$$u = e^{\sigma\sqrt{\Delta t}},$$

and

$$d = e^{-\sigma\sqrt{\Delta t}}.$$

Hence,

$$d = \frac{1}{u}.$$

The risk-neutral probability of an upward movement is

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

The probability of a downward movement is

$$1 - p.$$

4.6.1 Asset-Price Lattice

At time step i, after j upward movements,

$$S_{i,j} = S_0 u^j d^{\,i-j}.$$

The resulting lattice is

$$\boxed{S_0 \to \{S_0u, S_0d\} \to \left\{S_0u^2, S_0ud, S_0d^2\right\} \to \cdots}$$

4.6.2 Terminal Payoff

For a European call,

$$V_{N,j} = \max\left(S_{N,j} - K, 0\right).$$

For a European put,

$$V_{N,j} = \max\left(K - S_{N,j}, 0\right).$$

4.6.3 Backward Induction

Starting from maturity, the option price is calculated recursively as

$$V_{i,j} = e^{-r\Delta t}\left[pV_{i+1,j+1} + (1-p)V_{i+1,j}\right].$$

The desired option price is

$$V_{0,0}.$$

4.7 Binomial Convergence Study

The student shall calculate the binomial price for increasing values of N, for example,

$$N = 10, 25, 50, 100, 250, 500.$$

Define the error

$$E_{\text{Binomial}}(N) = \left| C^{(N)}_{\text{Binomial}} - C_{\text{BS}} \right|.$$

The student shall plot

$$E_{\text{Binomial}}$$

against

$$N.$$

The expected behaviour is

$$\boxed{E_{\text{Binomial}}(N) \longrightarrow 0 \qquad \text{as} \qquad N \longrightarrow \infty.}$$

4.8 Algorithm 6: Monte Carlo Pricing

Under the risk-neutral measure,

$$S_T = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z\right],$$

where

$$Z \sim N(0,1).$$

The Monte Carlo algorithm shall generate

$$Z_1, Z_2, \ldots, Z_M$$

and compute

$$S_T^{(j)} = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z_j\right].$$

For a call option, the simulated payoff is

$$H_j = \max\left(S_T^{(j)} - K, 0\right).$$

The Monte Carlo price is

$$\boxed{C_{\mathrm{MC}} = e^{-rT}\frac{1}{M}\sum_{j=1}^{M} H_j.}$$

4.9 Monte Carlo Statistical Error

The sample standard deviation of the discounted payoffs is

$$s_H = \sqrt{\frac{1}{M-1}\sum_{j=1}^{M}(e^{-rT}H_j - C_{\mathrm{MC}})^2}.$$

The Monte Carlo standard error is

$$\boxed{\mathrm{SE}_{\mathrm{MC}} = \frac{s_H}{\sqrt{M}}.}$$

Therefore, the sampling error decreases approximately as

$$\frac{1}{\sqrt{M}}.$$

The student shall verify this behaviour computationally.

4.10 Monte Carlo Convergence Experiment

The Monte Carlo price shall be calculated for

$$M = 10^2, 10^3, 10^4, 10^5, 10^6$$

or smaller upper limits if computational resources require.

Define

$$E_{\mathrm{MC}}(M) = \left| C_{\mathrm{MC}}^{(M)} - C_{\mathrm{BS}} \right|.$$

The student shall examine the behaviour of

$$E_{\mathrm{MC}}(M)$$

as the number of simulations increases.

4.10.1 Repeated Monte Carlo Runs

A single Monte Carlo run can produce an unusually good or bad answer by chance.

Therefore, selected experiments shall be repeated using different random seeds.

The student shall compare

mean Monte Carlo estimate,

standard deviation between runs,

and

theoretical Monte Carlo standard error.

This is important preparation for understanding uncertainty in machine-learning model evaluation.

4.11 Algorithm 7: Simulation of Complete GBM Paths

Although European option pricing requires only the terminal value S_T, complete GBM paths shall also be generated because they provide a useful visualization of the stochastic model.

For a discretization

$$t_0, t_1, \ldots, t_N,$$

with

$$\Delta t = \frac{T}{N},$$

the exact discrete GBM update is

$$\boxed{S_{i+1} = S_i \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_i\right].}$$

Under risk-neutral simulation,

$$\mu = r.$$

The student shall produce plots containing multiple simulated paths,

$$S_t^{(1)}, S_t^{(2)}, \ldots, S_t^{(M)}.$$

Key Idea

GBM path generation is included not simply for visualization.
The student should understand that the same stochastic model provides the foundation for

Black–Scholes, binomial approximation, Monte Carlo simulation.

4.12 Unified Cross-Validation of the Three Pricing Methods

After independently implementing all three approaches, the student shall construct a common test set.

For each parameter combination

$$(S_i, K_i, T_i, r_i, \sigma_i)\,,$$

calculate

$$C_{\mathrm{BS}}^{(i)},$$

$$C_{\mathrm{Binomial}}^{(i)},$$

and

$$C_{\mathrm{MC}}^{(i)}.$$

Define

$$e_{\mathrm{Binomial}}^{(i)} = C_{\mathrm{Binomial}}^{(i)} - C_{\mathrm{BS}}^{(i)},$$

and

$$e_{\mathrm{MC}}^{(i)} = C_{\mathrm{MC}}^{(i)} - C_{\mathrm{BS}}^{(i)}.$$

Summary error measures shall include

$$\mathrm{MAE}_{\mathrm{Binomial}} = \frac{1}{n}\sum_{i=1}^{n}\left|e_{\mathrm{Binomial}}^{(i)}\right|,$$

and

$$\mathrm{MAE}_{\mathrm{MC}} = \frac{1}{n}\sum_{i=1}^{n}\left|e_{\mathrm{MC}}^{(i)}\right|.$$

4.13 Parameter-Sensitivity Experiments

Before constructing the machine-learning dataset, the student shall study how the Black–Scholes price changes with each model input.

Each experiment shall vary only one variable while keeping the others fixed.

4.13.1 Underlying Asset Price

Study

$$C = C(S)$$

for a range of S.

The student shall plot

$$S \quad \text{versus} \quad C.$$

4.13.2 Strike Price

Study

$$C = C(K).$$

The expected behaviour for a call is

$$\frac{\partial C}{\partial K} < 0.$$

4.13.3 Volatility

Study

$$C = C(\sigma).$$

For standard European options,

$$\frac{\partial C}{\partial \sigma} = \nu > 0.$$

4.13.4 Time to Maturity

Study

$$C = C(T).$$

The student shall examine how option value changes as the expiry date approaches.

4.13.5 Interest Rate

Study

$$C = C(r).$$

These experiments provide the student with visual intuition regarding the nonlinear structure that the ML model will later attempt to learn.

4.14 Moneyness-Based Computational Study

Define log-moneyness

$$m = \ln\left(\frac{S}{K}\right).$$

The student shall generate option prices across a range such as

$$m \in [-0.5, 0.5]$$

or an appropriate range determined by the data.

Options shall be classified approximately as

$$m < -\delta \quad \Rightarrow \quad \text{OTM},$$

$$|m| \leq \delta \quad \Rightarrow \quad \text{ATM},$$

and

$$m > \delta \quad \Rightarrow \quad \text{ITM},$$

for a small chosen threshold δ.

These categories will later be retained during ML model evaluation.

4.15 Required Computational Plots

Before proceeding to machine learning, the student shall generate at least the following plots:

1. Black–Scholes call price against underlying price,

$$C_{\text{BS}} \text{ versus } S.$$

2. Black–Scholes call price against volatility,

$$C_{\text{BS}} \text{ versus } \sigma.$$

3. Delta against underlying price,

$$\Delta \text{ versus } S.$$

4. Gamma against underlying price,

$$\Gamma \text{ versus } S.$$

5. Vega against underlying price,

$$\nu \text{ versus } S.$$

6. Binomial pricing error against number of lattice steps,

$$E_{\text{Binomial}} \text{ versus } N.$$

7. Monte Carlo pricing error against number of simulations,

$$E_{\text{MC}} \text{ versus } M.$$

8. A collection of simulated geometric Brownian-motion paths.

4.16 Computational Validation Gate

The complete pre-ML computational stage may be summarized as

Input Parameters
(S, K, T, r, σ)
↓
Black–Scholes Analytical Price
↓
Analytical Greeks
↓
Finite-Difference Validation
↓
Implied-Volatility Solver
↓
Binomial Pricing
↓
Monte Carlo Pricing
↓
Cross-Method Validation
↓
Machine-Learning Dataset.

Project Milestone

The machine-learning stage shall begin only when the student can demonstrate all of the following:

1. Black–Scholes call and put prices are computed correctly;
2. put–call parity is satisfied numerically;
3. analytical and finite-difference Greeks agree;
4. implied volatility can be recovered from a known option price;

5. the binomial model converges toward the Black–Scholes price;
6. Monte Carlo prices approach the Black–Scholes price within the estimated statistical error;
7. complete GBM sample paths can be generated and interpreted; and
8. all algorithms have been organized into reusable functions suitable for subsequent dataset generation.

4.17 Output of the Computational Stage

The final output of this stage is a validated numerical engine

$$\boxed{\mathcal{P} : (S, K, T, r, \sigma) \longrightarrow [C, P, \Delta, \Gamma, \nu, \sigma_{\mathrm{IV}}]\,.}$$

This numerical engine will subsequently serve three purposes:

Generate Synthetic Training Data
Process Real Market Option Data
Provide a Benchmark for AI/ML Models.

Only after this computational foundation has been validated will the project move to the data-driven component.

5. AI/ML Formulation

The AI/ML component of the project begins only after the Black–Scholes and numerical pricing framework has been validated.

The purpose of this stage is not to discard the mathematical model. Instead, the project will formulate several supervised-learning problems and compare their behaviour against the classical Black–Scholes benchmark.

The overall data-driven architecture is

$$\boxed{\text{Financial Inputs} \longrightarrow \text{Feature Engineering} \longrightarrow \text{Machine-Learning Model} \longrightarrow \text{Prediction} \longrightarrow \text{Financial Validati}}$$

Three distinct learning formulations will be studied:

Direct Option-Price Learning
Black–Scholes Residual Learning
Implied-Volatility Learning.

5.1 Supervised-Learning Framework

Suppose that the dataset contains n observations.

For observation i, let

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{ip}]^T$$

denote the input feature vector and let

$$y_i$$

denote the corresponding target variable.

A supervised-learning model seeks a function

$$f_\theta$$

such that

$$\widehat{y}_i = f_\theta(\mathbf{x}_i).$$

Here,

$$\theta$$

denotes the parameters of the model.

The parameters are determined from the training dataset by minimizing a suitable loss function.

5.2 Choice of Financial Features

The raw option-pricing variables are

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma.$$

However, machine learning should not be applied blindly to raw variables when more meaningful financial representations are available.

An important derived variable is log-moneyness,

$$\boxed{m = \ln\left(\frac{S}{K}\right).}$$

The project will therefore consider a basic feature vector of the form

$$\mathbf{x} = [m, T, r, \sigma].$$

A richer feature vector may additionally include the Black–Scholes sensitivities,

$$\mathbf{x} = [m, T, r, \sigma, \Delta, \Gamma, \nu].$$

Where appropriate, additional market-derived volatility or return features may be introduced later.

Key Idea

Feature engineering must preserve financial meaning.
The objective is not to provide the ML algorithm with as many variables as possible, but to provide a compact set of variables that represent the structure of the option-pricing problem.

5.3 Learning Problem 1: Direct Option-Price Prediction

The first formulation is a purely data-driven benchmark.

The machine-learning model is asked to predict the option price directly,

$$\boxed{\widehat{C}_{\mathrm{ML}} = f_\theta(\mathbf{x}).}$$

For example,

$$\mathbf{x} = [m, T, r, \sigma]\,,$$

and

$$y = C.$$

For synthetic data, the target may initially be

$$y = C_{\mathrm{BS}}.$$

This experiment has a pedagogical purpose.

It tests whether the selected machine-learning models can approximate the nonlinear Black–Scholes mapping

$$(S, K, T, r, \sigma) \longrightarrow C_{\mathrm{BS}}.$$

However, reproducing a known analytical formula using machine learning is not the main research objective of the project.

Model-Risk Perspective

A neural network that reproduces Black–Scholes accurately does not constitute evidence that machine learning has improved option pricing.
It merely demonstrates that a flexible numerical model can approximate a known nonlinear function.

5.4 Learning Problem 2: Black–Scholes Residual Learning

The primary machine-learning formulation of the project is residual learning.

For every market observation, calculate

$$C_{\mathrm{BS}}^{(i)}$$

using the corresponding market variables.

Define the Black–Scholes residual as

$$\boxed{\varepsilon_i = C_{\mathrm{market}}^{(i)} - C_{\mathrm{BS}}^{(i)}.}$$

The machine-learning model is then trained using

$$\mathbf{x}_i \longrightarrow \varepsilon_i.$$

Thus,

$$\widehat{\varepsilon}_i = f_\theta(\mathbf{x}_i).$$

The hybrid price becomes

$$\boxed{\widehat{C}_{\mathrm{Hybrid}}^{(i)} = C_{\mathrm{BS}}^{(i)} + \widehat{\varepsilon}_i.}$$

5.5 Interpretation of Residual Learning

Residual learning separates the option-price prediction problem into two components,

$$C_{\mathrm{market}} = C_{\mathrm{BS}} + \varepsilon_{\mathrm{BS}}.$$

The first component,

$$C_{\mathrm{BS}},$$

contains the structure explained by the mathematical model.

The second component,

$$\varepsilon_{\mathrm{BS}},$$

contains the discrepancy between the mathematical model and the observed market value.

The machine-learning task is therefore reduced from

learn the complete pricing function

to

$$\boxed{\text{learn only the systematic part of the model error.}}$$

This is conceptually important because the machine-learning component does not need to relearn the entire option-pricing structure already provided by Black–Scholes.

5.6 Residual-Learning Architecture

The residual-learning framework can be represented as

$$\boxed{\begin{array}{ccc} (S,K,T,r,\sigma) & \longrightarrow & C_{\mathrm{BS}} \\ & & \downarrow \\ \mathbf{x} & \longrightarrow & \widehat{\varepsilon}_{\mathrm{ML}} \\ & & \downarrow \\ & & \widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}. \end{array}}$$

The residual model shall be treated as the principal AI/ML model in this project.

5.7 Learning Problem 3: Implied-Volatility Prediction

The third learning problem predicts implied volatility rather than option price.
For every observed option price,

$$C_{\mathrm{market}},$$

the numerical solver developed earlier provides

$$\sigma_{\mathrm{IV}}.$$

The training dataset therefore contains pairs

$$\mathbf{x}_i \longrightarrow \sigma_{\mathrm{IV},i}.$$

The ML model becomes

$$\boxed{\widehat{\sigma}_{\mathrm{IV}} = g_\theta(\mathbf{x}).}$$

A simple feature set is

$$\mathbf{x} = [m, T, r]\,,$$

where

$$m = \ln\left(\frac{S}{K}\right).$$

Additional volatility and market-state variables may be introduced if real market data permit.

5.8 Pricing Through Predicted Implied Volatility

Once

$$\widehat{\sigma}_{\mathrm{IV}}$$

has been obtained, it is inserted into the Black–Scholes formula,

$$\boxed{\widehat{C}_{\mathrm{IV-ML}} = C_{\mathrm{BS}}\left(S, K, T, r, \widehat{\sigma}_{\mathrm{IV}}\right).}$$

The complete architecture is therefore

$$\boxed{\text{Option Features} \longrightarrow \text{ML Model} \longrightarrow \widehat{\sigma}_{\mathrm{IV}} \longrightarrow \text{Black–Scholes} \longrightarrow \widehat{C}.}$$

This formulation retains the analytical pricing structure and assigns machine learning only the task of modelling the volatility surface.

5.9 Comparison of the Three Learning Formulations

The three approaches can now be summarized as follows.

Model A:	$\mathbf{x} \longrightarrow \widehat{C}_{\mathrm{ML}}$	Pure ML
Model B:	$\mathbf{x} \longrightarrow \widehat{\varepsilon}_{\mathrm{ML}}$	Residual ML
Model C:	$\mathbf{x} \longrightarrow \widehat{\sigma}_{\mathrm{IV}}$	Volatility ML.

The corresponding final option prices are

$$\widehat{C}_A = \widehat{C}_{\mathrm{ML}},$$

$$\widehat{C}_B = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}},$$

and

$$\widehat{C}_C = C_{\mathrm{BS}}\left(\widehat{\sigma}_{\mathrm{IV}}\right).$$

5.10 Training, Validation and Test Sets

The dataset shall be separated into three mutually distinct subsets,

$$\mathcal{D}_{\mathrm{train}},$$

$$\mathcal{D}_{\mathrm{val}},$$

and

$$\mathcal{D}_{\mathrm{test}}.$$

A representative split may be

$$70\% : 15\% : 15\%.$$

Thus,

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}.$$

The sets shall satisfy

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \emptyset,$$

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \emptyset,$$

and

$$\mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset.$$

5.10.1 Training Set

The training set is used to estimate the model parameters,

$$\theta.$$

5.10.2 Validation Set

The validation set is used for model selection and hyperparameter choice.

5.10.3 Test Set

The test set shall be used only after the final model has been selected.

Its purpose is to provide an unbiased estimate of out-of-sample performance.

Model-Risk Perspective

The test dataset must not be repeatedly inspected during model development.
Repeated use of the test dataset effectively converts it into another validation dataset and produces an overly optimistic estimate of model performance.

5.11 Time-Aware Data Splitting for Market Data

If real historical option data are used, purely random splitting may produce information leakage because observations from later dates can enter the training set while earlier observations appear in the test set.

For time-ordered market data, the preferred structure is

$$\boxed{\text{Earlier Dates} \longrightarrow \text{Training} \longrightarrow \text{Validation} \longrightarrow \text{Later Dates for Testing.}}$$

Thus,

$$t_{\text{train}} < t_{\text{validation}} < t_{\text{test}}.$$

This arrangement better represents the actual financial forecasting problem in which future market information is unavailable during model training.

5.12 Feature Scaling

Machine-learning algorithms can be sensitive to differences in the numerical scales of input variables.

For example,

$$T$$

may lie approximately between

$$0 \quad \text{and} \quad 2,$$

whereas other variables may have substantially different scales.

A standard transformation is

$$\boxed{z_j = \frac{x_j - \mu_j}{s_j},}$$

where

$$\mu_j$$

is the training-set mean of feature j, and

$$s_j$$

is the corresponding standard deviation.

The scaling parameters must be estimated using only

$$\mathcal{D}_{\text{train}}.$$

The same transformation is then applied to validation and test data.

5.13 Avoiding Data Leakage

Suppose the feature scaler is estimated using the complete dataset.

Then information from the test sample enters the training process.

This constitutes data leakage.

Therefore,

Fit Preprocessing on Training Data Only.

The correct workflow is

Split Data
↓
Fit Preprocessing on Training Set
↓
Transform Training Set
↓
Transform Validation/Test Using Same Parameters.

This rule applies to

scaling,

feature selection,

PCA,

and other data transformations.

5.14 Loss Function

For a regression problem, a standard loss function is mean squared error,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 .$$

For direct price prediction,

$$y_i = C_i.$$

For residual learning,

$$y_i = \varepsilon_i.$$

For implied-volatility learning,

$$y_i = \sigma_{\text{IV},i}.$$

Mean absolute error may alternatively be used,

$$\mathcal{L}_{\text{MAE}} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| .$$

5.15 Linear Regression as the Baseline Model

The first model shall be ordinary linear regression.

For p features,

$$y_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i.$$

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The ordinary least-squares estimate is

$$\boxed{\widehat{\boldsymbol{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y},}$$

provided the required inverse exists.

Linear regression provides an interpretable statistical baseline and will establish how much of the target can be explained by a simple linear model.

5.16 Ridge Regression

If the input features are correlated, the regression problem may become unstable.

A regularized alternative is ridge regression,

$$\boxed{\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^T\boldsymbol{\beta}\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2\right].}$$

The parameter

$$\lambda$$

controls the degree of regularization.

When

$$\lambda = 0,$$

ridge regression reduces to ordinary least squares.

Increasing

$$\lambda$$

penalizes excessively large coefficients and may reduce overfitting.

5.17 Nonlinear Machine-Learning Models

Black–Scholes option prices are nonlinear functions of the model inputs.

Therefore, after establishing the regression baseline, one nonlinear tree-based method and one small neural network shall be studied.

The model hierarchy is

$$\boxed{\text{Linear Regression} \longrightarrow \text{Ridge Regression} \longrightarrow \text{Tree-Based Model} \longrightarrow \text{Neural Network.}}$$

The objective is to determine whether increasing model flexibility produces a meaningful out-of-sample improvement.

5.18 Tree-Based Regression

A regression tree recursively divides the feature space into regions.

Within each terminal region R_m, the model predicts approximately

$$\widehat{y} = c_m, \qquad \mathbf{x} \in R_m.$$

A collection of trees may be combined through an ensemble model.

For example, a random-forest prediction may be represented as

$$\boxed{\widehat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}),}$$

where

$$T_b$$

is the prediction from tree b.

Alternatively, a gradient-boosted tree model may be used.

Only one principal tree-based ensemble need be retained for the final comparison.

5.19 Small Feed-Forward Neural Network

The final model class shall be a relatively small feed-forward neural network.

For one hidden layer,

$$\mathbf{h} = g\left(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1\right),$$

and

$$\widehat{y} = \mathbf{W}_2 \mathbf{h} + b_2.$$

Here,

$$g(\cdot)$$

is a nonlinear activation function.

For example,

$$g(z) = \max(0, z)$$

for the ReLU activation.

A two-hidden-layer network may be written as

$$\mathbf{h}_1 = g\left(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1\right),$$

$$\mathbf{h}_2 = g\left(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2\right),$$

and

$$\widehat{y} = \mathbf{W}_3\mathbf{h}_2 + b_3.$$

5.20 Restriction on Neural-Network Complexity

The project does not require a deep neural network.

A representative architecture may contain

$$2$$

hidden layers with a moderate number of neurons.

The objective is

$$\boxed{\text{controlled nonlinear approximation,}}$$

not

$$\boxed{\text{maximum possible network complexity.}}$$

Very large networks are inappropriate for a three-month project because they complicate interpretation, tuning and reproducibility.

5.21 Bias–Variance Trade-Off

A model that is too simple may fail to represent the relevant relationship.

This is known as underfitting.

A model that is excessively flexible may fit random fluctuations in the training dataset.

This is known as overfitting.

Conceptually,

$$\boxed{\text{Prediction Error} = \text{Bias} + \text{Variance} + \text{Irreducible Error.}}$$

The project therefore seeks a balance between model complexity and out-of-sample generalization.

5.22 Training Error and Validation Error

During model development, the student shall monitor

$$E_{\text{train}}$$

and

$$E_{\text{validation}}.$$

If both errors are large,

$$E_{\text{train}} \approx E_{\text{validation}} \gg 0,$$

the model may be underfitting.

If

$$E_{\text{train}} \ll E_{\text{validation}},$$

the model may be overfitting.

The desired situation is

$$E_{\text{train}} \approx E_{\text{validation}}$$

with both errors sufficiently small.

5.23 Hyperparameter Selection

Machine-learning models contain parameters determined from data and hyperparameters selected externally.

Examples include

$$\lambda$$

for ridge regression,

$$\text{tree depth},$$

$$\text{number of trees},$$

learning rate,

and

number of hidden neurons.

Hyperparameters shall be selected using

$$\mathcal{D}_{\text{validation}}$$

or cross-validation performed within the training dataset.

They shall not be selected using the final test set.

5.24 Model Comparison

For each learning formulation, the student shall compare at least

Linear Regression
Ridge Regression
One Tree-Based Ensemble
One Small Neural Network.

The goal is to determine whether increasingly nonlinear models improve performance sufficiently to justify their additional complexity.

5.25 Primary Research Comparison

The most important comparison in the project is not

Random Forest versus Neural Network.

The central comparison is

Black–Scholes
versus
Pure ML
versus
Black–Scholes + Residual ML
versus
ML Implied Volatility + Black–Scholes.

This comparison addresses the principal research question:

Is it better to replace the classical model,

or

to augment the classical model?

5.26 Generalization Beyond the Training Region

A machine-learning model may perform well on observations similar to its training data but fail when exposed to unfamiliar market conditions.

Therefore, the student shall distinguish between

Interpolation

and

Extrapolation.

Interpolation corresponds to predictions within the region well represented by the training sample.

Extrapolation corresponds to predictions outside or near the boundary of that region.

Financial risk management is particularly concerned with such boundary behaviour because market stress may push variables beyond their normal historical ranges.

5.27 Controlled Extrapolation Experiment

A simple experiment shall be performed.

Suppose the model is trained mainly on

$$m \in [-0.25, 0.25].$$

The student may evaluate the model on a wider range such as

$$m \in [-0.40, 0.40].$$

Likewise, a model trained over a volatility range

$$\sigma \in [0.10, 0.40]$$

may be evaluated at selected values outside or near the boundaries of this interval.

The purpose is not to claim reliable extrapolation.

The purpose is to observe where the ML model begins to lose stability.

Model-Risk Perspective

A financial ML model must not be judged solely by interpolation accuracy.
Poor extrapolation under unusual market conditions can represent a significant source of model risk even when average test-set error is small.

5.28 Financial Consistency Checks for ML Predictions

Predicted option prices shall be examined for basic financial consistency.

For a European call,

$$\widehat{C} \geq 0,$$

and

$$\widehat{C} \leq S.$$

The call price should generally increase with the underlying price,

$$\frac{\partial \widehat{C}}{\partial S} \geq 0,$$

and for standard European calls should increase with volatility,

$$\frac{\partial \widehat{C}}{\partial \sigma} \geq 0.$$

Thus, a model producing low RMSE but violating basic financial relationships must be treated cautiously.

5.29 Model Error as a Function of Market State

The student shall not report only one global error statistic.

For each model, define

$$e_i = C_i - \widehat{C}_i.$$

The errors shall be studied as functions of

$$m,$$

$$T,$$

and

$$\sigma.$$

Thus, plots such as

$$e \text{ versus } m,$$

$$e \text{ versus } T,$$

and

$$e \text{ versus } \sigma$$

shall be generated.

This allows systematic regions of model failure to be identified.

5.30 Research Hypotheses

The project may be organized around the following hypotheses.

5.30.1 Hypothesis 1

A nonlinear machine-learning model can approximate the Black–Scholes pricing function more accurately than a simple linear regression model.

Symbolically,

$$E_{\text{nonlinear}} < E_{\text{linear}}.$$

5.30.2 Hypothesis 2

For real market option prices, the Black–Scholes residual contains systematic structure that can be learned from observable option features.

That is,

$$\varepsilon_{\text{BS}} \neq \text{pure random noise.}$$

5.30.3 Hypothesis 3

A hybrid residual-learning model can outperform the uncorrected Black–Scholes benchmark,

$$E_{\text{Hybrid}} < E_{\text{BS}}.$$

5.30.4 Hypothesis 4

Predicting implied volatility and subsequently using Black–Scholes pricing provides greater financial interpretability than direct black-box option-price prediction.

5.30.5 Hypothesis 5

The model with the smallest average prediction error need not necessarily exhibit the best behaviour under market stress or extrapolation.

5.31 AI/ML Development Gate

The AI/ML stage shall proceed according to the sequence

Prepare Features and Targets
↓
Split Training/Validation/Test Data
↓
Fit Linear Regression
↓
Fit Ridge Regression
↓
Fit Tree-Based Model
↓
Fit Small Neural Network
↓
Compare Validation Performance
↓
Freeze Final Models
↓
Evaluate Once on Test Data
↓
Financial and Risk Validation.

Project Milestone

At the completion of the AI/ML formulation stage, the student must be able to state clearly, for every model,

1. what the input variables are;
2. what the prediction target is;
3. which observations form the training, validation and test sets;
4. which loss function is minimized;
5. which hyperparameters are selected;
6. which financial benchmark is used for comparison; and
7. what constitutes successful out-of-sample performance.

No machine-learning result shall be accepted without this specification.

5.32 Output of the AI/ML Formulation

At the end of this stage, the project will contain three explicit data-driven mappings:

$$\boxed{\mathcal{M}_1 : \mathbf{x} \longrightarrow \widehat{C}_{\mathrm{ML}},}$$

$$\boxed{\mathcal{M}_2 : \mathbf{x} \longrightarrow \widehat{\varepsilon}_{\mathrm{ML}},}$$

and

$$\boxed{\mathcal{M}_3 : \mathbf{x} \longrightarrow \widehat{\sigma}_{\mathrm{IV}}.}$$

These produce the three pricing outputs

$$\widehat{C}_{\mathrm{ML}},$$

$$\widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}},$$

and

$$\widehat{C}_{\mathrm{IV-ML}} = C_{\mathrm{BS}}\left(\widehat{\sigma}_{\mathrm{IV}}\right).$$

The next stage will determine how the datasets required for these models are generated, cleaned, transformed and organized.

6. Dataset Generation and Feature Engineering

The quality of a machine-learning model depends strongly on the quality of the dataset used for training and evaluation.

In the present project, data preparation will be carried out in two stages:

Stage I: Synthetic Black–Scholes Dataset

↓

Validate the ML Pipeline

Stage II: Real Market Option Dataset

↓

Study Model Residuals and Implied Volatility.

The first dataset provides a controlled environment in which the true pricing function is known.

The second dataset introduces the discrepancies between the idealized Black–Scholes assumptions and observed market behaviour.

6.1 Stage I: Synthetic Black–Scholes Dataset

The synthetic dataset shall be generated directly from the validated Black–Scholes pricing engine developed in the previous sections.

Each synthetic observation is defined by the parameter vector

$$\mathbf{z} = [S, K, T, r, \sigma] .$$

For each parameter combination, the computational framework shall calculate

$$C_{\mathrm{BS}},$$

$$P_{\mathrm{BS}},$$

$$\Delta,$$

$$\Gamma,$$

and

$$\nu.$$

Thus, one synthetic record may be represented as

$$\boxed{[S, K, T, r, \sigma, C_{\mathrm{BS}}, P_{\mathrm{BS}}, \Delta, \Gamma, \nu] .}$$

6.2 Selection of Parameter Ranges

The parameter ranges should represent financially reasonable option configurations rather than arbitrary numerical values.

A representative initial range may be chosen as

$$S \in [80, 120],$$

$$K \in [80, 120],$$

$$T \in [0.05, 2.0],$$

$$r \in [0.00, 0.10],$$

and

$$\sigma \in [0.10, 0.60].$$

These ranges are only starting values.

They may be modified later to reflect the range observed in the real market dataset.

6.3 Sampling Strategy

The student shall avoid generating all observations from a rigid uniform grid unless such a grid is required for visualization.

A random sampling strategy is preferable for ML training.

For example,

$$S_i \sim U(S_{\min}, S_{\max}),$$

$$K_i \sim U(K_{\min}, K_{\max}),$$

$$T_i \sim U(T_{\min}, T_{\max}),$$

$$r_i \sim U(r_{\min}, r_{\max}),$$

and

$$\sigma_i \sim U(\sigma_{\min}, \sigma_{\max}),$$

where $U(a, b)$ denotes the uniform distribution on the interval $[a, b]$.

The student may initially generate approximately

$$10^4$$

to

$$10^5$$

synthetic option observations.

The exact number is not critical provided that the dataset is sufficiently large to train the selected models without unnecessary computational burden.

6.4 Importance of Dimensionless Variables

Raw variables such as S and K contain redundant scale information.

A more informative variable is the ratio

$$\frac{S}{K}.$$

The preferred representation is log-moneyness,

$$\boxed{m = \ln\left(\frac{S}{K}\right).}$$

This variable naturally distinguishes between in-the-money, at-the-money and out-of-the-money options.

For call options,

$$m > 0$$

corresponds approximately to in-the-money conditions,

$$m = 0$$

corresponds to at-the-money conditions, and

$$m < 0$$

corresponds approximately to out-of-the-money conditions.

6.5 Basic Synthetic Feature Set

The first ML experiment shall use a compact feature set,

$$\boxed{\mathbf{x}^{(1)} = [m, T, r, \sigma].}$$

The target is

$$y = C_{\mathrm{BS}}.$$

Thus,

$$\boxed{[m, T, r, \sigma] \longrightarrow C_{\mathrm{BS}}.}$$

This experiment tests whether the machine-learning model can approximate the nonlinear Black–Scholes pricing function.

6.6 Extended Synthetic Feature Set

A second experiment may include the principal option sensitivities,

$$\boxed{\mathbf{x}^{(2)} = [m, T, r, \sigma, \Delta, \Gamma, \nu].}$$

This allows the student to determine whether sensitivity information improves the regression model.

However, the student shall interpret such an experiment carefully, because the Greeks themselves are derived from the same Black–Scholes model used to generate the target.

Model-Risk Perspective

Using Black–Scholes Greeks as features when the target is itself a Black–Scholes price can make the synthetic learning problem artificially easy.
Such experiments are useful for understanding feature relevance but must not be confused with evidence of improved market pricing.

6.7 Normalization by Strike

An alternative representation is to normalize the call price by the strike,

$$c = \frac{C}{K}.$$

Similarly,

$$s = \frac{S}{K}.$$

The machine-learning problem then becomes

$$\boxed{\left[\frac{S}{K}, T, r, \sigma\right] \longrightarrow \frac{C}{K}.}$$

This transformation may reduce scale dependence and improve numerical conditioning.

The student may compare the normalized and non-normalized representations.

6.8 Synthetic Dataset Validation

Before training an ML model, the synthetic dataset shall be subjected to basic consistency checks.

For every call option,

$$C_{\mathrm{BS}} \geq 0.$$

Also,

$$C_{\mathrm{BS}} \leq S.$$

For every put,

$$P_{\mathrm{BS}} \geq 0.$$

Put–call parity shall satisfy

$$C_{\mathrm{BS}} - P_{\mathrm{BS}} - S + Ke^{-rT} \approx 0.$$

Any rows violating basic financial consistency due to coding or data errors shall be investigated rather than silently removed.

6.9 Distributional Inspection of Synthetic Features

The student shall inspect the statistical distribution of each input variable.

For each feature x_j, calculate

$$\overline{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij},$$

and

$$s_j^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_{ij} - \overline{x}_j)^2 .$$

The student shall also plot histograms for

$$m, \qquad T, \qquad \sigma, \qquad C_{\mathrm{BS}}.$$

The purpose is to verify that the generated dataset covers the intended region of the parameter space.

6.10 Coverage of the Option Space

A machine-learning model may perform poorly if the training data are concentrated in only one option regime.

Therefore, the synthetic dataset shall contain sufficient examples of

ITM options,

ATM options,

and

OTM options.

A practical classification based on log-moneyness may use

$$m < -\delta \quad \Rightarrow \quad \text{OTM},$$

$$|m| \le \delta \quad \Rightarrow \quad \text{ATM},$$

and

$$m > \delta \quad \Rightarrow \quad \text{ITM},$$

where δ is a small threshold selected for the analysis.

The number of observations in each region shall be reported.

6.11 Maturity Buckets

Options shall also be grouped according to time to maturity.

For example,

$$T < T_1 \quad \Rightarrow \quad \text{Short Maturity},$$

$$T_1 \leq T < T_2 \quad \Rightarrow \quad \text{Medium Maturity},$$

and

$$T \geq T_2 \quad \Rightarrow \quad \text{Long Maturity}.$$

The exact thresholds may be selected according to the available dataset.

These categories will later be used for model-error analysis.

6.12 Stage II: Real Market Option Dataset

After the synthetic ML pipeline has been validated, the project shall move to real option data.

A real market dataset should contain, at minimum,

date,

underlying price,

strike price,

expiry date,

and

option price.

Where available, the dataset should also include

bid price,

ask price,

trading volume,

and

open interest.

The precise market and data source may be selected according to data availability during the project.

6.13 Choice of Market Price

If both bid and ask prices are available, the mid-price may be used as

$$\boxed{C_{\text{mid}} = \frac{C_{\text{bid}} + C_{\text{ask}}}{2}.}$$

The corresponding bid–ask spread is

$$\text{Spread} = C_{\text{ask}} - C_{\text{bid}}.$$

A relative spread may be defined as

$$\text{Relative Spread} = \frac{C_{\text{ask}} - C_{\text{bid}}}{C_{\text{mid}}}.$$

Large relative spreads may indicate illiquid options and should be treated cautiously.

6.14 Time to Maturity

Suppose the valuation date is

$$t$$

and the expiry date is

$$T_{\text{exp}}.$$

Time to maturity in years shall be calculated as

$$\boxed{\tau = \frac{\text{Number of days to expiry}}{365}.}$$

Alternatively, an appropriate trading-day convention may be used if the dataset is constructed consistently.

The variable

$$\tau$$

will replace T in the practical market implementation.

6.15 Risk-Free Interest Rate

Each market option observation requires an associated risk-free rate.

Ideally,

$$r = r(\tau),$$

so that the interest rate is maturity dependent.

If the project uses a simplified constant short-term interest rate, that approximation shall be stated explicitly.

More sophisticated yield-curve construction is outside the main scope of the three-month project unless reliable data are readily available.

6.16 Dividend Treatment

The basic Black–Scholes formulas developed earlier assume no dividend payment.

If options on a dividend-paying equity or equity index are used, the pricing model may need to incorporate a dividend yield q.

The call price becomes

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2),$$

with

$$d_1 = \frac{\ln(S/K) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

If dividend information is unavailable or deliberately excluded, the chosen market dataset should be selected so that the resulting simplification is clearly acknowledged.

Key Idea

The student shall not use real market option data mechanically.
Every real-data input must correspond to the assumptions of the pricing model being implemented.

6.17 Filtering Invalid or Unusable Market Observations

Market datasets frequently contain observations that are unsuitable for model estimation.

Examples include

missing prices,

zero bid prices,

negative or zero time to maturity,

clearly erroneous observations,

or extremely illiquid contracts.

The student shall define and document explicit filtering rules before model training.

The data-cleaning procedure should be reproducible.

6.18 No-Arbitrage Screening

For a non-dividend-paying European call,

$$C \geq \max\left(S - Ke^{-rT}, 0\right),$$

and

$$C \leq S.$$

For a European put,

$$P \geq \max\left(Ke^{-rT} - S, 0\right),$$

and

$$P \leq Ke^{-rT}.$$

Market observations violating basic no-arbitrage bounds shall be flagged and investigated.

6.19 Computing Market Implied Volatility

For each valid option observation, the implied volatility solver developed earlier shall be applied. Given

$$C_{\text{market}},$$

solve

$$C_{\text{BS}}\left(S, K, T, r, \sigma_{\text{IV}}\right) = C_{\text{market}}.$$

This produces

$$\boxed{\sigma_{\text{IV}}}$$

for each option.

The market dataset can then be augmented as

$$\boxed{[S, K, T, r, C_{\text{market}}, \sigma_{\text{IV}}]\,.}$$

6.20 Computing the Black–Scholes Benchmark Price

For every market observation, the student shall compute a Black–Scholes benchmark price

$$C_{\text{BS}}.$$

The precise volatility input used for this benchmark must be defined carefully.

Possible experiments include using

$$\text{historical volatility},$$

or another deliberately chosen baseline volatility estimate.

The benchmark must not use the same option's market-implied volatility if the purpose of the experiment is to measure the Black–Scholes pricing residual, because by definition

$$C_{\text{BS}}\left(\sigma_{\text{IV}}\right) = C_{\text{market}}.$$

Using the market-implied volatility in this case would force

$$\varepsilon_{\text{BS}} \approx 0$$

and eliminate the residual-learning problem.

Model-Risk Perspective

This is a critical methodological point.
If the market option price is first converted into its own implied volatility and that same implied volatility is immediately inserted back into Black–Scholes, the model will reproduce the market price by construction.
Such a calculation cannot be used to demonstrate that Black–Scholes predicts market prices accurately.

6.21 Historical Volatility Feature

Suppose daily underlying prices are

$$S_0, S_1, \ldots, S_n.$$

The daily log return is

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right).$$

Over a rolling window containing m returns, calculate

$$\overline{r} = \frac{1}{m}\sum_{i=1}^{m} r_i.$$

The sample standard deviation is

$$s_r = \sqrt{\frac{1}{m-1}\sum_{i=1}^{m}(r_i - \overline{r})^2}.$$

An annualized historical volatility estimate is

$$\boxed{\sigma_{\text{hist}} = s_r\sqrt{N_{\text{yr}}},}$$

where

$$N_{\text{yr}}$$

denotes the assumed number of trading periods per year.

For daily returns, a common convention is approximately

$$N_{\text{yr}} = 252.$$

6.22 Black–Scholes Residual Dataset

Using historical or another predetermined volatility estimate, calculate

$$C_{\text{BS}} = C_{\text{BS}}\left(S, K, T, r, \sigma_{\text{hist}}\right).$$

Then define

$$\boxed{\varepsilon_{\text{BS}} = C_{\text{market}} - C_{\text{BS}}.}$$

This creates the principal residual-learning dataset,

$$\boxed{\mathbf{x} \longrightarrow \varepsilon_{\text{BS}}.}$$

A representative feature vector is

$$\mathbf{x} = \left[m, T, r, \sigma_{\text{hist}}, \Delta, \Gamma, \nu\right].$$

6.23 Implied-Volatility Dataset

The second real-data learning problem uses

$$\sigma_{\text{IV}}$$

as the prediction target.

A basic feature vector is

$$\boxed{\mathbf{x}_{\mathrm{IV}} = [m, T, r]\,.}$$

The supervised-learning problem becomes

$$\boxed{[m, T, r] \longrightarrow \sigma_{\mathrm{IV}}.}$$

Where sufficient data are available, additional market-state variables may be included, such as

$$\sigma_{\mathrm{hist}},$$

$$\text{recent underlying return,}$$

or liquidity-related variables.

6.24 Volatility Surface Representation

Once implied volatilities have been computed, the student shall plot

$$\sigma_{\mathrm{IV}}$$

against moneyness for several maturities.

The basic relationship is

$$\boxed{\sigma_{\mathrm{IV}} = f(m, T).}$$

A three-dimensional visualization may be created with axes

$$m, \qquad T, \qquad \sigma_{\mathrm{IV}}.$$

This provides the empirical volatility surface that the ML model will attempt to approximate.

6.25 Feature Matrix

After preprocessing, the complete feature matrix may be represented as

$$\mathbf{X} = \begin{bmatrix} m_1 & T_1 & r_1 & \sigma_1 & \Delta_1 & \Gamma_1 & \nu_1 \\ m_2 & T_2 & r_2 & \sigma_2 & \Delta_2 & \Gamma_2 & \nu_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_n & T_n & r_n & \sigma_n & \Delta_n & \Gamma_n & \nu_n \end{bmatrix}.$$

The target vector depends on the learning problem.

6.25.1 Direct Price Learning

$$\mathbf{y}_C = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{bmatrix}.$$

6.25.2 Residual Learning

$$\mathbf{y}_\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

6.25.3 Implied-Volatility Learning

$$\mathbf{y}_{\mathrm{IV}} = \begin{bmatrix} \sigma_{\mathrm{IV},1} \\ \sigma_{\mathrm{IV},2} \\ \vdots \\ \sigma_{\mathrm{IV},n} \end{bmatrix}.$$

6.26 Feature Correlation Analysis

Before fitting the ML models, the student shall examine correlations between the numerical features.

For two variables X and Y, the sample correlation is

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

The feature correlation matrix shall be calculated and visualized.

Strong correlation between variables such as

$$\Delta, \qquad m,$$

or other derived quantities should be expected and interpreted.

This analysis also motivates the use of regularized regression.

6.27 Optional Principal Component Analysis

If the feature set is subsequently expanded and strong multicollinearity is observed, principal component analysis may be introduced as an optional exercise.

If

$$\boldsymbol{\Sigma}$$

is the covariance or correlation matrix, the eigenvalue problem is

$$\boldsymbol{\Sigma}\mathbf{v}_j = \lambda_j \mathbf{v}_j.$$

The principal components are linear combinations

$$PC_j = \mathbf{v}_j^T \mathbf{x}.$$

However, PCA is not mandatory for the core three-month project.

It should be introduced only if it solves an actual dimensionality or multicollinearity problem rather than simply because it is available.

6.28 Training, Validation and Test Split

For the synthetic dataset, a random split may initially be used.

For example,

$$70\% \quad \text{training},$$

$$15\% \quad \text{validation},$$

and

$$15\% \quad \text{testing}.$$

For real historical market data, the preferred split is chronological.

For example,

$$\begin{array}{|c|}\hline \text{Earlier Period} \\ \downarrow \\ \mathcal{D}_{\text{train}} \\ \text{Intermediate Period} \\ \downarrow \\ \mathcal{D}_{\text{validation}} \\ \text{Latest Period} \\ \downarrow \\ \mathcal{D}_{\text{test}}. \\ \hline \end{array}$$

This prevents future information from entering the training stage.

6.29 Standardization of Features

For ML methods requiring feature scaling, each variable shall be standardized using training-set statistics.

For feature x_j,

$$\boxed{z_{ij} = \frac{x_{ij} - \mu_j^{\text{train}}}{s_j^{\text{train}}}.}$$

The same

$$\mu_j^{\text{train}}$$

and

$$s_j^{\text{train}}$$

must then be applied to validation and test observations.

Under no circumstances shall scaling parameters be recomputed separately using the test dataset.

6.30 Missing Data

Missing observations shall be identified before model training.

The number and percentage of missing values in every field shall be reported.

Where missing observations are few and occur in essential variables such as

$$S, \qquad K, \qquad C_{\text{market}},$$

the corresponding records may be removed.

Complex data imputation is outside the principal scope of this project unless required by the selected dataset.

6.31 Outlier Inspection

Outliers shall not be removed automatically.

Instead, the student shall distinguish between

$$\text{data errors}$$

and

$$\text{genuine extreme market observations.}$$

A genuine extreme observation may be especially important for a market-risk study.

Therefore,

$$\boxed{\text{Extreme} \neq \text{Incorrect.}}$$

Any outlier-removal rule must be explicitly justified.

6.32 Final Dataset Structure

The final processed dataset should contain, where available,

$$\boxed{\begin{array}{c}\text{Date}\\ S\\ K\\ T\\ r\\ m=\ln(S/K)\\ \sigma_{\text{hist}}\\ C_{\text{market}}\\ C_{\text{BS}}\\ \varepsilon_{\text{BS}}\\ \sigma_{\text{IV}}\\ \Delta\\ \Gamma\\ \nu\\ \text{Moneyness Category}\\ \text{Maturity Category.}\end{array}}$$

6.33 Three Final ML Datasets

From the processed master dataset, three learning datasets shall be constructed.

6.33.1 Dataset A: Direct Pricing

$$\boxed{\mathbf{X}_A \longrightarrow C_{\text{market}}.}$$

6.33.2 Dataset B: Residual Learning

$$\boxed{\mathbf{X}_B \longrightarrow C_{\text{market}} - C_{\text{BS}}.}$$

6.33.3 Dataset C: Implied Volatility

$$\boxed{\mathbf{X}_C \longrightarrow \sigma_{\text{IV}}.}$$

6.34 Required Exploratory Plots

Before machine-learning model fitting, the student shall generate at least the following visualizations:

1. Histogram of log returns,

$$r_t.$$

2. Historical volatility through time,

$$\sigma_{\text{hist}}(t).$$

3. Market option price against strike,

$$C_{\text{market}} \text{ versus } K.$$

4. Market price against Black–Scholes price,

$$C_{\text{market}} \text{ versus } C_{\text{BS}}.$$

5. Black–Scholes residual against log-moneyness,

$$\varepsilon_{\text{BS}} \text{ versus } m.$$

6. Black–Scholes residual against maturity,

$$\varepsilon_{\text{BS}} \text{ versus } T.$$

7. Implied volatility against moneyness,

$$\sigma_{\text{IV}} \text{ versus } m.$$

8. Implied-volatility surface,

$$\sigma_{\text{IV}} = f(m, T).$$

9. Feature-correlation matrix.

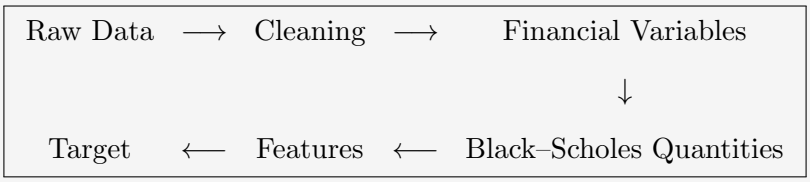
6.35 Dataset Quality Gate

Before model training begins, the student shall demonstrate that

1. the synthetic dataset satisfies the Black–Scholes consistency checks;
2. the selected parameter space contains ITM, ATM and OTM options;
3. short-, medium- and long-maturity contracts are represented;
4. real market observations have been cleaned using documented rules;
5. basic no-arbitrage bounds have been checked;
6. implied volatility has been computed reliably;
7. the Black–Scholes residual has been defined using an independent volatility benchmark rather than the same option's implied volatility;
8. the training, validation and test periods have been separated correctly;
9. preprocessing parameters have been estimated from the training data only; and
10. the final features and targets are explicitly documented.

Project Milestone

No machine-learning model shall be accepted if the student cannot trace every training observation back through the complete data pipeline,



6.36 Output of the Data Preparation Stage

The final output of this stage is a reproducible master dataset

$$\mathcal{D}$$

from which the three supervised-learning problems can be constructed:

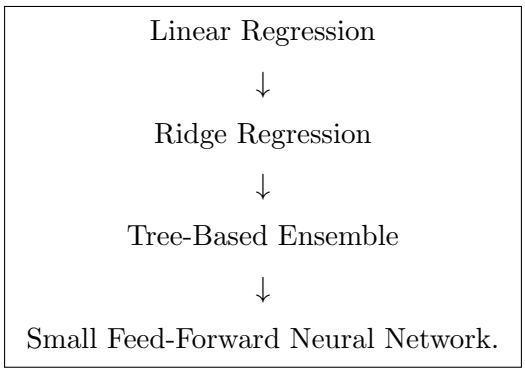
$$\boxed{\mathcal{D} \longrightarrow \{\mathcal{D}_{\text{Price}}, \mathcal{D}_{\text{Residual}}, \mathcal{D}_{\text{IV}}\}.}$$

The project can then proceed to the systematic implementation and comparison of the selected machine-learning algorithms.

7. Machine-Learning Algorithms

The purpose of this section is to define the machine-learning models that will be implemented and compared in the project.

The selected model hierarchy is deliberately limited to a small number of representative algorithms:



The objective is not to test as many algorithms as possible.

Instead, the project will examine whether increasing model complexity produces a genuine improvement over simpler and more interpretable models.

7.1 Common Supervised-Learning Formulation

For each observation i, let

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{ip}]^T$$

denote the feature vector.

The target is denoted by

$$y_i.$$

Depending on the experiment,

$$y_i$$

may represent

$$C_i,$$

or

$$\varepsilon_i = C_{\text{market},i} - C_{\text{BS},i},$$

or

$$\sigma_{\text{IV},i}.$$

The generic supervised-learning problem is

$$\boxed{\widehat{y}_i = f_\theta(\mathbf{x}_i).}$$

The model parameters

$$\theta$$

are estimated from the training dataset.

7.2 Algorithm 1: Ordinary Least-Squares Regression

The first model shall be ordinary least-squares regression.

For p input variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i.$$

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The least-squares estimate minimizes

$$J(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 .$$

Thus,

$$\boxed{\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .}$$

Provided that

$$\mathbf{X}^T\mathbf{X}$$

is invertible,

$$\boxed{\widehat{\boldsymbol{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}.}$$

The corresponding prediction is

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}.$$

7.3 Role of Linear Regression

Linear regression serves several purposes in the present project.

First, it provides an interpretable baseline.

Second, it determines whether the selected financial features already contain a strong approximately linear relationship with the target.

Third, it allows the student to understand the limitations of a linear model when approximating nonlinear option-pricing relationships.

For example, if

$$\mathbf{x} = [m, T, r, \sigma] ,$$

then the model takes the form

$$\widehat{y} = \beta_0 + \beta_1 m + \beta_2 T + \beta_3 r + \beta_4 \sigma.$$

This model should not be expected to reproduce the full nonlinear Black–Scholes structure exactly.

Its principal purpose is benchmarking.

7.4 Algorithm 2: Ridge Regression

Financial features may be correlated.

For example,

$$m, \qquad \Delta, \qquad \Gamma$$

may contain related information.

Strong correlation among explanatory variables can make ordinary least-squares coefficients unstable.

Ridge regression introduces an L_2 penalty.

The objective function is

$$\boxed{J_{\text{ridge}} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.}$$

Equivalently,

$$\boxed{\widehat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right].}$$

The closed-form solution is

$$\boxed{\widehat{\boldsymbol{\beta}}_{\text{ridge}} = \left(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}\right)^{-1} \mathbf{X}^T\mathbf{y}.}$$

7.5 Interpretation of the Regularization Parameter

The parameter

$$\lambda \geq 0$$

controls model regularization.

When

$$\lambda = 0,$$

ridge regression reduces to ordinary least squares.

As

$$\lambda$$

increases, large coefficients are penalized more strongly.

Thus,

$$\boxed{\lambda \text{ small} \rightarrow \text{less regularization},}$$

while

$$\boxed{\lambda \text{ large} \rightarrow \text{more regularization}.}$$

The value of

$$\lambda$$

shall be selected using validation data or cross-validation.

7.6 Algorithm 3: Tree-Based Ensemble Model

The next model shall be a nonlinear tree-based ensemble.

A single regression tree divides the feature space recursively into regions

$$R_1, R_2, \ldots, R_M.$$

Within each terminal region,

$$\widehat{y} = c_m, \qquad \mathbf{x} \in R_m.$$

The tree therefore constructs a piecewise nonlinear approximation to the target function.

7.7 Random Forest Regression

A suitable ensemble model for the project is random forest regression.

Suppose that

$$B$$

regression trees are constructed.

The prediction from tree b is

$$T_b(\mathbf{x}).$$

The random-forest prediction is

$$\boxed{\widehat{y}_{\mathrm{RF}} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}).}$$

Each tree is trained using a bootstrap sample of the training data, while only a subset of available features is considered at each split.

This introduces diversity among the individual trees.

7.8 Why Random Forest is Useful Here

Random forest provides several advantages for the present project.

It can represent nonlinear relationships,

$$y = f(\mathbf{x}),$$

without requiring an explicit analytical functional form.

It can also represent interactions such as

$$m \times T,$$

or

$$m \times \sigma,$$

without these interaction terms being specified manually.

This makes it a useful intermediate model between linear regression and neural networks.

7.9 Principal Random-Forest Hyperparameters

The principal hyperparameters to be investigated are

$$B = \text{number of trees},$$

$$d_{\max} = \text{maximum tree depth},$$

$$n_{\min} = \text{minimum samples required for a split},$$

and the number of candidate features considered at each split.

A small controlled grid shall be used rather than an excessively large hyperparameter search.

7.10 Alternative: Gradient-Boosted Trees

If preferred, gradient-boosted regression trees may replace random forest as the project's principal tree-based model.

A boosted model has the additive form

$$F_M(\mathbf{x}) = \sum_{m=0}^{M} \eta h_m(\mathbf{x}),$$

where

$$h_m$$

is a regression tree and

$$\eta$$

is the learning rate.

Only one principal tree-based ensemble shall be retained in the final model comparison.

Thus, the student should use either

Random Forest

or

Gradient Boosting,

rather than attempting to optimize many different tree algorithms.

7.11 Algorithm 4: Feed-Forward Neural Network

The final model class shall be a small feed-forward neural network.

Let the input vector be

$$\mathbf{x} \in \mathbb{R}^p.$$

For one hidden layer,

$$\mathbf{z}_1 = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1.$$

After applying a nonlinear activation function,

$$\mathbf{h}_1 = g(\mathbf{z}_1).$$

The output is then

$$\widehat{y} = \mathbf{W}_2\mathbf{h}_1 + b_2.$$

For two hidden layers,

$$\mathbf{h}_1 = g\left(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1\right),$$

$$\mathbf{h}_2 = g\left(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2\right),$$

and

$$\boxed{\widehat{y} = \mathbf{W}_3\mathbf{h}_2 + b_3.}$$

7.12 Activation Function

A suitable activation function is the rectified linear unit,

$$\boxed{g(z) = \max(0, z).}$$

This is commonly referred to as the ReLU activation.

A nonlinear activation function is essential because, without it, multiple linear layers would collapse mathematically into a single linear transformation.

7.13 Recommended Neural-Network Architecture

The neural network shall remain deliberately small.

A representative architecture is

$$\boxed{p \to 32 \to 16 \to 1.}$$

Thus,

$$p$$

input features are passed to a first hidden layer containing

$$32$$

neurons, then to a second hidden layer containing

$$16$$

neurons, and finally to a scalar regression output.

An even smaller network may be used if it provides comparable validation performance.

Model-Risk Perspective

The objective of the project is not to demonstrate that a large neural network can fit a sufficiently large dataset.
A smaller network that produces comparable out-of-sample performance is preferable because it is easier to train, interpret and validate.

7.14 Neural-Network Loss Function

For regression, the principal training loss shall be mean squared error,

$$\boxed{\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.}$$

The network parameters

$$\theta = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \ldots\}$$

are estimated by minimizing

$$\mathcal{L}(\theta).$$

7.15 Gradient-Based Training

The neural network is trained iteratively using gradient-based optimization.

Conceptually,

$$\boxed{\theta^{(k+1)} = \theta^{(k)} - \eta \nabla_\theta \mathcal{L}\left(\theta^{(k)}\right),}$$

where

$$\eta$$

is the learning rate.

An optimizer such as Adam may be used for implementation.

The mathematical details of advanced optimization algorithms are not a principal objective of this project.

7.16 Epochs and Mini-Batches

One complete pass through the training dataset is called an epoch.

The training data may be divided into mini-batches.

If the batch size is

$$B,$$

then the model parameters are updated after processing approximately

$$B$$

training observations.

The student shall monitor the training and validation loss as a function of epoch number.

7.17 Early Stopping

Training shall not automatically continue for a fixed very large number of epochs.

Instead, validation loss shall be monitored.

If

$$\mathcal{L}_{\text{val}}$$

does not improve for a predetermined number of epochs, training may be terminated.

This procedure is known as early stopping.

Conceptually,

$$\boxed{\text{Stop training when validation performance stops improving.}}$$

Early stopping helps control overfitting.

7.18 Algorithm Training Sequence

All models shall be trained using the same training and validation datasets.

The sequence is

Training Dataset
↓
OLS
↓
Ridge
↓
Tree Ensemble
↓
Small ANN
↓
Validation Comparison
↓
Freeze Final Models
↓
Final Test Evaluation.

7.19 Common Evaluation Metrics

For each model, the mean absolute error shall be calculated as

$$\boxed{\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \widehat{y}_i| \,.}$$

The root mean square error is

$$\boxed{\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2}.}$$

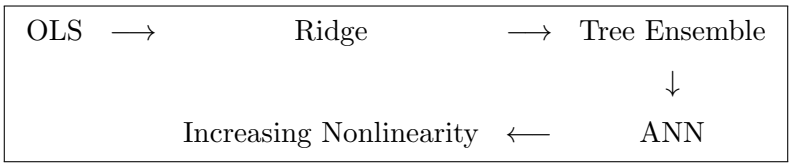
The coefficient of determination is

$$\boxed{R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \overline{y})^2}.}$$

7.20 Model-Complexity Comparison

The student shall compare model performance against model complexity.

The hierarchy may be represented as



The student shall determine whether

$$\mathrm{RMSE}_{\mathrm{ANN}} < \mathrm{RMSE}_{\mathrm{RF}}$$

or similar improvements are sufficiently large to justify additional complexity.

A difference that is numerically very small should not automatically be interpreted as a meaningful improvement.

7.21 Feature Importance

For the tree-based model, feature importance shall be examined.

The student should determine whether the most influential variables include quantities such as

$$m, \qquad T, \qquad \sigma,$$

or other engineered features.

Feature importance shall be interpreted cautiously.

It indicates which variables are useful to the fitted predictive model; it does not automatically establish economic causality.

7.22 Residual Analysis

For every fitted model, define the prediction residual

$$\boxed{e_i = y_i - \widehat{y}_i.}$$

The student shall examine

$$e_i$$

against

$$m_i,$$

$$T_i,$$

and

$$\sigma_i.$$

A good model should not leave strong systematic structure in its residuals.

If, for example,

$$e$$

changes systematically with moneyness, this indicates that the model has not fully captured the moneyness dependence of the target.

7.23 Model Performance by Moneyness

The overall test dataset shall be divided into

$$\text{ITM},$$

$$\text{ATM},$$

and

$$\text{OTM}$$

subsets.

For every subset, the student shall separately calculate

$$\text{MAE},$$

and

$$\text{RMSE}.$$

Thus,

$$\text{RMSE}_{\text{ITM}},$$

$$\text{RMSE}_{\text{ATM}},$$

and

$$\text{RMSE}_{\text{OTM}}$$

shall be reported.

7.24 Model Performance by Maturity

Similarly, model errors shall be evaluated for

$$\text{short maturity},$$

$$\text{medium maturity},$$

and

$$\text{long maturity}.$$

This gives

$$\text{RMSE}_{\text{short}},$$

$$\text{RMSE}_{\text{medium}},$$

and

$$\text{RMSE}_{\text{long}}.$$

The purpose is to identify regions where a model performs poorly even if its overall error appears small.

7.25 Training Stability

For the neural network, a single training run shall not be treated as conclusive.

Selected experiments shall be repeated using several random seeds.

If

$$E_1, E_2, \ldots, E_R$$

represent test errors from R independent runs, the student shall report

$$\overline{E} = \frac{1}{R} \sum_{i=1}^{R} E_i,$$

and the corresponding standard deviation.

This gives an indication of training variability.

7.26 Model Selection Principle

The final model shall not be selected solely according to minimum training error.

The selection criterion shall consider

Validation Accuracy
+
Test Accuracy
+
Stability
+
Financial Consistency
+
Interpretability.

This is particularly important in financial applications, where a model may have a small average error but behave poorly under unusual market conditions.

7.27 Required Plots for Each Model

For each major ML model, the student shall produce at least the following plots:

1. Predicted value against true value,

$$\widehat{y} \text{ versus } y.$$

2. Prediction residual against predicted value,

$$e \text{ versus } \widehat{y}.$$

3. Residual against log-moneyness,

$$e \text{ versus } m.$$

4. Residual against maturity,

$$e \text{ versus } T.$$

5. Training and validation loss for the neural network.
6. Feature importance for the selected tree-based ensemble.

7.28 Algorithmic Comparison Table

The final report shall summarize the ML models in a table containing at least

Model
Number of Features
Principal Hyperparameters
Training MAE
Validation MAE
Test MAE
Test RMSE
R^2
Training Time.

This allows both predictive performance and computational complexity to be compared.

7.29 Machine-Learning Algorithm Gate

The machine-learning algorithm stage shall be considered complete only when the student can demonstrate that

1. OLS has been implemented as the statistical baseline;
2. ridge regression has been fitted with validation-based regularization;
3. one nonlinear tree-based ensemble has been trained;
4. one small feed-forward neural network has been trained;
5. all models use the same predefined train/validation/test datasets;
6. model hyperparameters are selected without using the final test set;
7. prediction errors are reported globally and by moneyness and maturity;
8. model residuals have been inspected visually;
9. neural-network training stability has been examined; and
10. the final model choice is justified by more than a single error metric.

Project Milestone

At the end of this stage, the student shall possess a controlled hierarchy of models

$$\mathcal{M}_{\text{OLS}}, \quad \mathcal{M}_{\text{Ridge}}, \quad \mathcal{M}_{\text{Tree}}, \quad \mathcal{M}_{\text{ANN}},$$

that can be applied consistently to each of the three learning problems:

Direct Price Prediction,	Residual Prediction,	Implied-Volatility Prediction.

These trained models provide the input to the hybrid Black–Scholes–ML framework developed in the next section.

8. Hybrid Black–Scholes–ML Framework

The central objective of this project is not merely to compare different machine-learning algorithms.

The principal research question is whether a classical mathematical pricing model can be improved through a carefully designed data-driven correction while preserving financial interpretability.

The four pricing frameworks to be compared are

Black–Scholes
Pure Machine Learning
Black–Scholes + Residual ML
ML Implied Volatility + Black–Scholes.

These models differ fundamentally in the way financial structure and data-driven learning are combined.

8.1 Model 1: Classical Black–Scholes Benchmark

The first model is the classical analytical benchmark,

$$\boxed{\widehat{C}_{\mathrm{BS}} = C_{\mathrm{BS}}\left(S, K, T, r, \sigma_{\mathrm{base}}\right).}$$

Here,

$$\sigma_{\mathrm{base}}$$

denotes the volatility input selected independently for the benchmark, for example a historical-volatility estimate.

The pricing architecture is

$$\boxed{\text{Market Variables} \longrightarrow \text{Black–Scholes} \longrightarrow \widehat{C}_{\mathrm{BS}}.}$$

This model contains no machine-learning component.

It provides the reference against which all data-driven models will be evaluated.

8.2 Model 2: Pure Machine-Learning Pricing

The second model ignores the analytical pricing formula during prediction.

The feature vector is

$$\mathbf{x} = \left[m, T, r, \sigma_{\mathrm{base}}, \ldots\right],$$

where

$$m = \ln\left(\frac{S}{K}\right).$$

The machine-learning model directly predicts

$$\boxed{\widehat{C}_{\mathrm{ML}} = f_\theta(\mathbf{x}).}$$

The corresponding architecture is

$$\boxed{\text{Market Features} \longrightarrow \text{Machine Learning} \longrightarrow \widehat{C}_{\mathrm{ML}}.}$$

This model provides a useful black-box benchmark.

Model-Risk Perspective

The pure ML model is deliberately included as a comparison model.
It is not automatically considered superior simply because it produces a smaller pricing error.
Unlike Black–Scholes, it does not explicitly preserve the analytical structure of option pricing.

8.3 Model 3: Black–Scholes Residual Learning

The third model is the principal hybrid architecture of the project.

For each market observation, first calculate

$$C_{\mathrm{BS}} = C_{\mathrm{BS}}\left(S, K, T, r, \sigma_{\mathrm{base}}\right).$$

The Black–Scholes residual is then

$$\boxed{\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}.}$$

A machine-learning algorithm is trained to approximate this residual,

$$\boxed{\widehat{\varepsilon}_{\mathrm{ML}} = f_\theta(\mathbf{x}).}$$

The corrected hybrid price is

$$\boxed{\widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}.}$$

8.4 Interpretation of the Residual Model

The market price is decomposed as

$$C_{\mathrm{market}} = C_{\mathrm{BS}} + \varepsilon_{\mathrm{BS}}.$$

The hybrid model approximates this as

$$C_{\mathrm{market}} \approx C_{\mathrm{BS}} + f_\theta(\mathbf{x}).$$

Thus, Black–Scholes is responsible for the primary pricing structure,

$$C_{\mathrm{BS}},$$

while machine learning is responsible only for the systematic component of the pricing discrepancy,

$$\varepsilon_{\mathrm{BS}}.$$

The learning problem is therefore reduced from

learn the entire option-pricing function

to

$$\boxed{\text{learn the systematic model error.}}$$

8.5 Residual-Learning Architecture

The complete residual-learning workflow is

$$\boxed{\begin{array}{c}\text{Market Inputs}\\ \downarrow\\ \text{Black–Scholes Benchmark}\\ \downarrow\\ C_{\mathrm{BS}}\\ \downarrow\\ \varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}\\ \downarrow\\ \text{Train ML Residual Model}\\ \downarrow\\ \widehat{\varepsilon}_{\mathrm{ML}}\\ \downarrow\\ \widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}.\end{array}}$$

8.6 Choice of Residual-Learning Features

A basic residual-learning feature vector is

$$\boxed{\mathbf{x}_{\varepsilon} = [m, T, r, \sigma_{\mathrm{base}}]\,.}$$

A richer model may include Black–Scholes sensitivities,

$$\mathbf{x}_{\varepsilon} = [m, T, r, \sigma_{\mathrm{base}}, \Delta_{\mathrm{BS}}, \Gamma_{\mathrm{BS}}, \nu_{\mathrm{BS}}]\,.$$

The Greeks in this feature vector shall be calculated using the same independent benchmark volatility

$$\sigma_{\text{base}}$$

used to obtain

$$C_{\text{BS}}.$$

8.7 Model 4: ML Implied Volatility + Black–Scholes

The fourth framework retains Black–Scholes pricing but uses machine learning to estimate the implied-volatility surface.

For each market observation, implied volatility is first recovered numerically from

$$C_{\text{BS}}\left(S, K, T, r, \sigma_{\text{IV}}\right) = C_{\text{market}}.$$

This produces the supervised-learning target

$$\sigma_{\text{IV}}.$$

The ML model is then trained as

$$\boxed{\widehat{\sigma}_{\text{IV}} = g_\theta(\mathbf{x}_{\text{IV}}).}$$

A representative feature vector is

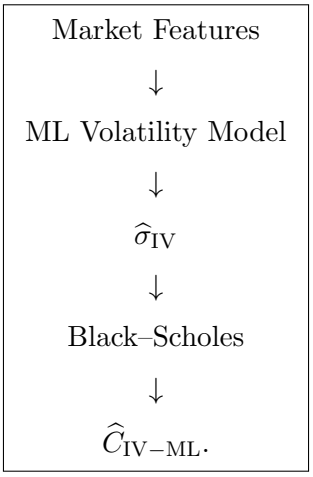
$$\mathbf{x}_{\text{IV}} = \left[m, T, r\right].$$

The predicted implied volatility is then inserted into the Black–Scholes formula,

$$\boxed{\widehat{C}_{\text{IV}-\text{ML}} = C_{\text{BS}}\left(S, K, T, r, \widehat{\sigma}_{\text{IV}}\right).}$$

8.8 Implied-Volatility Hybrid Architecture

The complete model is



This architecture differs fundamentally from direct price learning.

Machine learning estimates a financially meaningful intermediate quantity,

$$\sigma_{\mathrm{IV}},$$

while Black–Scholes retains responsibility for the final option valuation.

8.9 Why the Two Hybrid Models are Different

The residual model uses the decomposition

$$\boxed{C_{\mathrm{market}} = C_{\mathrm{BS}} + \varepsilon.}$$

The implied-volatility model uses

$$\boxed{C_{\mathrm{market}} = C_{\mathrm{BS}}\left(\sigma_{\mathrm{IV}}\right).}$$

Thus, the residual model learns

$$\varepsilon,$$

whereas the volatility model learns

$$\sigma_{\mathrm{IV}}.$$

These are not equivalent learning problems.

The residual model corrects the price directly.

The implied-volatility model modifies one of the inputs to the analytical pricing model.

8.10 Critical Separation of the Two Experiments

The residual-learning and implied-volatility-learning experiments must be constructed separately.

For residual learning, define

$$C_{\mathrm{BS}} = C_{\mathrm{BS}}\left(\sigma_{\mathrm{base}}\right),$$

where

$$\sigma_{\mathrm{base}}$$

is independent of the current option's market price.

Then,

$$\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}\left(\sigma_{\mathrm{base}}\right).$$

For implied-volatility learning, the target is instead

$$\sigma_{\mathrm{IV}},$$

obtained from the market price itself.

Model-Risk Perspective

The same option's implied volatility must not be used as the Black–Scholes volatility input in the residual-learning experiment.
If

$$\sigma_{\mathrm{base}} = \sigma_{\mathrm{IV}},$$

then

$$C_{\mathrm{BS}}\left(\sigma_{\mathrm{IV}}\right) = C_{\mathrm{market}}$$

by construction, and therefore

$$\varepsilon_{\mathrm{BS}} \approx 0.$$

This would destroy the residual-learning problem.

8.11 Training Architecture for the Four Models

The comparison shall use a common dataset split.

Let

$$\mathcal{D}_{\mathrm{train}}, \qquad \mathcal{D}_{\mathrm{val}}, \qquad \mathcal{D}_{\mathrm{test}}$$

denote the training, validation and test sets.

The models shall be trained as follows.

8.11.1 Black–Scholes

No training is required.

The benchmark produces

$$\widehat{C}_{\mathrm{BS}}.$$

8.11.2 Pure ML

Train

$$\mathbf{x} \longrightarrow C_{\mathrm{market}}.$$

8.11.3 Residual ML

Train

$$\mathbf{x}_{\varepsilon} \longrightarrow \varepsilon_{\mathrm{BS}}.$$

8.11.4 Implied-Volatility ML

Train

$$\mathbf{x}_{\mathrm{IV}} \longrightarrow \sigma_{\mathrm{IV}}.$$

The resulting

$$\widehat{\sigma}_{\mathrm{IV}}$$

is then passed through Black–Scholes to obtain the final predicted option price.

8.12 Unified Pricing Comparison

For every test observation i, compute

$$C_{\mathrm{market},i},$$

$$\widehat{C}_{\mathrm{BS},i},$$

$$\widehat{C}_{\mathrm{ML},i},$$

$$\widehat{C}_{\mathrm{Hybrid},i},$$

and

$$\widehat{C}_{\mathrm{IV-ML},i}.$$

Thus, each observation produces the comparison vector

$$\boxed{\left[C_{\text{market}}, \widehat{C}_{\text{BS}}, \widehat{C}_{\text{ML}}, \widehat{C}_{\text{Hybrid}}, \widehat{C}_{\text{IV-ML}}\right].}$$

8.13 Pricing Errors

Define

$$e_{\text{BS},i} = C_{\text{market},i} - \widehat{C}_{\text{BS},i},$$

$$e_{\text{ML},i} = C_{\text{market},i} - \widehat{C}_{\text{ML},i},$$

$$e_{\text{Hybrid},i} = C_{\text{market},i} - \widehat{C}_{\text{Hybrid},i},$$

and

$$e_{\text{IV-ML},i} = C_{\text{market},i} - \widehat{C}_{\text{IV-ML},i}.$$

The objective is to compare both the magnitude and the structure of these errors.

8.14 Improvement Relative to Black–Scholes

For any model M, define the RMSE improvement relative to Black–Scholes as

$$\boxed{I_M = 1 - \frac{\text{RMSE}_M}{\text{RMSE}_{\text{BS}}}.}$$

If

$$I_M > 0,$$

the model improves on the Black–Scholes RMSE.

For example,

$$I_M = 0.20$$

corresponds to an approximate 20% reduction in RMSE relative to the Black–Scholes benchmark.

Similarly, one may define

$$I_M^{\text{MAE}} = 1 - \frac{\text{MAE}_M}{\text{MAE}_{\text{BS}}}.$$

8.15 Error Reduction by Market Region

Overall improvement may conceal important differences across option regimes.

Therefore, the improvement measure shall also be calculated for

$$\text{ITM}, \qquad \text{ATM}, \qquad \text{OTM}$$

options.

For example,

$$I^{\text{ATM}}_{\text{Hybrid}} = 1 - \frac{\text{RMSE}_{\text{Hybrid,ATM}}}{\text{RMSE}_{\text{BS,ATM}}}.$$

Similarly, improvement shall be evaluated for

$$\text{short}, \qquad \text{medium}, \qquad \text{long}$$

maturities.

8.16 Residual Structure Before and After ML Correction

An important experiment is to compare

$$\varepsilon_{\text{BS}}$$

with the residual remaining after hybrid correction,

$$\varepsilon_{\text{Hybrid}} = C_{\text{market}} - \widehat{C}_{\text{Hybrid}}.$$

If the ML correction is successful, systematic dependence on variables such as

$$m$$

and

$$T$$

should be reduced.

Thus, compare

$$\varepsilon_{\text{BS}} \text{ versus } m$$

with

$$\varepsilon_{\text{Hybrid}} \text{ versus } m.$$

Likewise, compare

$$\varepsilon_{\mathrm{BS}} \text{ versus } T$$

with

$$\varepsilon_{\mathrm{Hybrid}} \text{ versus } T.$$

Key Idea

The hybrid model should not be judged only by whether its RMSE is smaller.
A successful correction should also remove systematic structure from the Black–Scholes residuals.

8.17 Financial Consistency of the Hybrid Price

The corrected price must remain financially meaningful.

For a European call,

$$\widehat{C}_{\mathrm{Hybrid}} \geq 0,$$

and

$$\widehat{C}_{\mathrm{Hybrid}} \leq S.$$

Similarly, the no-arbitrage lower bound shall be checked,

$$\widehat{C}_{\mathrm{Hybrid}} \geq \max\left(S - Ke^{-rT}, 0\right).$$

A hybrid correction that materially violates such bounds must be flagged even if its aggregate statistical error is small.

8.18 Monotonicity Check

A European call price should increase with the underlying price.

Therefore, numerically,

$$\frac{\partial \widehat{C}}{\partial S} \geq 0$$

should normally hold.

The student shall investigate this property for

$$\widehat{C}_{\mathrm{ML}},$$

$$\widehat{C}_{\mathrm{Hybrid}},$$

and

$$\widehat{C}_{\mathrm{IV-ML}}.$$

A finite-difference approximation may be used,

$$\frac{\widehat{C}(S+h)-\widehat{C}(S-h)}{2h}.$$

8.19 Volatility Monotonicity

For standard European options, option value increases with volatility.

Thus,

$$\frac{\partial C}{\partial \sigma}=\nu>0.$$

The student shall test whether ML-based pricing models preserve this basic relationship over the principal region of the dataset.

8.20 Hybrid Greeks

For the residual model,

$$\widehat{C}_{\mathrm{Hybrid}}=C_{\mathrm{BS}}+f_\theta(\mathbf{x}).$$

Therefore, formally,

$$\frac{\partial \widehat{C}_{\mathrm{Hybrid}}}{\partial S}=\frac{\partial C_{\mathrm{BS}}}{\partial S}+\frac{\partial f_\theta}{\partial S}.$$

Hence,

$$\boxed{\Delta_{\mathrm{Hybrid}}=\Delta_{\mathrm{BS}}+\Delta_{\mathrm{ML}}.}$$

Similarly,

$$\boxed{\Gamma_{\mathrm{Hybrid}}=\Gamma_{\mathrm{BS}}+\Gamma_{\mathrm{ML}}.}$$

These additional ML sensitivities need not have simple analytical expressions for every algorithm.

They may therefore be estimated numerically.

8.21 Numerical Hybrid Delta

For a small perturbation h,

$$\boxed{\Delta_{\mathrm{Hybrid}}\approx\frac{\widehat{C}_{\mathrm{Hybrid}}(S+h)-\widehat{C}_{\mathrm{Hybrid}}(S-h)}{2h}.}$$

Similarly,

$$\boxed{\Gamma_{\text{Hybrid}} \approx \frac{\widehat{C}_{\text{Hybrid}}(S+h) - 2\widehat{C}_{\text{Hybrid}}(S) + \widehat{C}_{\text{Hybrid}}(S-h)}{h^2}.}$$

This allows the risk sensitivities of the hybrid model to be compared with the classical Black–Scholes Greeks.

8.22 Risk Interpretation of the IV-ML Model

For the implied-volatility architecture,

$$\widehat{C}_{\text{IV}-\text{ML}} = C_{\text{BS}}\left(S, K, T, r, \widehat{\sigma}_{\text{IV}}\right).$$

Changes in the predicted price therefore arise from both direct changes in the Black–Scholes variables and changes in the ML-predicted volatility.

Conceptually,

$$d\widehat{C} = \Delta_{\text{BS}}\, dS + \nu_{\text{BS}}\, d\widehat{\sigma}_{\text{IV}} + \cdots$$

This provides a natural connection between volatility prediction and market-risk analysis.

8.23 Out-of-Sample Hybrid Evaluation

The hybrid models shall be evaluated only after all model choices have been frozen using training and validation data.

The final test workflow is

Unseen Test Observation

↓

Generate Required Financial Features

↓

Apply Frozen ML Model

↓

Construct Final Option Price

↓

Compare with Market Price

↓

Perform Financial Consistency Checks.

8.24 Stress Testing the Four Pricing Models

The four models shall also be compared under controlled perturbations.

For example,

$$S \longrightarrow S(1 + \delta_S),$$

and

$$\sigma \longrightarrow \sigma + \delta_\sigma.$$

The student shall compute

$$\widehat{C}_{\mathrm{BS}},$$

$$\widehat{C}_{\mathrm{ML}},$$

$$\widehat{C}_{\mathrm{Hybrid}},$$

and

$$\widehat{C}_{\mathrm{IV-ML}}$$

under each stress scenario.

The resulting price changes are

$$\Delta C_M = C_M^{\mathrm{stressed}} - C_M^{\mathrm{base}}.$$

These quantities will be used in the later market-risk analysis.

8.25 Hybrid Model Selection Criteria

The preferred hybrid architecture shall be selected using a combination of

Pricing Accuracy
+
Residual Structure
+
Financial Consistency
+
Stress Behaviour
+
Interpretability.

The model with the smallest RMSE will therefore not necessarily be selected automatically.

8.26 Required Hybrid-Model Plots

The student shall generate at least the following comparisons:

1. Market price versus Black–Scholes price,
$$C_{\text{market}} \text{ versus } C_{\text{BS}}.$$
2. Market price versus pure ML price,
$$C_{\text{market}} \text{ versus } \widehat{C}_{\text{ML}}.$$
3. Market price versus residual-corrected price,
$$C_{\text{market}} \text{ versus } \widehat{C}_{\text{Hybrid}}.$$
4. Market price versus IV-ML price,
$$C_{\text{market}} \text{ versus } \widehat{C}_{\text{IV}-\text{ML}}.$$
5. Black–Scholes residual versus moneyness.
6. Hybrid residual versus moneyness.
7. Black–Scholes residual versus maturity.
8. Hybrid residual versus maturity.
9. True implied volatility versus predicted implied volatility.
10. Classical and hybrid prices under selected stress scenarios.

8.27 Hybrid Framework Gate

The hybrid stage shall be considered complete only when the student can demonstrate that

1. a classical Black–Scholes benchmark has been defined using an independent volatility input;
2. a pure ML pricing model has been trained;
3. a Black–Scholes residual-learning model has been trained;
4. an implied-volatility ML model has been trained;
5. the residual and implied-volatility experiments have been kept methodologically separate;
6. all four models have been evaluated on the same unseen test period;
7. pricing improvements have been reported globally and by moneyness and maturity;
8. no-arbitrage and monotonicity checks have been performed;
9. hybrid sensitivities have been investigated numerically; and
10. the final preferred model has been justified on both statistical and financial grounds.

Project Milestone

The central deliverable of this stage is a unified comparison

Black–Scholes
Pure ML
Black–Scholes + Residual ML
ML Implied Volatility + Black–Scholes

for which pricing accuracy, model residuals, financial consistency, risk sensitivities and stress behaviour can all be evaluated within the same computational framework.

9. Model Validation and Performance Metrics

The purpose of model validation is to determine whether the proposed machine-learning and hybrid models produce genuine out-of-sample improvements over the classical Black–Scholes benchmark.

Model evaluation shall therefore not be based on a single accuracy measure.

The validation framework will combine

Statistical Accuracy
+
Residual Diagnostics
+
Performance by Market Regime
+
Financial Consistency
+
Stress and Extrapolation Behaviour.

The final objective is to determine whether the machine-learning component improves the Black–Scholes framework in a way that is both statistically measurable and financially meaningful.

9.1 Models to be Evaluated

The following pricing models shall be compared on the same test set:

Classical Black–Scholes
Pure Machine Learning
Black–Scholes + Residual ML
ML Implied Volatility + Black–Scholes.

For each test observation i, the corresponding predictions are

$$\widehat{C}_{\mathrm{BS},i},$$

$$\widehat{C}_{\mathrm{ML},i},$$

$$\widehat{C}_{\mathrm{Hybrid},i},$$

and

$$\widehat{C}_{\mathrm{IV-ML},i}.$$

The observed market price is

$$C_{\mathrm{market},i}.$$

9.2 Prediction Error

For any pricing model M, define the prediction error as

$$\boxed{e_{M,i} = C_{\mathrm{market},i} - \widehat{C}_{M,i}.}$$

A positive error implies that the model underprices relative to the market,

$$e_{M,i} > 0,$$

while a negative error implies overpricing,

$$e_{M,i} < 0.$$

The complete residual vector is

$$\mathbf{e}_M = [e_{M,1}, e_{M,2}, \ldots, e_{M,n}]^T.$$

9.3 Mean Error

The mean pricing error is

$$\boxed{\mathrm{ME}_M = \frac{1}{n}\sum_{i=1}^{n} e_{M,i}.}$$

This statistic measures systematic pricing bias.

If

$$\mathrm{ME}_M > 0,$$

the model tends to underprice.

If

$$\mathrm{ME}_M < 0,$$

the model tends to overprice.

A value close to zero is desirable, although a small mean error alone does not imply that individual prediction errors are small.

9.4 Mean Absolute Error

The mean absolute error is

$$\boxed{\mathrm{MAE}_M = \frac{1}{n}\sum_{i=1}^{n} |e_{M,i}| \,.}$$

MAE measures the average magnitude of the pricing error without allowing positive and negative errors to cancel.

It is easy to interpret because it is expressed in the same monetary units as the option price.

9.5 Mean Squared Error

The mean squared error is

$$\boxed{\mathrm{MSE}_M = \frac{1}{n}\sum_{i=1}^{n} e_{M,i}^2.}$$

Large errors receive a greater penalty because they enter quadratically.

9.6 Root Mean Square Error

The root mean square error is

$$\boxed{\mathrm{RMSE}_M = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_{M,i}^2}.}$$

RMSE is one of the principal performance metrics in the project.

Because large errors receive disproportionate weight, RMSE is particularly useful when occasional large pricing failures are important.

9.7 Coefficient of Determination

For models predicting a continuous target, the coefficient of determination is

$$\boxed{R^2 = 1 - \frac{\sum_{i=1}^{n} \left(C_{\mathrm{market},i} - \widehat{C}_{M,i}\right)^2}{\sum_{i=1}^{n} \left(C_{\mathrm{market},i} - \overline{C}_{\mathrm{market}}\right)^2}.}$$

Here,

$$\overline{C}_{\text{market}} = \frac{1}{n}\sum_{i=1}^{n} C_{\text{market},i}.$$

A value close to

$$1$$

indicates that a large fraction of the variation in market prices is explained by the model.

However, R^2 shall not be used as the sole criterion for model selection.

9.8 Normalized Pricing Error

Absolute pricing errors can be difficult to compare when option prices vary substantially in magnitude.

A normalized error may therefore be defined as

$$\boxed{e^{\text{rel}}_{M,i} = \frac{C_{\text{market},i} - \widehat{C}_{M,i}}{C_{\text{market},i} + \epsilon},}$$

where

$$\epsilon > 0$$

is a small numerical constant.

The corresponding mean absolute relative error is

$$\boxed{\text{MARE}_M = \frac{1}{n}\sum_{i=1}^{n} \left| e^{\text{rel}}_{M,i} \right|.}$$

This metric shall be interpreted carefully for options with very small market prices.

9.9 Error Relative to Bid–Ask Spread

Where bid and ask prices are available, model error may also be compared with market transaction uncertainty.

Define the bid–ask spread as

$$s_i = C_{\text{ask},i} - C_{\text{bid},i}.$$

A spread-normalized prediction error is

$$\boxed{e^{\text{spread}}_{M,i} = \frac{\left| C_{\text{market},i} - \widehat{C}_{M,i} \right|}{s_i + \epsilon}.}$$

If

$$e_{M,i}^{\text{spread}} < 1,$$

the prediction error lies within approximately one observed bid–ask spread.

This gives a more market-oriented interpretation of pricing accuracy.

9.10 Improvement Relative to Black–Scholes

For any model M, define the relative RMSE improvement over Black–Scholes as

$$\boxed{I_M^{\text{RMSE}} = 1 - \frac{\text{RMSE}_M}{\text{RMSE}_{\text{BS}}}.}$$

Likewise,

$$\boxed{I_M^{\text{MAE}} = 1 - \frac{\text{MAE}_M}{\text{MAE}_{\text{BS}}}.}$$

If

$$I_M^{\text{RMSE}} > 0,$$

the model improves upon Black–Scholes in RMSE terms.

If

$$I_M^{\text{RMSE}} < 0,$$

the model performs worse than Black–Scholes.

9.11 Percentage Error Reduction

The improvement measure may be expressed as a percentage,

$$\boxed{\%\text{Improvement} = 100 \left[1 - \frac{\text{RMSE}_M}{\text{RMSE}_{\text{BS}}}\right].}$$

For example,

$$25\%$$

improvement means that the model reduces Black–Scholes RMSE by one quarter.

9.12 Predicted Versus Observed Price Plot

For every model, the student shall construct a scatter plot of

$$\widehat{C}_M$$

against

$$C_{\text{market}}.$$

The reference line is

$$\boxed{\widehat{C} = C_{\text{market}}.}$$

Predictions lying close to this line indicate good agreement.
Separate plots shall be produced for

Black–Scholes,

Pure ML,

Residual Hybrid,

and

IV-ML Hybrid.

9.13 Residual Diagnostics

A model may have a low aggregate RMSE while still displaying systematic residual structure.
Therefore, residual analysis is mandatory.
For model M,

$$e_M = C_{\text{market}} - \widehat{C}_M.$$

The student shall examine

e_M versus m,

e_M versus T,

e_M versus σ,

and

e_M versus $\widehat{C}_M$.

9.14 Desired Residual Behaviour

An ideal residual should display no obvious systematic pattern.

Conceptually,

$$\boxed{\mathbb{E}\left[e_M \mid \mathbf{x}\right] \approx 0.}$$

Thus, residuals should fluctuate around zero rather than exhibiting clear curvature, trend or regime dependence.

If residual structure remains, it indicates that the model has failed to capture part of the relevant relationship.

9.15 Comparison of Black–Scholes and Hybrid Residuals

An important validation test is to determine whether the hybrid model removes systematic structure present in Black–Scholes errors.

Define

$$e_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}},$$

and

$$e_{\mathrm{Hybrid}} = C_{\mathrm{market}} - \widehat{C}_{\mathrm{Hybrid}}.$$

The student shall compare

$$e_{\mathrm{BS}} \text{ versus } m$$

with

$$e_{\mathrm{Hybrid}} \text{ versus } m,$$

and similarly for maturity.

A successful hybrid model should reduce both

$$\text{error magnitude}$$

and

$$\text{systematic error structure.}$$

9.16 Performance by Moneyness

Global accuracy measures shall be supplemented by regime-specific analysis.

Using log-moneyness,

$$m = \ln\left(\frac{S}{K}\right),$$

the option dataset shall be divided into

$$\text{OTM}, \qquad \text{ATM}, \qquad \text{ITM}$$

regions.

For each model, calculate

$$\text{RMSE}_{\text{OTM}},$$

$$\text{RMSE}_{\text{ATM}},$$

and

$$\text{RMSE}_{\text{ITM}}.$$

Similarly,

$$\text{MAE}_{\text{OTM}},$$

$$\text{MAE}_{\text{ATM}},$$

and

$$\text{MAE}_{\text{ITM}}$$

shall be reported.

9.17 Performance by Maturity

The test set shall also be divided into

$$\text{Short Maturity},$$

$$\text{Medium Maturity},$$

and

$$\text{Long Maturity}$$

subsets.

For each model,

$$\mathrm{RMSE}_{\mathrm{short}},$$

$$\mathrm{RMSE}_{\mathrm{medium}},$$

and

$$\mathrm{RMSE}_{\mathrm{long}}$$

shall be computed.

This allows the student to identify maturity regions where a model may be weak despite good overall performance.

9.18 Two-Dimensional Error Map

A more informative diagnostic combines moneyness and maturity.

Partition the option space into cells

$$R_{jk} = \{(m, T) : m \in M_j,\ T \in T_k\}\,.$$

For each region, calculate

$$\boxed{\mathrm{RMSE}_{jk} = \sqrt{\frac{1}{n_{jk}} \sum_{i \in R_{jk}} e_i^2}.}$$

The student may visualize

$$\mathrm{RMSE}(m, T)$$

as a heat map.

This provides a direct picture of where each pricing model succeeds or fails.

9.19 Implied-Volatility Model Validation

For the implied-volatility learning model, validation shall first be performed in volatility space.

Define

$$e_{\sigma,i} = \sigma_{\mathrm{IV},i} - \widehat{\sigma}_{\mathrm{IV},i}.$$

The volatility RMSE is

$$\boxed{\mathrm{RMSE}_{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} e_{\sigma,i}^2}.}$$

The volatility MAE is

$$\boxed{\mathrm{MAE}_\sigma = \frac{1}{n}\sum_{i=1}^{n} |e_{\sigma,i}| \,.}$$

The student shall compare

$$\widehat{\sigma}_{\mathrm{IV}}$$

against

$$\sigma_{\mathrm{IV}}$$

before converting the predicted volatility back into option prices.

9.20 Volatility-Surface Error

Since

$$\sigma_{\mathrm{IV}} = f(m, T),$$

the student shall compare the true and predicted volatility surfaces.
Define

$$\Delta\sigma(m, T) = \sigma_{\mathrm{IV}}(m, T) - \widehat{\sigma}_{\mathrm{IV}}(m, T).$$

The error surface

$$\Delta\sigma(m, T)$$

shall be visualized.
This can reveal systematic failures in specific strike–maturity regions.

9.21 Training Versus Validation Performance

For every trainable model, the student shall compare

$$E_{\mathrm{train}}$$

with

$$E_{\mathrm{validation}}.$$

If

$$E_{\mathrm{train}} \approx E_{\mathrm{validation}}$$

and both are small, the model is likely to generalize reasonably.

If

$$E_{\text{train}} \ll E_{\text{validation}},$$

overfitting may be present.

If both are large, the model may be underfitting.

9.22 Final Test Evaluation

The final test dataset shall be evaluated only after

features,

preprocessing,

model class,

and

hyperparameters

have been frozen.

The correct sequence is

Training
↓
Validation and Model Selection
↓
Freeze Model
↓
Evaluate Once on Test Data.

9.23 Chronological Out-of-Sample Validation

For real market data, the preferred test design is chronological.

If

$$t_1 < t_2 < t_3,$$

then a representative structure is

$$\boxed{\begin{array}{c} t \le t_1 \\ \text{Training} \\ t_1 < t \le t_2 \\ \text{Validation} \\ t_2 < t \le t_3 \\ \text{Testing.} \end{array}}$$

This better represents the actual financial use case in which future observations are unavailable at training time.

9.24 Rolling Validation as an Optional Extension

If the real dataset is sufficiently large, the student may perform a small rolling-window experiment.

For example,

$$\boxed{\begin{array}{lcl} \text{Train}_1 & \to & \text{Test}_1 \\ \text{Train}_2 & \to & \text{Test}_2 \\ \text{Train}_3 & \to & \text{Test}_3. \end{array}}$$

This permits examination of whether model performance remains stable through time.

Rolling validation is optional and shall only be introduced if the core project has already been completed.

9.25 Financial No-Arbitrage Checks

Statistical accuracy is not sufficient.

Predicted option prices shall also satisfy basic financial bounds.

For a European call,

$$\boxed{\max\left(S - Ke^{-rT}, 0\right) \le \widehat{C} \le S.}$$

For a European put,

$$\boxed{\max\left(Ke^{-rT} - S, 0\right) \le \widehat{P} \le Ke^{-rT}.}$$

The number and percentage of predictions violating these bounds shall be reported.

9.26 Monotonicity with Respect to the Underlying

For a European call,

$$\frac{\partial C}{\partial S} \ge 0.$$

For a fitted model, this may be checked numerically using

$$\boxed{\Delta_{\text{num}} = \frac{\widehat{C}(S+h) - \widehat{C}(S-h)}{2h}.}$$

Predictions for which

$$\Delta_{\text{num}} < 0$$

shall be flagged.

9.27 Convexity with Respect to the Underlying

For a standard European option,

$$\Gamma = \frac{\partial^2 C}{\partial S^2} \geq 0.$$

A numerical approximation is

$$\boxed{\Gamma_{\text{num}} = \frac{\widehat{C}(S+h) - 2\widehat{C}(S) + \widehat{C}(S-h)}{h^2}.}$$

The student shall investigate whether the pure ML and hybrid models preserve this convexity property.

9.28 Volatility Sensitivity Check

For standard European options,

$$\nu = \frac{\partial C}{\partial \sigma} > 0.$$

Numerically,

$$\boxed{\nu_{\text{num}} = \frac{\widehat{C}(\sigma+h_\sigma) - \widehat{C}(\sigma-h_\sigma)}{2h_\sigma}.}$$

A negative value over ordinary regions of the option space indicates financially questionable model behaviour.

9.29 Stress-Test Validation

The test dataset shall be subjected to controlled market shocks.

Representative underlying-price shocks may include

$$\delta_S = -10\%, \quad -5\%, \quad +5\%, \quad +10\%.$$

Thus,

$$S^{\text{stress}} = S(1 + \delta_S).$$

Representative volatility shocks may include

$$\Delta\sigma = -0.05, \quad +0.05, \quad +0.10.$$

Thus,

$$\sigma^{\text{stress}} = \sigma + \Delta\sigma.$$

The exact stress levels may be adjusted to the selected market and dataset.

9.30 Model Response Under Stress

For each pricing model M, define the stress response

$$\boxed{\Delta C_M = C_M^{\text{stress}} - C_M^{\text{base}}.}$$

The student shall compare

$$\Delta C_{\text{BS}},$$

$$\Delta C_{\text{ML}},$$

$$\Delta C_{\text{Hybrid}},$$

and

$$\Delta C_{\text{IV}-\text{ML}}.$$

Large or irregular differences between these responses shall be investigated.

9.31 Interpolation and Extrapolation Tests

The student shall distinguish between

interpolation,

where the test observation lies within a region strongly represented in training data, and

extrapolation,

where the observation lies near or outside the training range.

Suppose training data cover

$$m \in [-0.25, 0.25].$$

A controlled extrapolation experiment may test

$$m \in [-0.40, -0.25) \cup (0.25, 0.40].$$

A similar experiment may be performed for volatility or maturity.

Model-Risk Perspective

A model that performs well under interpolation may fail severely under extrapolation. This issue is particularly important in market-risk applications, because stressed conditions may lie outside the region represented densely in historical training data.

9.32 Stability Across Random Seeds

For stochastic ML algorithms, selected training experiments shall be repeated using different random seeds.

Suppose the RMSE values are

$$E_1, E_2, \ldots, E_R.$$

The average performance is

$$\boxed{\overline{E} = \frac{1}{R} \sum_{i=1}^{R} E_i.}$$

The sample standard deviation is

$$\boxed{s_E = \sqrt{\frac{1}{R-1} \sum_{i=1}^{R} \left(E_i - \overline{E}\right)^2}.}$$

A model whose performance varies strongly with random initialization shall be treated cautiously.

9.33 Model Ranking

Each model shall be evaluated according to several dimensions.

A final comparison table should contain

Test MAE
Test RMSE
R^2
Bias
No-Arbitrage Violations
Monotonicity Violations
Stress Stability
Extrapolation Behaviour
Interpretability.

The final ranking shall therefore not be based on a single scalar metric.

9.34 Acceptance Criterion for ML Improvement

The project shall claim that machine learning improves upon the Black–Scholes benchmark only if the selected ML or hybrid model satisfies all of the following broad conditions:

1. The model reduces out-of-sample pricing error relative to Black–Scholes.
2. The improvement is observed on the final unseen test data, not only on training data.
3. The improvement is not restricted to a very small subset of options.
4. Systematic Black–Scholes residual structure is reduced.
5. Basic no-arbitrage relationships are not materially violated.
6. Price sensitivities remain financially reasonable.
7. The model does not exhibit severe instability under moderate stress.

Key Idea

The central question is not

"Does ML reduce RMSE?"

but rather

Does ML reduce out-of-sample pricing error
while preserving financially credible behaviour?

9.35 Required Validation Figures

The final report shall contain at least the following validation figures:

1. Market price versus Black–Scholes price.
2. Market price versus pure ML prediction.
3. Market price versus hybrid residual prediction.

4. True versus predicted implied volatility.
5. Black–Scholes residual versus moneyness.
6. Hybrid residual versus moneyness.
7. Black–Scholes residual versus maturity.
8. Hybrid residual versus maturity.
9. RMSE by moneyness category.
10. RMSE by maturity category.
11. Error heat map in the (m, T) plane.
12. Training and validation loss for the ANN.
13. Model response under selected underlying-price shocks.
14. Model response under selected volatility shocks.

9.36 Validation Gate

The validation stage shall be considered complete only when the student has demonstrated that

1. all models are evaluated on the same final test set;
2. MAE, RMSE, mean error and R^2 have been reported;
3. normalized or spread-adjusted errors have been examined where appropriate;
4. errors are analysed by moneyness and maturity;
5. residual structures have been inspected;
6. implied-volatility prediction has been evaluated separately;
7. financial no-arbitrage conditions have been checked;
8. numerical delta, gamma and volatility sensitivity have been inspected;
9. stress tests have been performed;
10. limited extrapolation experiments have been conducted; and
11. any claim of improvement over Black–Scholes is supported by out-of-sample evidence.

Project Milestone

At the end of the validation stage, the student shall be able to produce a defensible model comparison of the form

Classical Accuracy
↓
ML Accuracy
↓
Hybrid Improvement
↓
Residual Diagnostics
↓
Financial Consistency
↓
Stress and Extrapolation Behaviour
↓
Final Model Selection.

Only after this validation shall the project proceed to the final market-risk analysis.

10. Market-Risk Analysis

The final technical stage of the project extends the pricing framework to market-risk analysis.

The objective is to determine not only whether a model predicts option prices accurately, but also whether it produces financially reasonable changes in portfolio value when the underlying market variables are perturbed.

The risk-analysis workflow is

Option Portfolio
↓
Portfolio Greeks
↓
Delta–Gamma–Vega Approximation
↓
Full Repricing
↓
Stress Scenarios
↓
Classical versus Hybrid Risk Comparison.

10.1 Construction of a Simple Option Portfolio

Consider a portfolio containing N European options.

Let

$$n_i$$

denote the position in option i, and let

$$C_i$$

denote the corresponding option price.

The total portfolio value is

$$\boxed{V_P = \sum_{i=1}^{N} n_i C_i.}$$

The position

$$n_i > 0$$

represents a long position, while

$$n_i < 0$$

represents a short position.

The portfolio should remain deliberately small so that each source of risk can be interpreted individually.

A representative portfolio may contain options with different

$$K, \qquad T, \qquad \text{and moneyness.}$$

10.2 Portfolio Delta

For option i, define

$$\Delta_i = \frac{\partial C_i}{\partial S}.$$

The portfolio delta is

$$\boxed{\Delta_P = \sum_{i=1}^{N} n_i \Delta_i.}$$

For a small change in the underlying asset price,

$$\Delta S,$$

the first-order approximation to the portfolio value change is

$$\boxed{\Delta V_P \approx \Delta_P \Delta S.}$$

This is the delta approximation.

10.3 Interpretation of Portfolio Delta

If

$$\Delta_P > 0,$$

the portfolio value tends to increase when the underlying asset price increases.

If

$$\Delta_P < 0,$$

the portfolio value tends to decrease when the underlying price increases.

A portfolio satisfying

$$\Delta_P \approx 0$$

is approximately delta neutral for sufficiently small changes in the underlying price.

10.4 Portfolio Gamma

The gamma of option i is

$$\Gamma_i = \frac{\partial^2 C_i}{\partial S^2}.$$

The portfolio gamma is

$$\boxed{\Gamma_P = \sum_{i=1}^{N} n_i \Gamma_i.}$$

Gamma measures the curvature of the portfolio value with respect to the underlying price.

Including the second-order term gives

$$\boxed{\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2.}$$

This is the delta–gamma approximation.

10.5 Portfolio Vega

The vega of option i is

$$\nu_i = \frac{\partial C_i}{\partial \sigma}.$$

The portfolio vega is

$$\boxed{\nu_P = \sum_{i=1}^{N} n_i \nu_i.}$$

For a small volatility change

$$\Delta\sigma,$$

the first-order volatility contribution to portfolio P&L is

$$\nu_P \Delta\sigma.$$

10.6 Delta–Gamma–Vega Approximation

Combining the principal underlying-price and volatility sensitivities gives

$$\boxed{\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.}$$

This approximation provides a compact local representation of nonlinear option-portfolio risk.

The three components are

$$\Delta_P \Delta S = \text{first-order underlying-price effect},$$

$$\frac{1}{2}\Gamma_P (\Delta S)^2 = \text{curvature effect},$$

and

$$\nu_P \Delta\sigma = \text{volatility effect}.$$

Key Idea

The delta–gamma–vega approximation provides the connection between the mathematical derivatives of the option-pricing model and the financial P&L of the portfolio.
It therefore converts the Greeks from abstract sensitivities into direct risk measures.

10.7 Full Repricing

The Taylor approximation is valid locally.

For larger market movements, the entire portfolio should be repriced.

Let

$$S^{\text{stress}} = S + \Delta S,$$

and

$$\sigma^{\text{stress}} = \sigma + \Delta\sigma.$$

The stressed portfolio value is

$$V_P^{\text{stress}} = \sum_{i=1}^{N} n_i C_i \left(S^{\text{stress}}, \sigma^{\text{stress}}\right).$$

The exact repricing P&L is

$$\boxed{\Delta V_P^{\text{full}} = V_P^{\text{stress}} - V_P^{\text{base}}.}$$

10.8 Taylor Approximation Error

Define

$$\Delta V_P^{\text{Taylor}} = \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.$$

The approximation error is

$$\boxed{E_{\text{Taylor}} = \Delta V_P^{\text{full}} - \Delta V_P^{\text{Taylor}}.}$$

The absolute approximation error is

$$\left|E_{\text{Taylor}}\right|.$$

A relative approximation error may be defined as

$$\boxed{E_{\text{Taylor}}^{\text{rel}} = \frac{\left|\Delta V_P^{\text{full}} - \Delta V_P^{\text{Taylor}}\right|}{\left|\Delta V_P^{\text{full}}\right| + \epsilon}.}$$

10.9 Local Versus Nonlocal Risk

For very small market movements, one expects

$$\Delta V_P^{\text{Taylor}} \approx \Delta V_P^{\text{full}}.$$

As the stress magnitude increases,

$$|\Delta S| \uparrow$$

or

$$|\Delta\sigma| \uparrow,$$

higher-order terms become increasingly important.

Thus,

$$E_{\text{Taylor}}$$

may increase.

This provides a direct computational demonstration of the local nature of Taylor-based risk approximations.

10.10 Underlying-Price Stress Scenarios

The first set of deterministic stress scenarios shall perturb the underlying asset price.

Representative shocks may be

$$\delta_S = -10\%, \quad -5\%, \quad -2\%, \quad +2\%, \quad +5\%, \quad +10\%.$$

The stressed underlying price is

$$\boxed{S^{\text{stress}} = S(1 + \delta_S).}$$

For each scenario, the student shall calculate

$$\Delta V_P^{\text{Delta}},$$

$$\Delta V_P^{\text{Delta-Gamma}},$$

and

$$\Delta V_P^{\text{Full}}.$$

10.11 Volatility Stress Scenarios

The second set of stress scenarios shall perturb volatility.

Representative shocks may include

$$\Delta\sigma = -0.10, \quad -0.05, \quad +0.05, \quad +0.10.$$

Thus,

$$\boxed{\sigma^{\text{stress}} = \sigma + \Delta\sigma.}$$

The student shall calculate the corresponding portfolio P&L using

$$\nu_P \Delta\sigma$$

and full option repricing.

10.12 Combined Price–Volatility Stress

An option portfolio may experience simultaneous movements in the underlying price and volatility.
Therefore, selected combined scenarios shall also be considered.
For example,

$$\delta_S = -10\%, \qquad \Delta\sigma = +0.10,$$

or

$$\delta_S = +10\%, \qquad \Delta\sigma = -0.05.$$

The corresponding approximate P&L is

$$\boxed{\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.}$$

The result shall again be compared with full repricing.

10.13 Stress Matrix

The combined stress experiment may be represented using a matrix.
Let

$$\delta_S^{(1)}, \delta_S^{(2)}, \ldots, \delta_S^{(m)}$$

represent underlying-price shocks, and let

$$\Delta\sigma^{(1)}, \Delta\sigma^{(2)}, \ldots, \Delta\sigma^{(k)}$$

represent volatility shocks.
Then define

$$\boxed{\mathcal{L}_{jk} = \Delta V_P \left(\delta_S^{(j)}, \Delta\sigma^{(k)}\right).}$$

The resulting stress matrix can be visualized as a heat map.
This provides a compact view of the portfolio P&L surface.

10.14 Black–Scholes Portfolio Risk

Using the classical pricing model,

$$C_i = C_{\mathrm{BS},i},$$

the portfolio value is

$$\boxed{V_P^{\mathrm{BS}} = \sum_{i=1}^{N} n_i C_{\mathrm{BS},i}.}$$

The corresponding portfolio sensitivities are

$$\Delta_P^{\mathrm{BS}},$$

$$\Gamma_P^{\mathrm{BS}},$$

and

$$\nu_P^{\mathrm{BS}}.$$

These quantities form the classical risk benchmark.

10.15 Hybrid Portfolio Risk

For the residual-learning model,

$$C_i^{\mathrm{Hybrid}} = C_{\mathrm{BS},i} + \widehat{\varepsilon}_{\mathrm{ML},i}.$$

Therefore,

$$\boxed{V_P^{\mathrm{Hybrid}} = \sum_{i=1}^{N} n_i C_i^{\mathrm{Hybrid}}.}$$

The hybrid portfolio delta may be calculated numerically as

$$\boxed{\Delta_P^{\mathrm{Hybrid}} \approx \frac{V_P^{\mathrm{Hybrid}}(S+h) - V_P^{\mathrm{Hybrid}}(S-h)}{2h}.}$$

The hybrid gamma may similarly be estimated by

$$\boxed{\Gamma_P^{\mathrm{Hybrid}} \approx \frac{V_P^{\mathrm{Hybrid}}(S+h) - 2V_P^{\mathrm{Hybrid}}(S) + V_P^{\mathrm{Hybrid}}(S-h)}{h^2}.}$$

10.16 Comparison of Classical and Hybrid Sensitivities

Define

$$\Delta_{\mathrm{diff}} = \Delta_P^{\mathrm{Hybrid}} - \Delta_P^{\mathrm{BS}},$$

and

$$\Gamma_{\mathrm{diff}} = \Gamma_P^{\mathrm{Hybrid}} - \Gamma_P^{\mathrm{BS}}.$$

These quantities measure the change in estimated risk exposure produced by the ML correction. Likewise,

$$\nu_{\text{diff}} = \nu_P^{\text{Hybrid}} - \nu_P^{\text{BS}}$$

may be estimated if the hybrid model has a volatility-dependent correction.

10.17 Risk Difference Between Models

For a given stress scenario, let

$$\Delta V_P^{\text{BS}}$$

denote the Black–Scholes portfolio P&L and let

$$\Delta V_P^{\text{Hybrid}}$$

denote the hybrid portfolio P&L.

Define the model-risk difference

$$\boxed{R_{\text{model}} = \Delta V_P^{\text{Hybrid}} - \Delta V_P^{\text{BS}}.}$$

A large value of

$$|R_{\text{model}}|$$

indicates that the model choice materially changes the estimated portfolio response under stress.

Model-Risk Perspective

Model risk is not only the possibility that one model produces a different price from another.
For a risk manager, the more important question may be whether different models imply materially different portfolio losses under the same market shock.

10.18 Risk Contribution of Individual Positions

For option i, the first-order delta contribution is

$$RC_{\Delta,i} = n_i \Delta_i \Delta S.$$

The gamma contribution is

$$RC_{\Gamma,i} = \frac{1}{2} n_i \Gamma_i (\Delta S)^2.$$

The vega contribution is

$$RC_{\nu,i} = n_i \nu_i \Delta\sigma.$$

Thus,

$$\boxed{RC_i = RC_{\Delta,i} + RC_{\Gamma,i} + RC_{\nu,i}.}$$

The student shall identify which portfolio positions contribute most strongly to the total stressed P&L.

10.19 Risk Decomposition

The total Taylor-based P&L may be decomposed as

$$\Delta V_P = P\&L_\Delta + P\&L_\Gamma + P\&L_\nu,$$

where

$$P\&L_\Delta = \Delta_P \Delta S,$$

$$P\&L_\Gamma = \frac{1}{2}\Gamma_P (\Delta S)^2,$$

and

$$P\&L_\nu = \nu_P \Delta\sigma.$$

The student shall visualize these components separately for selected stress scenarios.

10.20 Delta-Hedged Portfolio Experiment

A particularly useful experiment is to construct an approximately delta-neutral portfolio.

Suppose a portfolio initially has delta

$$\Delta_P.$$

A position in the underlying asset can be added so that

$$\Delta_P^{\text{hedged}} \approx 0.$$

For such a portfolio, the first-order price sensitivity is largely removed.

The remaining local P&L is then dominated by

$$\boxed{\Delta V_P \approx \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.}$$

This experiment demonstrates directly why nonlinear option risk cannot be represented by delta alone.

10.21 Comparison Before and After Delta Hedging

The student shall compare

$$\Delta V_P^{\text{unhedged}}$$

with

$$\Delta V_P^{\text{hedged}}$$

under the same underlying-price stress.

This comparison should illustrate the reduction in first-order directional exposure.

However, the hedged portfolio may still retain substantial

$$\Gamma$$

and

$$\nu$$

risk.

10.22 Optional Monte Carlo Portfolio-Risk Experiment

If time permits, the deterministic stress analysis may be supplemented by a small Monte Carlo risk experiment.

Under risk-neutral simulation,

$$S_{t+\Delta t} = S_t \exp\left[\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right].$$

For each simulated market state,

$$S^{(j)},$$

the portfolio may be repriced,

$$V_P^{(j)}.$$

The simulated P&L is

$$\Delta V_P^{(j)} = V_P^{(j)} - V_P^{(0)}.$$

This produces an empirical portfolio P&L distribution.

10.23 Empirical P&L Distribution

From

$$M$$

simulated P&L observations,

$$\Delta V_P^{(1)}, \ldots, \Delta V_P^{(M)},$$

the student may calculate

$$\overline{\Delta V}_P$$

and

$$s_{\Delta V}.$$

A histogram of simulated portfolio P&L shall also be generated.

This experiment is optional because a full Value-at-Risk study would extend beyond the primary scope of the three-month project.

10.24 Why Full VaR is Not the Core Objective

Although market-risk analysis often includes Value-at-Risk, the core objective of this project is

$$\boxed{\text{pricing-model risk and sensitivity analysis,}}$$

rather than the development of a complete regulatory VaR system.

Therefore, the project shall prioritize

$$\Delta, \qquad \Gamma, \qquad \nu,$$

Taylor approximation,

full repricing,

and deterministic stress testing.

A simple simulated P&L distribution may be added only if the main project is completed successfully.

10.25 Stress Behaviour of the ML Model

Machine-learning models may behave unpredictably when the stressed inputs move away from the training distribution.

Therefore, under every stress scenario the student shall check whether

$$\widehat{C}_{\mathrm{ML}}$$

and

$$\widehat{C}_{\text{Hybrid}}$$

remain financially meaningful.

The following properties shall be monitored:

$$\widehat{C} \geq 0,$$

$$\widehat{C} \leq S,$$

$$\frac{\partial \widehat{C}}{\partial S} \geq 0,$$

and, where appropriate,

$$\frac{\partial^2 \widehat{C}}{\partial S^2} \geq 0.$$

10.26 Stress Distance from the Training Domain

Suppose the training dataset covers

$$m \in [m_{\min}, m_{\max}],$$

and

$$\sigma \in [\sigma_{\min}, \sigma_{\max}].$$

A stress scenario may move the option to

$$m^{\text{stress}}$$

or

$$\sigma^{\text{stress}}$$

outside these intervals.

Such scenarios shall be explicitly identified as extrapolation.

The student shall distinguish

$$\boxed{\text{Stress within training support}}$$

from

$$\boxed{\text{Stress outside training support.}}$$

This distinction is essential when interpreting ML-based risk estimates.

10.27 Model-Risk Map

A useful final diagnostic is to construct a model-risk surface.

Define

$$R_{\text{model}}(\delta_S, \Delta\sigma) = \Delta V_P^{\text{Hybrid}} - \Delta V_P^{\text{BS}}.$$

The student may visualize

$$R_{\text{model}}$$

over a grid of underlying-price and volatility shocks.

Large absolute values identify regions in which classical and hybrid risk estimates differ substantially.

10.28 Required Market-Risk Figures

The final report shall contain at least the following risk-analysis figures:

1. Portfolio value versus underlying asset price.
2. Portfolio delta versus underlying asset price.
3. Portfolio gamma versus underlying asset price.
4. Portfolio vega versus underlying asset price.
5. Delta approximation versus full repricing.
6. Delta–gamma approximation versus full repricing.
7. Delta–gamma–vega approximation versus full repricing.
8. Portfolio P&L under underlying-price stresses.
9. Portfolio P&L under volatility stresses.
10. Combined price–volatility stress heat map.
11. Black–Scholes versus hybrid stressed P&L.
12. Model-risk difference,

    $$R_{\text{model}},$$

    under selected scenarios.

10.29 Market-Risk Comparison Table

For each stress scenario, the student shall prepare a comparison table containing

Stress Scenario
ΔS
$\Delta\sigma$
ΔV_{Δ}
$\Delta V_{\Delta\Gamma}$
$\Delta V_{\Delta\Gamma\nu}$
$\Delta V_{\mathrm{BS,Full}}$
$\Delta V_{\mathrm{Hybrid,Full}}$
R_{model}.

This table will provide the principal quantitative summary of the market-risk component.

10.30 Central Risk Question

The final risk-analysis question is not simply

"Which model gives the best option price?"

but

Does the choice of pricing model materially change the estimated portfolio exposure under market stress?

This question connects machine learning directly with model risk.

10.31 Market-Risk Analysis Gate

The market-risk stage shall be considered complete only when the student can demonstrate that

1. a small option portfolio has been constructed;
2. portfolio delta, gamma and vega have been calculated;
3. delta, delta–gamma and delta–gamma–vega approximations have been implemented;
4. Taylor approximations have been compared with full repricing;
5. underlying-price stress scenarios have been performed;
6. volatility stress scenarios have been performed;
7. at least one combined price–volatility stress has been studied;
8. Black–Scholes and hybrid portfolio P&L have been compared;
9. the difference in risk estimates between models has been quantified;
10. ML predictions have been checked for financially unreasonable behaviour under stress; and
11. stress scenarios outside the training domain have been identified explicitly as extrapolation.

Project Milestone

At the completion of this stage, the student shall be able to connect

Option Pricing
↓
Greeks
↓
Portfolio Risk
↓
Taylor Approximation
↓
Full Repricing
↓
Stress Testing
↓
Black–Scholes versus Hybrid Model Risk.

This completes the main technical framework of the project.

11. Three-Month Work Plan

The project duration is twelve weeks.

The work plan is deliberately structured so that the student first develops the mathematical and numerical foundation, then constructs the machine-learning models, and finally performs hybrid-model validation and market-risk analysis.

The overall progression is

Weeks 1–4
Mathematical and Black–Scholes Foundation
↓
Weeks 5–6
Numerical Validation and Dataset Construction
↓
Weeks 7–9
Machine Learning and Hybrid Modelling
↓
Weeks 10–11
Validation and Market-Risk Analysis
↓
Week 12
Final Integration, Report and Presentation.

11.1 Week 1: Financial Returns, Probability and Data Handling

Reading focus

The student shall study the sections dealing with

simple returns, log returns, normal and lognormal distributions,

and the basic statistical description of financial data.

The student should understand the distinction between

$$R_t = \frac{S_t - S_{t-1}}{S_{t-1}}$$

and

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right).$$

Mathematical objective

Understand why log returns are useful and verify numerically that

$$r_t \approx R_t$$

for sufficiently small returns.

Computational task

Using Python,

1. import a sample asset-price time series;
2. compute simple returns;
3. compute log returns;
4. calculate mean, variance, skewness and kurtosis;
5. plot the price series;
6. plot the return series; and
7. plot a return histogram.

Mandatory output

`week01_returns_analysis.ipynb`

containing all calculations and plots.

Project Milestone

The student must be able to explain why asset prices and asset returns are different modelling objects and why log returns are preferred in many continuous-time financial models.

11.2 Week 2: Brownian Motion and Geometric Brownian Motion

Reading focus

Study the stochastic-process material required to understand

$$dW_t$$

and geometric Brownian motion,

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.$$

Mathematical objective

Understand

$$\Delta W = \sqrt{\Delta t}\, Z, \qquad Z \sim N(0,1),$$

and the GBM solution

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, Z\right].$$

Computational task

The student shall

1. generate standard normal random variables;
2. simulate Brownian-motion increments;
3. simulate complete Brownian-motion paths;
4. simulate GBM asset-price paths; and
5. investigate the effect of changing μ and σ.

Mandatory plots

$$W_t \text{ versus } t,$$

and multiple

$$S_t \text{ versus } t$$

GBM paths.

Mandatory output

`week02_gbm_simulation.ipynb`

Project Milestone

The student must be able to generate a GBM path independently and explain the separate roles of drift and volatility.

11.3 Week 3: Black–Scholes Mathematics and Analytical Pricing

Reading focus

Review the option-pricing framework, option sensitivities and the mathematical assumptions connecting GBM with the Black–Scholes–Merton model.

Mathematical objective

Understand the logical sequence

GBM → Itô's Lemma → Delta Hedging → No Arbitrage → Black–Scholes PDE.

The student shall understand the PDE

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0.$$

Computational task

Implement the European call and put formulas,

$$C_{\mathrm{BS}} = SN(d_1) - Ke^{-rT}N(d_2),$$

and

$$P_{\mathrm{BS}} = Ke^{-rT}N(-d_2) - SN(-d_1).$$

Validation task

Verify put–call parity,

$$C - P = S - Ke^{-rT}.$$

Mandatory output

`black_scholes.py`

with reusable functions for call and put pricing.

Project Milestone

The student must demonstrate correct pricing over multiple combinations of

$$S, \quad K, \quad T, \quad r, \quad \sigma.$$

11.4 Week 4: Greeks and Implied Volatility

Reading focus

Study option sensitivities, numerical differentiation and iterative root-finding methods.

Mathematical objective

Understand

$$\Delta = \frac{\partial C}{\partial S},$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2},$$

and

$$\nu = \frac{\partial C}{\partial \sigma}.$$

Computational task

Implement

1. analytical delta;
2. analytical gamma;
3. analytical vega;
4. finite-difference delta;
5. finite-difference gamma;
6. finite-difference vega;
7. Newton–Raphson implied volatility; and
8. bisection fallback for implied volatility.

Validation task

Generate a price using a known volatility

$$\sigma_{\text{true}},$$

then recover

$$\sigma_{\mathrm{IV}}$$

and verify

$$\sigma_{\mathrm{IV}} \approx \sigma_{\mathrm{true}}.$$

Mandatory output

```
greeks.py
implied_volatility.py
```

Project Milestone

By the end of Week 4, the student must possess a complete analytical Black–Scholes engine returning

$$C, \quad P, \quad \Delta, \quad \Gamma, \quad \nu, \quad \sigma_{\mathrm{IV}}.$$

11.5 Week 5: Binomial and Monte Carlo Validation

Reading focus

Study the numerical-method sections on

binomial lattices

and

Monte Carlo simulation.

Computational task A: Binomial pricing

Implement

$$u = e^{\sigma\sqrt{\Delta t}},$$

$$d = e^{-\sigma\sqrt{\Delta t}},$$

and

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Use backward induction to calculate the European option price.

Computational task B: Monte Carlo pricing

Generate

$$S_T^{(j)} = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z_j\right]$$

and calculate

$$C_{\mathrm{MC}} = e^{-rT}\frac{1}{M}\sum_{j=1}^{M}\max\left(S_T^{(j)} - K, 0\right).$$

Required convergence studies

The student shall verify

$$C_{\mathrm{Binomial}} \longrightarrow C_{\mathrm{BS}}$$

as the number of lattice steps increases, and

$$C_{\mathrm{MC}} \longrightarrow C_{\mathrm{BS}}$$

as the number of simulations increases.

Mandatory output

```
binomial.py
monte_carlo.py
week05_validation.ipynb
```

Project Milestone

The student shall demonstrate

$$\boxed{C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}.}$$

No ML work shall begin before this comparison is successful.

11.6 Week 6: Synthetic Dataset and Feature Engineering

Mathematical objective

Understand the transformation from financial variables to machine-learning features.

The primary derived feature is

$$m = \ln\left(\frac{S}{K}\right).$$

Computational task

Generate a synthetic option dataset by sampling

$$S, \quad K, \quad T, \quad r, \quad \sigma.$$

For every observation, calculate

$$C_{\mathrm{BS}}, \quad \Delta, \quad \Gamma, \quad \nu.$$

Construct the feature set

$$\mathbf{x} = [m, T, r, \sigma] .$$

An extended feature set may contain

$$\mathbf{x} = [m, T, r, \sigma, \Delta, \Gamma, \nu] .$$

Data task

Split the dataset into

70% training,

15% validation,

and

15% testing.

Mandatory output

`synthetic_option_dataset.csv`

and

`week06_data_generation.ipynb`.

Project Milestone

The student must demonstrate complete traceability

$$\text{Input Parameters} \rightarrow \text{BS Quantities} \rightarrow \text{Features} \rightarrow \text{Target}.$$

11.7 Week 7: Regression Baselines

Reading focus

Study simple and multivariate regression and the interpretation of regression residuals.

Computational task

Implement

OLS

and

Ridge Regression.

Initially train

$$[m, T, r, \sigma] \longrightarrow C_{\mathrm{BS}}.$$

The student shall calculate

MAE,

RMSE,

and

$$R^2.$$

Required analysis

Compare

OLS

with

Ridge

and inspect residuals against

$$m$$

and

$$T.$$

Mandatory output

`week07_regression.ipynb`

Project Milestone

The student must understand why a linear model cannot generally reproduce the full nonlinear Black–Scholes structure.

11.8 Week 8: Tree-Based Model and Neural Network

Computational task A

Train one tree-based ensemble,

$$\text{Random Forest}$$

or

$$\text{Gradient Boosting.}$$

Computational task B

Train a small neural network such as

$$\boxed{p \to 32 \to 16 \to 1.}$$

Required comparison

Compare

$$\text{OLS}, \qquad \text{Ridge}, \qquad \text{Tree}, \qquad \text{ANN}.$$

Evaluate

$$\text{MAE}, \qquad \text{RMSE}, \qquad R^2.$$

Mandatory plots

1. predicted versus true option price;
2. residual versus moneyness;
3. residual versus maturity;
4. feature importance for the tree model; and
5. training and validation loss for the ANN.

Mandatory output

$$\boxed{\texttt{week08_ml_models.ipynb}}$$

Project Milestone

The student shall identify whether additional nonlinear model complexity produces a meaningful improvement over regression.

11.9 Week 9: Real Market Data and Hybrid Models

Data objective

Construct a cleaned real option dataset containing, where available,

$$S, \quad K, \quad T, \quad r, \quad C_{\text{market}}.$$

Compute

$$m = \ln\left(\frac{S}{K}\right),$$

$$\sigma_{\text{hist}},$$

$$C_{\text{BS}},$$

$$\varepsilon_{\text{BS}} = C_{\text{market}} - C_{\text{BS}},$$

and

$$\sigma_{\text{IV}}.$$

Hybrid Model A

Train

$$\boxed{\mathbf{x} \longrightarrow \varepsilon_{\text{BS}}.}$$

Construct

$$\widehat{C}_{\text{Hybrid}} = C_{\text{BS}} + \widehat{\varepsilon}_{\text{ML}}.$$

Hybrid Model B

Train

$$\boxed{\mathbf{x} \longrightarrow \sigma_{\text{IV}}.}$$

Then calculate

$$\widehat{C}_{\text{IV}-\text{ML}} = C_{\text{BS}}\left(\widehat{\sigma}_{\text{IV}}\right).$$

Mandatory output

`week09_hybrid_models.ipynb`

Project Milestone

The student must clearly distinguish

$$\text{Residual Learning}$$

from

$$\text{Implied-Volatility Learning.}$$

The same option's implied volatility shall not be used to define the residual-learning benchmark.

11.10 Week 10: Out-of-Sample Model Validation

Validation objective

Evaluate

$$\text{Black–Scholes},$$

$$\text{Pure ML},$$

$$\text{Residual Hybrid},$$

and

$$\text{IV-ML Hybrid}$$

on the same unseen test dataset.

Required performance metrics

Calculate

$$\text{ME}, \quad \text{MAE}, \quad \text{RMSE}, \quad R^2.$$

Evaluate errors separately for

$$\text{ITM}, \quad \text{ATM}, \quad \text{OTM},$$

and for

$$\text{short}, \quad \text{medium}, \quad \text{long maturity}.$$

Required financial checks

Verify

$$\widehat{C} \geq 0,$$

$$\widehat{C} \leq S,$$

and numerical monotonicity and convexity where appropriate.

Mandatory output

`week10_model_validation.ipynb`

Project Milestone

A claim that ML improves Black–Scholes may be made only if the improvement occurs on unseen out-of-sample data.

11.11 Week 11: Portfolio Risk and Stress Testing

Portfolio construction

Construct a small portfolio,

$$V_P = \sum_{i=1}^{N} n_i C_i.$$

Calculate

$$\Delta_P, \qquad \Gamma_P, \qquad \nu_P.$$

Taylor risk approximation

Evaluate

$$\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.$$

Full repricing

Compare Taylor approximation with

$$\Delta V_P^{\text{full}} = V_P^{\text{stress}} - V_P^{\text{base}}.$$

Stress scenarios

Perform

$$S \longrightarrow S(1 + \delta_S)$$

and

$$\sigma \longrightarrow \sigma + \Delta\sigma.$$

Compare

$$\Delta V_P^{\mathrm{BS}}$$

with

$$\Delta V_P^{\mathrm{Hybrid}}.$$

Mandatory output

`week11_market_risk.ipynb`

Project Milestone

The student must demonstrate whether the choice of pricing model materially changes the estimated portfolio P&L under selected stress scenarios.

11.12 Week 12: Final Integration and Project Completion

The final week is reserved for integration, verification and documentation.

The student shall not introduce a new major algorithm during this week.

Code integration

The student shall organize the final code into reusable modules,

`black_scholes.py`
`greeks.py`
`implied_volatility.py`
`binomial.py`
`monte_carlo.py`
`data_generation.py`
`ml_models.py`
`risk_analysis.py`.

Final model comparison

Prepare the consolidated comparison

Black–Scholes
Pure ML
BS + Residual ML
ML Implied Volatility + BS.

Final report

The report shall contain

1. mathematical foundation;
2. computational methodology;

3. dataset description;
4. ML algorithms;
5. hybrid architecture;
6. validation results;
7. market-risk analysis;
8. limitations;
9. conclusions; and
10. possible future extensions.

Final presentation

The student shall prepare a concise technical presentation explaining

Problem → Mathematics → Computation → ML → Hybrid Model → Risk Analysis → Conclusion.

11.13 Monthly Stage Gates

For supervision purposes, the twelve-week plan shall contain three major stage gates.

11.13.1 Stage Gate 1: End of Week 4

The student must demonstrate

GBM Understanding
Black–Scholes Pricing
Greeks
Implied Volatility.

If these components are incomplete, the project shall not proceed to machine learning.

11.13.2 Stage Gate 2: End of Week 8

The student must demonstrate

BS / Binomial / Monte Carlo Validation
Synthetic Dataset
OLS and Ridge
Tree-Based Model
Small ANN.

11.13.3 Stage Gate 3: End of Week 12

The student must demonstrate

Real-Data Processing Residual Hybrid Model Implied-Volatility Hybrid Model Out-of-Sample Comparison Portfolio Stress Analysis Final Report and Code.

11.14 Weekly Supervisor Review

A short review shall be conducted at the end of every week.

The student shall present

1. the concepts studied during the week;
2. the code developed;
3. the numerical results obtained;
4. at least one relevant plot;
5. any discrepancy or failure encountered; and
6. the planned work for the following week.

The supervisor review should focus on whether the student understands the mathematical and financial meaning of the calculations rather than whether the code merely executes successfully.

11.15 Work-Plan Discipline

The project shall follow the sequence

Understand ↓ Derive ↓ Implement ↓ Validate ↓ Interpret.

The student shall not proceed directly from

formula

to

software output

without numerical verification and financial interpretation.

Project Milestone

The twelve-week project is considered successful only when the student can explain the complete chain

Financial Returns
↓
GBM
↓
Black–Scholes
↓
Greeks and Implied Volatility
↓
Binomial and Monte Carlo Validation
↓
Machine Learning
↓
Hybrid Pricing
↓
Out-of-Sample Validation
↓
Portfolio Market-Risk Analysis.

This sequence defines the complete three-month learning and research trajectory of the project.

12. Expected Deliverables

The project shall produce a complete and reproducible computational package rather than a collection of disconnected notebooks or isolated plots.

The expected deliverables are grouped into

Mathematical Deliverables
Computational Deliverables
Dataset Deliverables
Machine-Learning Deliverables
Validation Deliverables
Market-Risk Deliverables
Documentation and Presentation.

12.1 Deliverable 1: Mathematical Foundation Note

The student shall prepare a concise mathematical note explaining the principal equations used in the project.

The note shall include

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t,$$

the Black–Scholes PDE,

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0,$$

the European call and put formulas,

$$C_{\mathrm{BS}}, \qquad P_{\mathrm{BS}},$$

the principal Greeks,

$$\Delta, \qquad \Gamma, \qquad \nu,$$

and the implied-volatility problem,

$$C_{\mathrm{BS}}\left(S, K, T, r, \sigma_{\mathrm{IV}}\right) = C_{\mathrm{market}}.$$

The purpose is to demonstrate that the student understands the mathematical model implemented in the code.

12.2 Deliverable 2: Core Python Pricing Library

The student shall submit a modular Python implementation containing at least the following files:

```
black_scholes.py
greeks.py
implied_volatility.py
binomial.py
monte_carlo.py
data_generation.py
ml_models.py
risk_analysis.py.
```

Each module shall contain reusable functions rather than code that depends entirely on notebook execution order.

12.3 Deliverable 3: Black–Scholes Pricing Engine

The pricing engine shall accept

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma$$

and return

$$\boxed{C, \qquad P, \qquad \Delta, \qquad \Gamma, \qquad \nu.}$$

The implementation shall satisfy numerical consistency checks such as

$$C - P = S - Ke^{-rT}.$$

The code shall also support vectorized evaluation over multiple option observations where practical.

12.4 Deliverable 4: Implied-Volatility Solver

The student shall submit a robust numerical solver for

$$F(\sigma) = C_{\mathrm{BS}}(\sigma) - C_{\mathrm{market}} = 0.$$

The implementation shall include

$$\boxed{\text{Newton–Raphson} + \text{Bisection Fallback.}}$$

The solver shall report convergence failure rather than silently returning an invalid volatility estimate.

12.5 Deliverable 5: Numerical Pricing Validation

The student shall submit independent implementations of

Binomial Pricing

and

Monte Carlo Pricing.

The final validation shall demonstrate

$$\boxed{C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}.}$$

The deliverable shall include convergence studies for both numerical methods.

12.6 Deliverable 6: Synthetic Option Dataset

A reproducible synthetic dataset shall be generated from the validated pricing engine.

The dataset shall contain, at minimum,

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma,$$

together with

$$C_{\mathrm{BS}}, \qquad P_{\mathrm{BS}}, \qquad \Delta, \qquad \Gamma, \qquad \nu.$$

Derived variables shall include

$$m = \ln\left(\frac{S}{K}\right).$$

The dataset shall be stored in a portable format such as

`synthetic_option_dataset.csv`.

12.7 Deliverable 7: Cleaned Real-Market Dataset

Where real option data are available, the student shall produce a cleaned master dataset.

The processed dataset should contain, where applicable,

Date
S
K
T
r
C_{market}
m
σ_{hist}
C_{BS}
$\varepsilon_{\mathrm{BS}}$
σ_{IV}
Δ
Γ
ν.

The data-cleaning rules shall be documented explicitly.

12.8 Deliverable 8: Machine-Learning Models

The student shall implement and compare

OLS Regression
Ridge Regression
One Tree-Based Ensemble
One Small Neural Network.

The final code shall permit these algorithms to be applied consistently to

Direct Pricing,

Residual Learning,

and

Implied-Volatility Learning.

12.9 Deliverable 9: Pure ML Pricing Model

The student shall construct

$$\boxed{\widehat{C}_{\mathrm{ML}} = f_\theta(\mathbf{x}).}$$

The purpose of this model is to provide a direct data-driven benchmark against which the hybrid methods can be compared.

12.10 Deliverable 10: Residual Hybrid Model

The primary hybrid model shall be

$$\boxed{\widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}.}$$

The student shall submit

$$\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}},$$

the trained residual model,

$$\widehat{\varepsilon}_{\mathrm{ML}} = f_\theta(\mathbf{x}),$$

and the final corrected pricing output.

12.11 Deliverable 11: Implied-Volatility Hybrid Model

The student shall also construct

$$\boxed{\widehat{\sigma}_{\mathrm{IV}} = g_\theta(\mathbf{x}).}$$

The predicted volatility shall be passed through the analytical pricing model,

$$\boxed{\widehat{C}_{\mathrm{IV-ML}} = C_{\mathrm{BS}}\left(S, K, T, r, \widehat{\sigma}_{\mathrm{IV}}\right).}$$

The student shall provide both

$$\sigma_{\mathrm{IV}} \text{ versus } \widehat{\sigma}_{\mathrm{IV}}$$

and the resulting option-price comparison.

12.12 Deliverable 12: Unified Model Comparison

The final computational package shall produce the comparison

$$\boxed{\begin{array}{c} C_{\mathrm{market}} \\ C_{\mathrm{BS}} \\ \widehat{C}_{\mathrm{ML}} \\ \widehat{C}_{\mathrm{Hybrid}} \\ \widehat{C}_{\mathrm{IV-ML}}. \end{array}}$$

For each model, the student shall report

$$\mathrm{ME}, \qquad \mathrm{MAE}, \qquad \mathrm{RMSE}, \qquad R^2.$$

The comparison shall also be reported by moneyness and maturity.

12.13 Deliverable 13: Mandatory Validation Figures

The final project shall contain at least the following figures:

1. Sample GBM paths.
2. Black–Scholes option price versus underlying price.
3. Delta versus underlying price.
4. Gamma versus underlying price.
5. Vega versus underlying price.
6. Binomial convergence toward Black–Scholes.
7. Monte Carlo convergence toward Black–Scholes.
8. Market price versus Black–Scholes price.
9. Market price versus pure ML price.
10. Market price versus residual-hybrid price.
11. True versus predicted implied volatility.
12. Implied-volatility surface.
13. Black–Scholes residual versus moneyness.
14. Hybrid residual versus moneyness.
15. Black–Scholes residual versus maturity.

16. Hybrid residual versus maturity.
17. RMSE by moneyness category.
18. RMSE by maturity category.
19. Error heat map in the (m, T) plane.

12.14 Deliverable 14: Market-Risk Analysis Package

The student shall construct a small option portfolio,

$$V_P = \sum_{i=1}^{N} n_i C_i,$$

and calculate

$$\Delta_P, \qquad \Gamma_P, \qquad \nu_P.$$

The risk-analysis code shall compute

$$\Delta V_P^{\Delta},$$

$$\Delta V_P^{\Delta\Gamma},$$

$$\Delta V_P^{\Delta\Gamma\nu},$$

and

$$\Delta V_P^{\text{full}}.$$

The student shall compare Taylor approximations against full repricing.

12.15 Deliverable 15: Stress-Test Results

The project shall contain deterministic stress scenarios involving

$$S \longrightarrow S(1 + \delta_S),$$

and

$$\sigma \longrightarrow \sigma + \Delta\sigma.$$

At least one combined price–volatility stress matrix shall be produced.

The student shall compare

$$\Delta V_P^{\text{BS}}$$

with

$$\Delta V_P^{\text{Hybrid}}.$$

The resulting model-risk difference,

$$\boxed{R_{\text{model}} = \Delta V_P^{\text{Hybrid}} - \Delta V_P^{\text{BS}},}$$

shall be reported.

12.16 Deliverable 16: Mandatory Risk Figures

The project shall contain at least

1. portfolio value versus underlying price;
2. portfolio delta versus underlying price;
3. portfolio gamma versus underlying price;
4. portfolio vega versus underlying price;
5. Taylor approximation versus full repricing;
6. portfolio P&L under price stress;
7. portfolio P&L under volatility stress;
8. combined stress heat map; and
9. Black–Scholes versus hybrid stressed P&L.

12.17 Deliverable 17: Jupyter Notebooks

The development process shall be documented through a sequence of notebooks.

A recommended set is

```
week01_returns_analysis.ipynb
week02_gbm_simulation.ipynb
week05_validation.ipynb
week06_data_generation.ipynb
week07_regression.ipynb
week08_ml_models.ipynb
week09_hybrid_models.ipynb
week10_model_validation.ipynb
week11_market_risk.ipynb.
```

Equivalent organization is acceptable provided that the workflow remains clear and reproducible.

12.18 Deliverable 18: Reproducibility Package

The student shall submit sufficient information for another student or supervisor to reproduce the principal results.

The package shall contain

Source Code
Notebooks
Datasets or Data-Preparation Instructions
Package Requirements
Random Seeds
Model Hyperparameters
Saved Final Results.

A requirements file such as

`requirements.txt`

or an equivalent environment specification should be included.

12.19 Deliverable 19: Final Technical Report

The student shall prepare a structured technical report.

A recommended organization is

1. Introduction
2. Mathematical Foundation
3. Black–Scholes Model
4. Numerical Methods
5. Dataset and Feature Engineering
6. Machine-Learning Algorithms
7. Hybrid BS–ML Framework
8. Model Validation
9. Market-Risk Analysis
10. Results and Discussion
11. Limitations
12. Conclusions and Future Work.

The report shall emphasize interpretation rather than presenting large volumes of raw software output.

12.20 Deliverable 20: Final Presentation

The student shall prepare a technical presentation summarizing the project.

The presentation shall follow the logical sequence

Problem
↓
GBM and Black–Scholes
↓
Numerical Validation
↓
Machine Learning
↓
Hybrid Model
↓
Validation
↓
Market-Risk Analysis
↓
Conclusions.

The presentation should emphasize the research question and principal results rather than explaining every line of code.

12.21 Minimum Acceptable Final Package

The minimum acceptable final submission shall therefore contain

Validated Black–Scholes Code
Binomial and Monte Carlo Code
Greeks and Implied-Volatility Solver
Synthetic Dataset
Processed Real-Market Dataset
OLS / Ridge / Tree / ANN Models
Residual Hybrid Model
IV-ML Hybrid Model
Out-of-Sample Validation
Portfolio Risk and Stress Analysis
Technical Report
Final Presentation.

12.22 Deliverable Acceptance Principle

A deliverable shall not be considered complete merely because a file exists.

Every computational deliverable must satisfy three conditions:

Correctness
+
Reproducibility
+
Interpretability.

Thus, the student must be able to explain

what was computed,

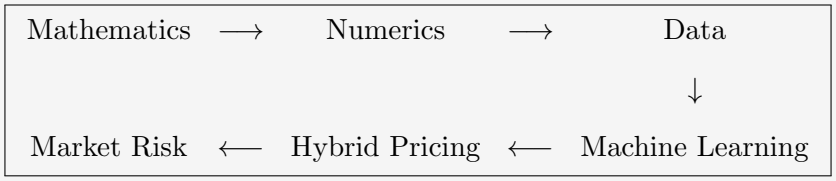
why it was computed,

and

how the result was validated.

Project Milestone

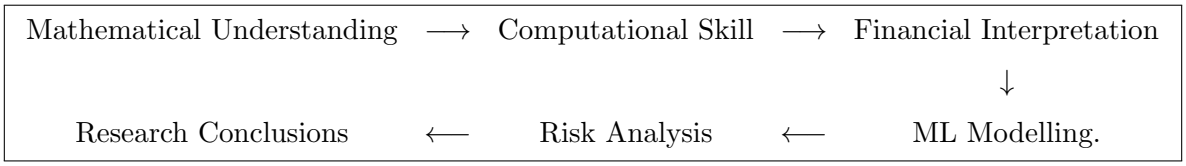
The final project shall not be judged solely by the predictive performance of the neural network.
Successful completion requires an integrated and reproducible framework connecting



13. Assessment and Project Milestones

The project shall be evaluated continuously over the twelve-week period.

The assessment is designed to ensure that the student develops



The final grade shall therefore not be determined solely by the prediction accuracy of the machine-learning models.

13.1 Assessment Components

The recommended assessment distribution is

Component	Weight
Mathematical Foundation	15%
Black–Scholes and Numerical Implementation	15%
Data Preparation and Feature Engineering	10%
Machine-Learning Models	15%
Hybrid BS–ML Framework	15%
Validation and Market-Risk Analysis	15%
Final Report	10%
Presentation and Viva	5%
Total	**100%**

13.2 Component 1: Mathematical Foundation

The student shall demonstrate understanding of

Returns and Log Returns
Brownian Motion
Geometric Brownian Motion
Itô's Lemma
Black–Scholes PDE
Delta Hedging
Greeks
Implied Volatility.

Assessment shall emphasize understanding rather than memorization of formulas.

The student should be able to explain the sequence

$$\begin{array}{ccccc} \text{GBM} & \longrightarrow & \text{Itô's Lemma} & \longrightarrow & \text{Delta Hedging} \\ & & & & \downarrow \\ \text{Option Price} & \longleftarrow & \text{BS Solution} & \longleftarrow & \text{BS PDE.} \end{array}$$

13.3 Component 2: Black–Scholes and Numerical Implementation

The student shall demonstrate correct implementation of

$$C_{\mathrm{BS}}, \qquad P_{\mathrm{BS}}, \qquad \Delta, \qquad \Gamma, \qquad \nu,$$

and

$$\sigma_{\mathrm{IV}}.$$

The analytical implementation shall be independently checked using

Binomial Pricing

and

Monte Carlo Pricing.

A central numerical requirement is

$$\boxed{C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}.}$$

13.4 Component 3: Data Preparation and Feature Engineering

Assessment shall include the student's ability to transform raw financial data into modelling variables.

The student shall demonstrate the construction of quantities such as

$$m = \ln\left(\frac{S}{K}\right),$$

$$T, \qquad r, \qquad \sigma_{\mathrm{hist}},$$

and, where required,

$$C_{\mathrm{BS}}, \qquad \Delta, \qquad \Gamma, \qquad \nu, \qquad \sigma_{\mathrm{IV}}.$$

Special attention shall be given to avoiding data leakage.

13.5 Component 4: Machine-Learning Models

The student shall demonstrate implementation and comparison of

OLS
Ridge Regression
Tree-Based Ensemble
Small Neural Network.

The assessment shall consider

model selection,

hyperparameter choice,

overfitting,

and

out-of-sample performance.

13.6 Component 5: Hybrid Black–Scholes–ML Framework

This component represents one of the principal research elements of the project.

The student shall demonstrate the two hybrid architectures

$$\boxed{\widehat{C}_{\text{Hybrid}} = C_{\text{BS}} + \widehat{\varepsilon}_{\text{ML}}}$$

and

$$\boxed{\widehat{C}_{\text{IV}-\text{ML}} = C_{\text{BS}}\left(\widehat{\sigma}_{\text{IV}}\right).}$$

The student must clearly explain why

Residual Learning

and

Implied-Volatility Learning

represent different modelling strategies.

13.7 Component 6: Validation and Market-Risk Analysis

The student shall compare

Black–Scholes
Pure ML
Residual Hybrid
IV-ML Hybrid.

The comparison shall include

$$\text{MAE}, \qquad \text{RMSE}, \qquad R^2,$$

together with residual analysis.

The student shall also demonstrate portfolio-risk calculations using

$$\Delta_P, \qquad \Gamma_P, \qquad \nu_P,$$

and

$$\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.$$

Full repricing shall be used to validate the Taylor approximation.

13.8 Formal Review Gate 1: End of Week 4

The first formal review evaluates the mathematical and analytical foundation.

The student must demonstrate

GBM Simulation
↓
Black–Scholes Pricing
↓
Greeks
↓
Implied Volatility.

The student shall present working code and explain the mathematics behind each calculation.

Model-Risk Perspective

If the student cannot explain the Black–Scholes model and implied volatility at this stage, the project shall not proceed directly to machine learning.
The mathematical and numerical foundation must first be corrected.

13.9 Formal Review Gate 2: End of Week 8

The second review evaluates numerical validation and ML readiness.

The student must demonstrate

$$C_{\text{BS}} \approx C_{\text{Binomial}} \approx C_{\text{MC}}$$
↓
Validated Dataset
↓
ML Models.

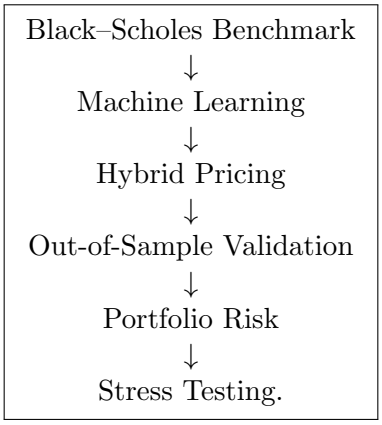
The review shall include

$$\text{OLS}, \quad \text{Ridge}, \quad \text{Tree Model}, \quad \text{ANN}.$$

The student must demonstrate that the train, validation and test sets have been constructed correctly.

13.10 Formal Review Gate 3: End of Week 12

The final review shall demonstrate the complete project architecture,



The student shall present both successful and unsuccessful modelling results.

Unexpected results shall not be removed merely because they do not support the initial hypothesis.

13.11 Final Viva Questions

The final viva shall test conceptual understanding.

The student should be able to answer questions such as

1. Why is geometric Brownian motion used in the classical Black–Scholes framework?
2. Why does the drift μ disappear from the Black–Scholes PDE?
3. What is the financial meaning of delta hedging?
4. Why is implied volatility obtained through numerical inversion?
5. Why should Black–Scholes be retained as a benchmark rather than replaced immediately by ML?
6. Why can residual learning be easier than learning the entire option-price function?
7. What is the difference between residual learning and implied-volatility learning?
8. Why must financial data be split chronologically?
9. Why can a model with lower RMSE still be financially unacceptable?
10. What happens to an ML model under extrapolation?
11. Why does delta alone fail to describe nonlinear option risk?
12. When does the delta–gamma approximation become inaccurate?
13. Does the hybrid model materially change portfolio risk under stress?

13.12 Criteria for Successful Completion

Successful completion requires more than obtaining a high

$$R^2.$$

The student must demonstrate

Mathematical Understanding
↓
Correct Implementation
↓
Independent Numerical Validation
↓
Responsible ML Modelling
↓
Financial Interpretation
↓
Reproducible Conclusions.

Key Idea

The purpose of the project is not to demonstrate that machine learning is automatically superior to Black–Scholes.
The purpose is to investigate, through controlled numerical experiments, where data-driven models improve classical pricing, where they fail, and whether these differences materially affect market-risk estimates.

13.13 Final Project Success Criterion

The project shall be regarded as successful if the student can defend, using both mathematical reasoning and numerical evidence, the complete relationship

Black–Scholes ⟶ ML Correction ⟶ Improved Pricing?
↓
Model-Risk Insight ⟵ Stress Testing ⟵ Portfolio Risk.

The question mark in

Improved Pricing?

is intentional.

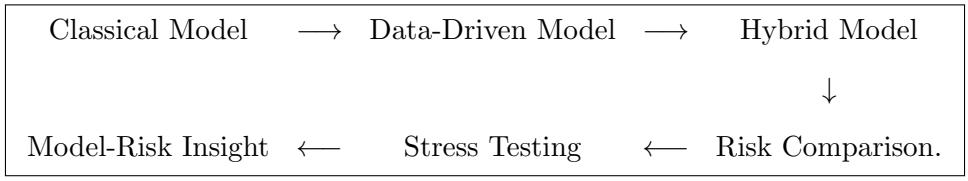
Whether machine learning genuinely improves the classical model is a result to be established by the project, not an assumption made at the beginning.

14. Expected Outcomes, Limitations and Future Extensions

The project is designed as a three-month investigation into the interaction between classical mathematical finance and modern machine-learning methods.

The objective is not to replace the Black–Scholes framework, but to use it as a mathematically interpretable benchmark around which data-driven models can be constructed, tested and critically evaluated.

The expected progression is



14.1 Expected Outcome 1: Validated Black–Scholes Framework

The first expected outcome is a complete computational implementation of the Black–Scholes framework.

The student shall be able to calculate

$$C_{\mathrm{BS}}, \qquad P_{\mathrm{BS}},$$

together with

$$\Delta, \qquad \Gamma, \qquad \nu,$$

and implied volatility,

$$\sigma_{\mathrm{IV}}.$$

The implementation shall be independently checked using

$$C_{\mathrm{Binomial}}$$

and

$$C_{\mathrm{MC}}.$$

Thus, the first expected computational result is

$$\boxed{C_{\mathrm{BS}} \approx C_{\mathrm{Binomial}} \approx C_{\mathrm{MC}}.}$$

14.2 Expected Outcome 2: Understanding the Black–Scholes Error

When real market prices are introduced, the student shall calculate

$$\boxed{\varepsilon_{\mathrm{BS}} = C_{\mathrm{market}} - C_{\mathrm{BS}}.}$$

The project shall investigate whether this residual exhibits systematic dependence on variables such as

$$m = \ln\left(\frac{S}{K}\right),$$

$$T,$$

and the selected volatility measure.

The objective is to determine whether Black–Scholes pricing errors contain structure that may be learned from data.

14.3 Expected Outcome 3: Pure Machine-Learning Benchmark

The student shall construct a direct machine-learning pricing model,

$$\boxed{\widehat{C}_{\mathrm{ML}} = f_\theta(\mathbf{x}).}$$

This model provides an important comparison because it asks whether the option-pricing relationship can be learned directly from the selected features without explicitly retaining the analytical pricing formula.

The performance of this model shall be compared with

$$C_{\mathrm{BS}}.$$

14.4 Expected Outcome 4: Residual Hybrid Model

A principal expected outcome is the construction of

$$\boxed{\widehat{C}_{\mathrm{Hybrid}} = C_{\mathrm{BS}} + \widehat{\varepsilon}_{\mathrm{ML}}.}$$

This architecture preserves the classical pricing model while allowing machine learning to estimate systematic pricing discrepancies.

The project shall determine whether

$$\mathrm{RMSE}_{\mathrm{Hybrid}} < \mathrm{RMSE}_{\mathrm{BS}}$$

on genuinely unseen data.

However, this inequality is an experimental question and shall not be assumed in advance.

14.5 Expected Outcome 5: ML Implied-Volatility Model

The second hybrid architecture shall estimate

$$\boxed{\widehat{\sigma}_{\mathrm{IV}} = g_\theta(m, T, \ldots).}$$

The predicted volatility is then passed through the classical pricing formula,

$$\boxed{\widehat{C}_{\mathrm{IV-ML}} = C_{\mathrm{BS}}\left(S, K, T, r, \widehat{\sigma}_{\mathrm{IV}}\right).}$$

This provides a different mechanism for combining financial structure with machine learning.

14.6 Expected Outcome 6: Identification of Where ML Helps

The project should not report only a single overall RMSE.

Instead, the student shall determine whether ML improvement depends on

Moneyness Maturity Volatility Region Market Regime.

For example, it is possible that

$$\mathrm{RMSE}_{\mathrm{Hybrid}} < \mathrm{RMSE}_{\mathrm{BS}}$$

for one region of the option surface, while

$$\mathrm{RMSE}_{\mathrm{Hybrid}} \geq \mathrm{RMSE}_{\mathrm{BS}}$$

in another.

Such a result is scientifically meaningful and shall not be treated as a failure of the project.

14.7 Expected Outcome 7: Financial Consistency Assessment

The project shall determine whether improved statistical accuracy is accompanied by financially sensible predictions.

The student shall examine

$$\widehat{C} \geq 0,$$

$$\widehat{C} \leq S,$$

and the lower no-arbitrage bound

$$\widehat{C} \geq \max\left(S - Ke^{-rT}, 0\right).$$

The model shall also be checked for

$$\frac{\partial \widehat{C}}{\partial S} \geq 0$$

and

$$\frac{\partial^2 \widehat{C}}{\partial S^2} \geq 0$$

where appropriate.

Thus, the project distinguishes

$$\boxed{\text{Statistical Accuracy} \neq \text{Financial Validity.}}$$

14.8 Expected Outcome 8: Portfolio-Risk Comparison

The student shall construct a small option portfolio and calculate

$$\Delta_P, \qquad \Gamma_P, \qquad \nu_P.$$

The local portfolio-risk approximation is

$$\boxed{\Delta V_P \approx \Delta_P \Delta S + \frac{1}{2}\Gamma_P (\Delta S)^2 + \nu_P \Delta\sigma.}$$

This approximation shall be compared with full repricing.

The project shall therefore demonstrate the transition

$$\boxed{\begin{array}{ccccc} \text{Option Price} & \longrightarrow & \text{Greeks} & \longrightarrow & \text{Portfolio Risk} \\ & & & & \downarrow \\ \text{Model Comparison} & \longleftarrow & \text{Stress P\&L} & \longleftarrow & \text{Full Repricing.} \end{array}}$$

14.9 Expected Outcome 9: Quantification of Model Risk

For each stress scenario, the student shall compare

$$\Delta V_P^{\mathrm{BS}}$$

with

$$\Delta V_P^{\mathrm{Hybrid}}.$$

The model-risk difference is

$$\boxed{R_{\mathrm{model}} = \Delta V_P^{\mathrm{Hybrid}} - \Delta V_P^{\mathrm{BS}}.}$$

The magnitude

$$|R_{\mathrm{model}}|$$

provides a simple measure of how strongly the estimated portfolio response depends on the pricing model.

This is one of the principal links between the pricing and market-risk components of the project.

14.10 Expected Outcome 10: Understanding ML Extrapolation Risk

Machine-learning models learn from the region represented in their training data.

The project shall therefore distinguish between

$$\boxed{\text{Interpolation}}$$

and

$$\boxed{\text{Extrapolation.}}$$

A stress scenario may move an observation outside the region covered during training.

The student shall investigate whether ML predictions become unstable under such conditions.

This is particularly important because stress testing deliberately examines unusual market states.

14.11 Limitations of the Study

The project is intentionally restricted so that the complete framework can be developed within twelve weeks.

The principal limitations shall be stated explicitly.

14.11.1 Black–Scholes Assumptions

The classical Black–Scholes framework is based on restrictive assumptions.

The project shall not attempt to remove all of these assumptions.

In particular, the benchmark retains the classical continuous-time diffusion structure

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.$$

Real financial markets may exhibit behaviour that is not fully represented by this model.

14.11.2 Constant-Volatility Benchmark

The basic Black–Scholes model assumes a volatility input that does not depend explicitly on strike and maturity.

Market implied volatility may instead display structure of the form

$$\boxed{\sigma_{\mathrm{IV}} = f(K, T),}$$

or equivalently,

$$\sigma_{\mathrm{IV}} = f(m, T).$$

The ML implied-volatility experiment investigates this structure but does not constitute a complete stochastic-volatility model.

14.11.3 Limited Market Dataset

The quality of the real-data experiment depends on the availability and cleanliness of option-market data.

Potential difficulties include

missing observations,

illiquid contracts,

bid–ask effects,

and

inconsistent timestamps.

These limitations shall be documented rather than hidden through excessive preprocessing.

14.11.4 Transaction Costs and Liquidity

The basic Black–Scholes derivation assumes frictionless trading.

The core project does not explicitly model

transaction costs,

market impact,

or

liquidity constraints.

Consequently, the delta-hedging and portfolio-risk calculations should be interpreted as model-based risk estimates rather than complete trading strategies.

14.11.5 Restricted Machine-Learning Complexity

The project deliberately restricts the ML algorithms to

Linear / Ridge Regression Tree-Based Ensemble Small Neural Network.

Large deep-learning architectures are outside the core scope.

The objective is to understand the interaction between financial structure and machine learning rather than to maximize neural-network complexity.

14.11.6 Limited Hyperparameter Search

A three-month project does not require exhaustive optimization over large hyperparameter spaces.

The student shall use a controlled and documented search over a small number of meaningful hyperparameters.

This reduces the possibility of obtaining apparently strong results through excessive tuning to the validation data.

14.11.7 Stress Scenarios are Not Forecasts

The deterministic stress scenarios used in this project are

$$\boxed{\text{“What-if” experiments,}}$$

not forecasts of future market movements.

For example,

$$S \to 0.90S$$

does not imply that a 10% market decline is predicted.

It measures the model and portfolio response conditional on such a movement.

14.11.8 No Complete Regulatory Risk System

The project does not attempt to construct a complete regulatory market-risk platform.

A full treatment of

$$\text{VaR}, \qquad \text{Expected Shortfall}, \qquad \text{backtesting},$$

and advanced regulatory capital calculations lies beyond the primary scope of this three-month study.

Model-Risk Perspective

The limited scope is intentional.
A smaller framework that is mathematically understood, independently validated and reproducible is preferable to a large collection of advanced algorithms that the student cannot properly interpret or verify.

14.12 Future Extension 1: Volatility-Surface Modelling

A natural extension is to develop a more complete representation of

$$\sigma_{\text{IV}}(m, T).$$

The student could investigate

volatility smile,

volatility skew,

and

term structure.

Machine learning could then be used to construct a smooth two-dimensional implied-volatility surface.

14.13 Future Extension 2: Stochastic Volatility

The constant-volatility assumption may be replaced by a model in which volatility itself evolves randomly.

Conceptually,

$$\boxed{\begin{aligned} dS_t &= \text{Price Dynamics} \\ d\sigma_t &= \text{Volatility Dynamics.} \end{aligned}}$$

This would allow comparison between

Black–Scholes,

Stochastic-Volatility Models,

and

ML Hybrid Models.

14.14 Future Extension 3: Jump Models

Large market movements may not be represented adequately by continuous GBM paths.

A future model could introduce jumps,

$$\boxed{\text{Diffusion} + \text{Jump Component.}}$$

The effect of jump risk on option pricing and hybrid-model residuals could then be investigated.

14.15 Future Extension 4: Physics-Informed or Finance-Informed Neural Networks

A more advanced ML model could incorporate the Black–Scholes PDE directly into the learning objective.

Suppose the neural-network approximation is

$$V_\theta(S,t).$$

Define the Black–Scholes PDE residual

$$\boxed{\mathcal{R}_{\mathrm{BS}} = \frac{\partial V_\theta}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V_\theta}{\partial S^2} + rS\frac{\partial V_\theta}{\partial S} - rV_\theta.}$$

A future loss function could combine data error with the PDE residual,

$$\boxed{\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \lambda \mathcal{L}_{\mathrm{BS}}.}$$

This would create a finance-informed neural network in which the analytical model acts as a structural constraint rather than merely as an input feature.

14.16 Future Extension 5: No-Arbitrage-Constrained ML

Financial restrictions could be incorporated directly into the ML objective.

For example, penalties could be introduced when

$$\frac{\partial \widehat{C}}{\partial S} < 0$$

or

$$\frac{\partial^2 \widehat{C}}{\partial S^2} < 0.$$

Conceptually,

$$\boxed{\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{pricing}} + \lambda_1 \mathcal{L}_{\mathrm{arbitrage}} + \lambda_2 \mathcal{L}_{\mathrm{smoothness}}.}$$

This would move the methodology from purely data-driven learning toward financially constrained machine learning.

14.17 Future Extension 6: Dynamic Hedging

The current project examines sensitivities and static stress scenarios.

A future study could simulate repeated delta hedging over

$$t_0 < t_1 < \cdots < t_N = T.$$

At each time step, the hedge could be recalculated,

$$\Delta_{t_i} = \frac{\partial V}{\partial S}\left(S_{t_i}, t_i\right).$$

The resulting hedging error could then be compared between

Black–Scholes

and

Hybrid ML

models.

14.18 Future Extension 7: Value-at-Risk and Expected Shortfall

The portfolio framework can subsequently be extended to loss distributions.
If

$$L = -\Delta V_P,$$

then a future study may estimate

$$\mathrm{VaR}_\alpha$$

and

$$\mathrm{ES}_\alpha\,.$$

This would extend the present project from

pricing-model risk

toward a broader market-risk measurement framework.

14.19 Future Extension 8: Regime-Aware Machine Learning

Financial markets may exhibit different behaviour during

low-volatility periods,

normal periods,

and

high-volatility periods.

A future architecture could therefore be

Market Data
↓
Regime Identification
↓
Regime-Specific ML Model
↓
Option Pricing and Risk.

This would permit investigation of whether model performance changes systematically across market regimes.

14.20 Future Extension 9: Model Uncertainty

The present project primarily compares point predictions.

A future study could estimate predictive uncertainty,

$$\widehat{C} \pm \text{uncertainty}.$$

The objective would then change from predicting only

"What is the option price?"

to also asking

"How uncertain is the model about that price?"

This is especially relevant when the model operates outside its training domain.

14.21 Future Extension 10: Explainable AI

For tree-based models, feature importance can provide an initial interpretation of the learned relationship.

More advanced work could investigate how predictions depend on

$$m, \qquad T, \qquad r, \qquad \sigma, \qquad \Delta, \qquad \Gamma, \qquad \nu.$$

The objective would be to determine whether the ML model learns relationships that are financially interpretable rather than merely statistically predictive.

14.22 Research Path Beyond the Three-Month Project

The possible longer-term research path can therefore be summarized as

BS + Basic ML ⟶ Hybrid ML ⟶ Constrained ML
↓
Advanced Risk ⟵ Dynamic Hedging ⟵ Advanced Volatility.

The present project occupies the first two stages of this progression.

14.23 Final Expected Research Outcome

At the end of the project, the student should be able to answer four central questions.

1. How accurately does the classical Black–Scholes model describe the selected option-price dataset?
2. Can machine learning identify systematic structure in the discrepancy between Black–Scholes and observed market prices?
3. Does a hybrid Black–Scholes–ML model improve genuinely out-of-sample pricing performance while maintaining reasonable financial behaviour?
4. Does the choice between classical and hybrid pricing models materially alter estimated portfolio risk under market stress?

Key Idea

The strongest possible conclusion is not necessarily

"Machine learning beats Black–Scholes."

A scientifically stronger conclusion may be

where ML improves the classical model,
where it does not,
and what the difference means for market risk.

14.24 Closing Perspective

The complete project can now be summarized compactly as

$$\begin{array}{ccccc} \text{Black–Scholes} & \longrightarrow & \text{ML} & \longrightarrow & \text{Hybrid Pricing} \\ & & & & \downarrow \\ \text{Research Insight} & \longleftarrow & \text{Model Risk} & \longleftarrow & \text{Market Risk.} \end{array}$$

The project therefore moves beyond the question of whether machine learning can reproduce option prices.

It asks the more important question of whether combining mathematical finance with data-driven learning can produce a pricing and risk framework that is simultaneously

Accurate
Interpretable
Financially Consistent
Robust under Stress.

Project Milestone

The final outcome of the three-month project shall be a reproducible Black–Scholes–AI/ML framework in which classical mathematical finance provides the structural benchmark, machine learning provides the data-driven correction, and market-risk analysis determines whether that correction is financially meaningful.

References

[1] Carol Alexander. *Market Risk Analysis, Volume I: Quantitative Methods in Finance*. John Wiley & Sons, Chichester, 2008.

[2] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

[3] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

[4] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[5] John C. Cox, Stephen A. Ross, and Mark Rubinstein. Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3):229–263, 1979.

[6] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, New York, 2004.

[7] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, Cambridge, Massachusetts, 2016.

[8] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2 edition, 2009.

[9] Desmond J. Higham. *An Introduction to Financial Option Valuation: Mathematics, Stochastics and Computation*. Cambridge University Press, Cambridge, 2004.

[10] Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

[11] John C. Hull. *Options, Futures, and Other Derivatives*. Pearson, 11 edition, 2022.

[12] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *International Conference on Learning Representations*, 2015.

[13] Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

[14] Steven E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, New York, 2004.